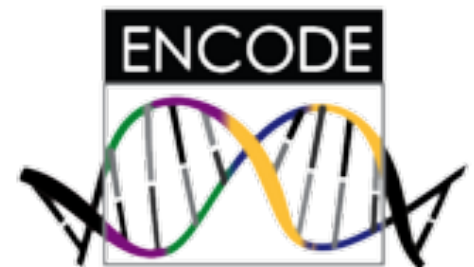


The Encode Project

I-II

<https://www.encodeproject.org/help/project-overview/>



What is



standing for ?

- an ENCyclopedia Of Dna Elements
- an international project launched by the US National Human Genome Research Institute (NHGRI), who also headed the HGP (Human Genome Project).
- a consortium of 442 scientists from all over the world
- a repository of functional elements of the genome
- a **goal** to build a comprehensive parts list of functional elements in the human genome, including elements that act at the protein and RNA levels, and regulatory elements that control cells and circumstances in which a gene is active.

Why ENCODE?

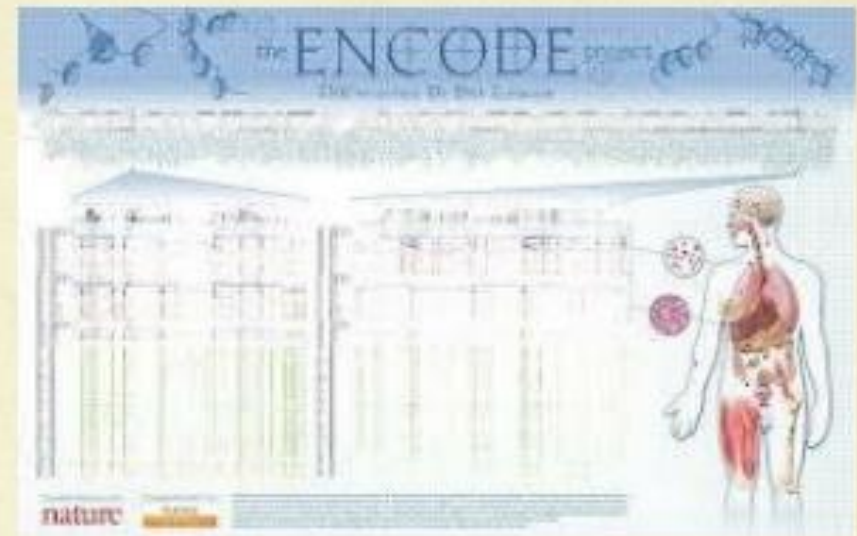
Moving us from “Here’s the genome” towards “Here’s what the genome does”.

~ by some guy



1990 ~ 2001, Human Genome Project

As, Ts, Cs, Gs



2003 ~ 2012, ENCODE

living organisms



Introduction

- Coding region accounts for about 2% of the human genome.
- Non-coding region actually has a role in regulating the activity of gene expression.

The importance of Non-coding DNA

- Carries out regulatory functions of genome and is important for disease and gene regulation:
 - ❑ Genes are regulated throughout time and space responding to environmental exposures.
 - ❑ Majority of common disease-associated genetic variants lie outside of protein-coding regions: about 90% of GWAS findings lie outside of protein-coding regions.

The goal of ENCODE

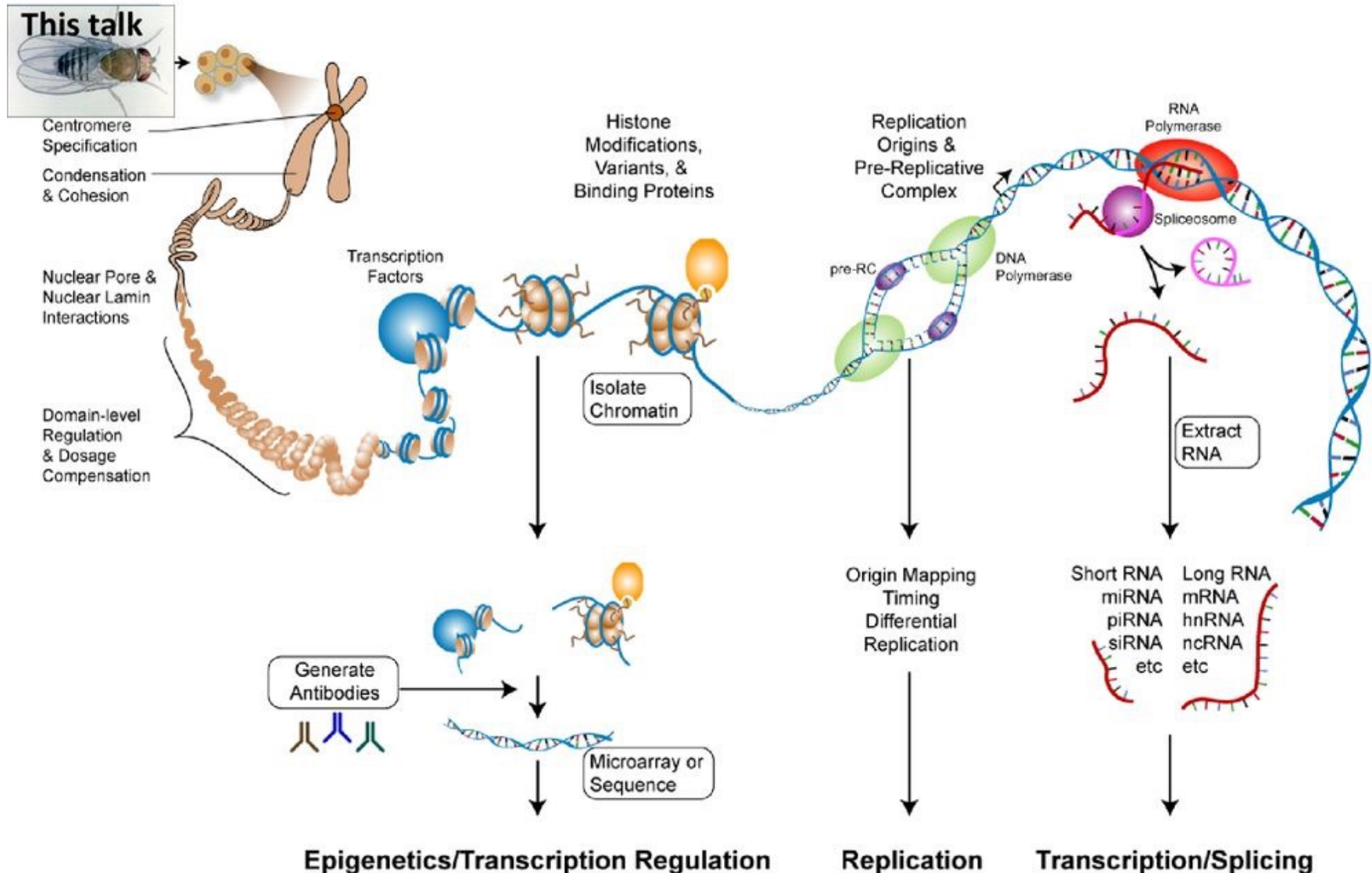
- The goal of ENCODE is to build a comprehensive parts list of **functional elements** in the **human genome and select model organism genomes**, including elements that act at the **protein and RNA levels**, and **regulatory elements** that control cells and circumstances in which a gene is active.
- Develop **freely available resource** for research community to enhance understanding of **regulation of gene expression and genetic basis of disease**



The introduction of Encode

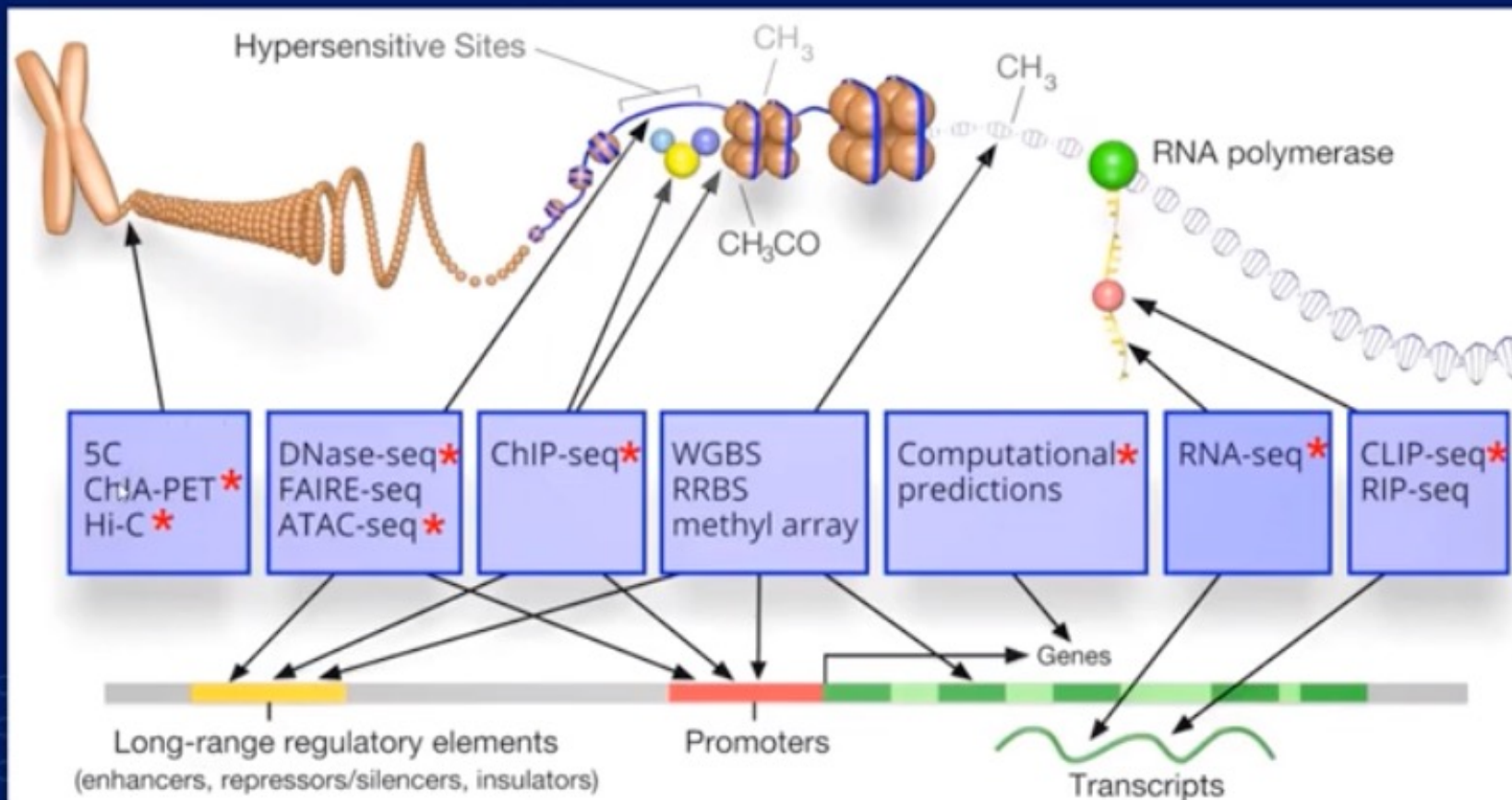
- Catalog all functional elements in the human and select model organism genomes
- Develop freely available resource for research community to enhance understanding of
 - ☐ Regulation of gene expression
 - ☐ Genetic basis of disease
- Project components:
 - ☐ Data generation
 - ☐ Data analysis
 - ☐ Data repository
 - ☐ Outreach

mod/ENCODE: (aka. everything you wanted to know about gene regulation but were afraid to ask)



Encode Project components: Data generation

ENCODE Assays



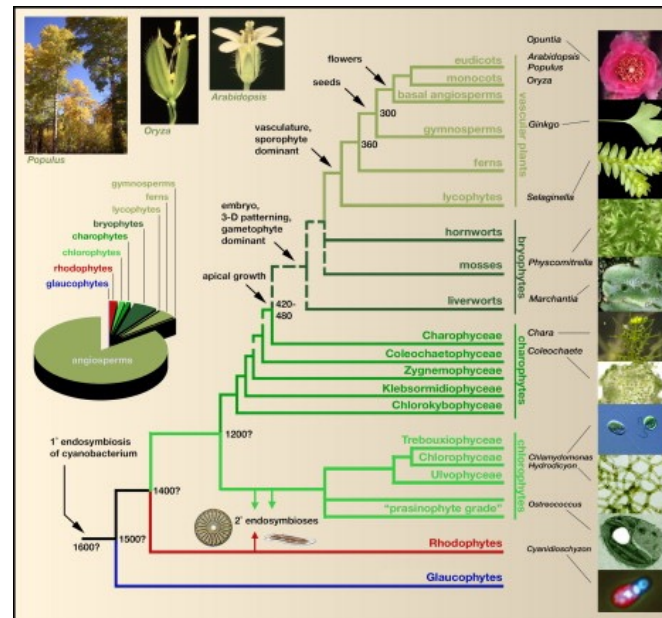
Biochemical
Features

Assays

Genomic
Elements

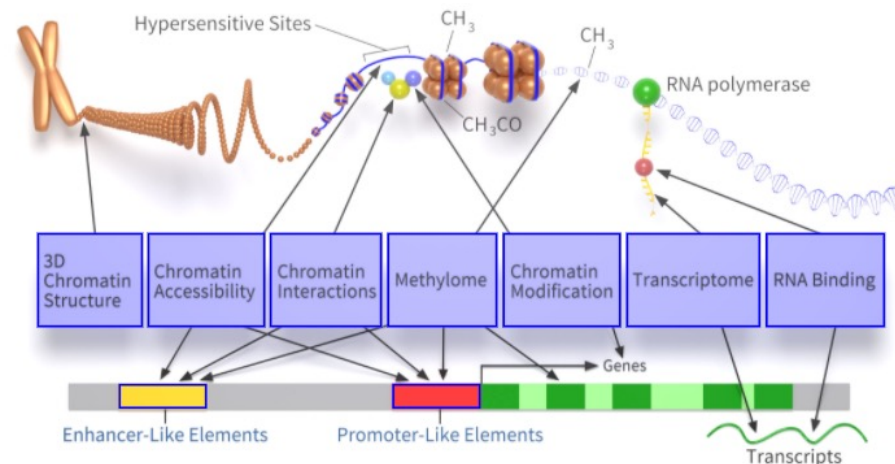
The assays and methods for identifying functional element

- The discovery and annotation of **gene elements** is accomplished primarily by **sequencing** a diverse range of **RNA sources**, **comparative genomics**, integrative **bioinformatic methods**, and human curation.



The assays and methods for identifying functional element

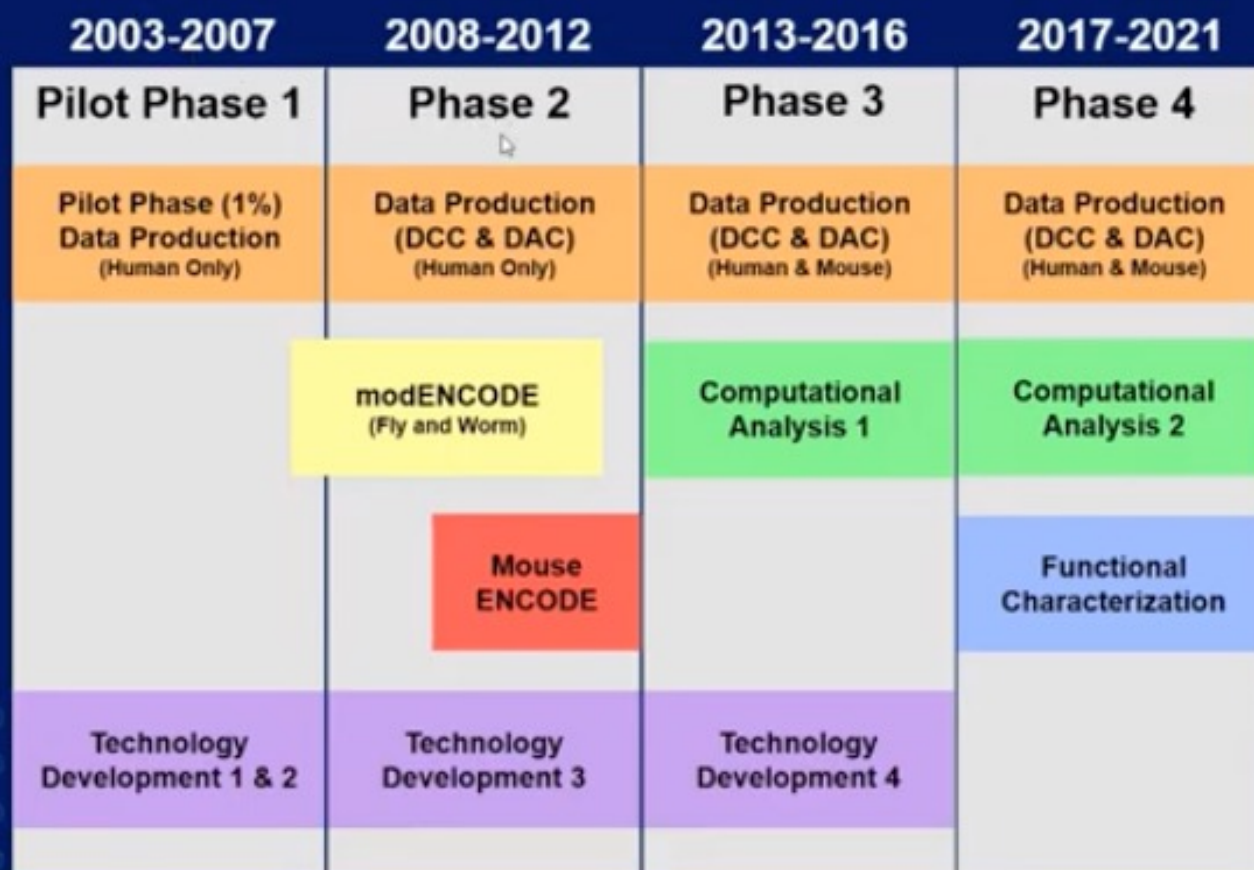
- **Regulatory elements** are typically investigated through **DNA hypersensitivity assays**, assays of DNA methylation, and **immunoprecipitation (IP)** of proteins that interact with DNA and RNA, i.e., modified histones, transcription factors, chromatin regulators, and RNA-binding proteins, followed by sequencing.



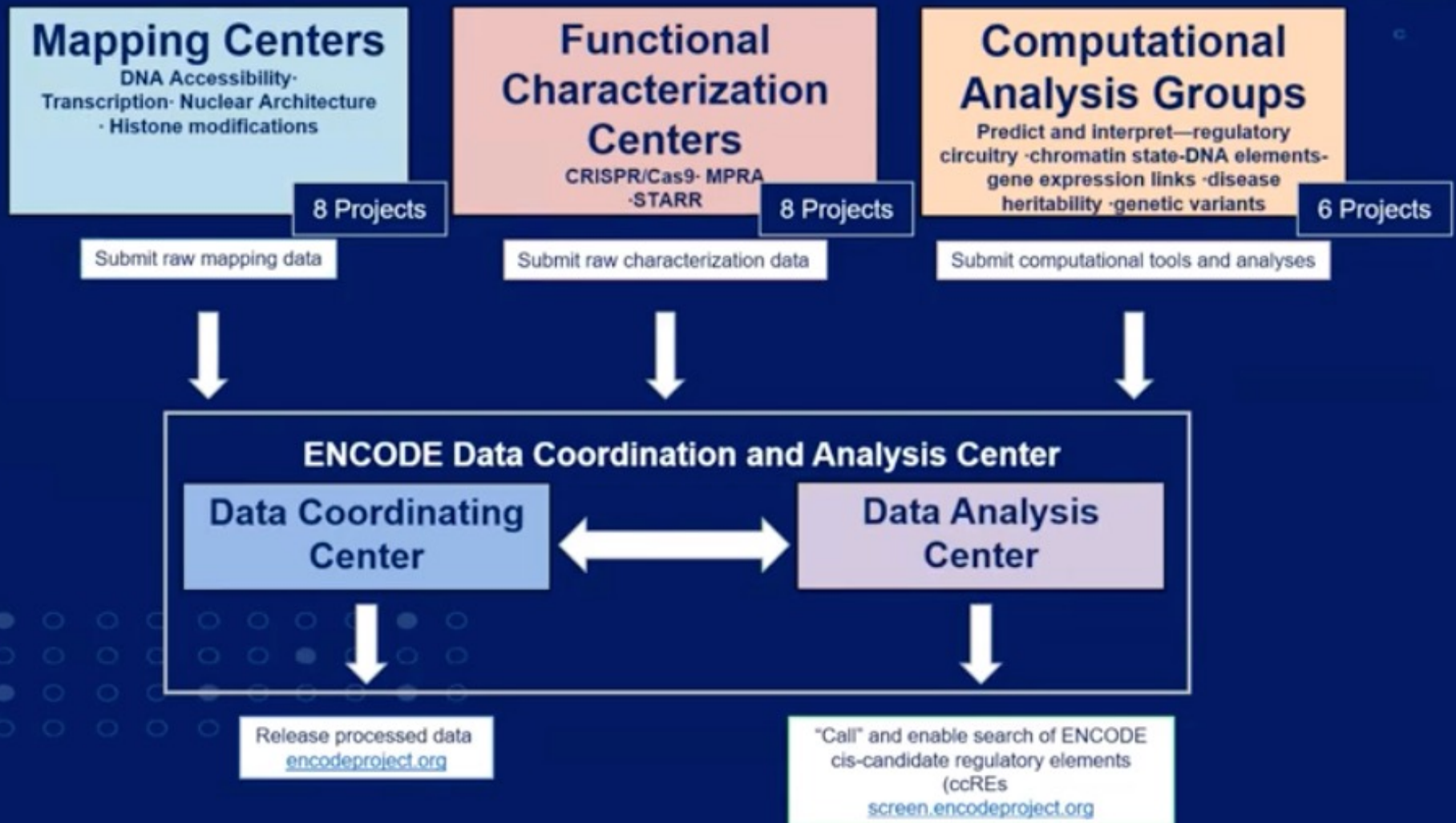
ENCODE Timeline

Project components

Data generation
Data analysis
Data repository

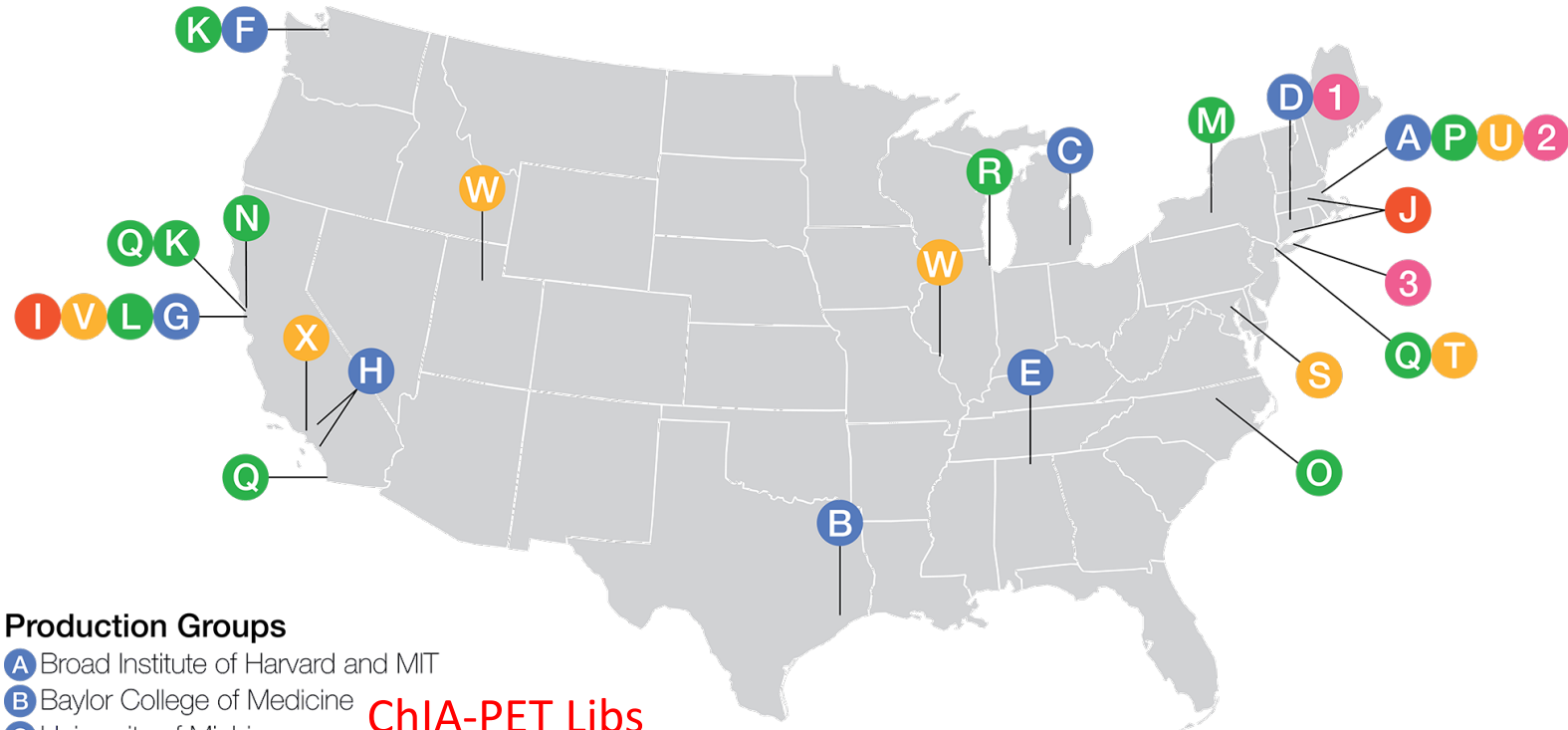


ENCODE 4 Organization



Current ENCODE Consortium

Participants and Projects_ENCODE 4



Production Groups

- A** Broad Institute of Harvard and MIT
- B** Baylor College of Medicine
- C** University of Michigan
- D** The Jackson Laboratory
- E** HudsonAlpha Institute for Biotechnology, University of Alabama in Huntsville
- F** Altius Institute for Biomedical Sciences
- G** Stanford University
- H** California Institute of Technology, University of California, Irvine

Data Coordination Center

- I** Stanford University

Data Analysis Center

- J** University of Massachusetts Medical School; Yale University

ChIA-PET Libs

Characterization Centers

- K** University of California, San Francisco; University of Washington
- L** Stanford University
- M** Cornell University
- N** Lawrence Berkeley National Laboratory
- O** Duke University
- P** Broad Institute of Harvard and MIT
- Q** University of California, San Francisco; University of California, San Diego; Ludwig Institute for Cancer Research
- R** University of Chicago

Computational Analysis Groups

- S** Johns Hopkins University
- T** Memorial Sloan Kettering Cancer Center
- U** Harvard University; Brigham and Women's Hospital
- V** Stanford University
- W** Washington University; University of Utah
- X** University of California, Los Angeles

Affiliated Groups

- 1** University of Connecticut Health Center
- 2** Dana-Farber Cancer Institute
- 3** Cold Spring Harbor Laboratory

Current ENCODE Consortium

Participants and Projects_ENCODE 4

- ENCODE 4 is composed primarily of scientists funded under Requests for Application (RFAs) released by **NHGRI in 2011**.
- NHGRI provides **a list of current ENCODE projects** and participants.
- The ENCODE **External Consultants Panel oversees** the activities of the Consortium and provides advice and feedback on the Consortium's goals, progress and membership.
- **The consortium is open** to any investigator willing to abide by the criteria for participation established for the ENCODE Project by NHGRI and has brought in additional participants when appropriate.

the ENCODE program directors



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Program Director **Genome Analysis**

Division of Genome Sciences

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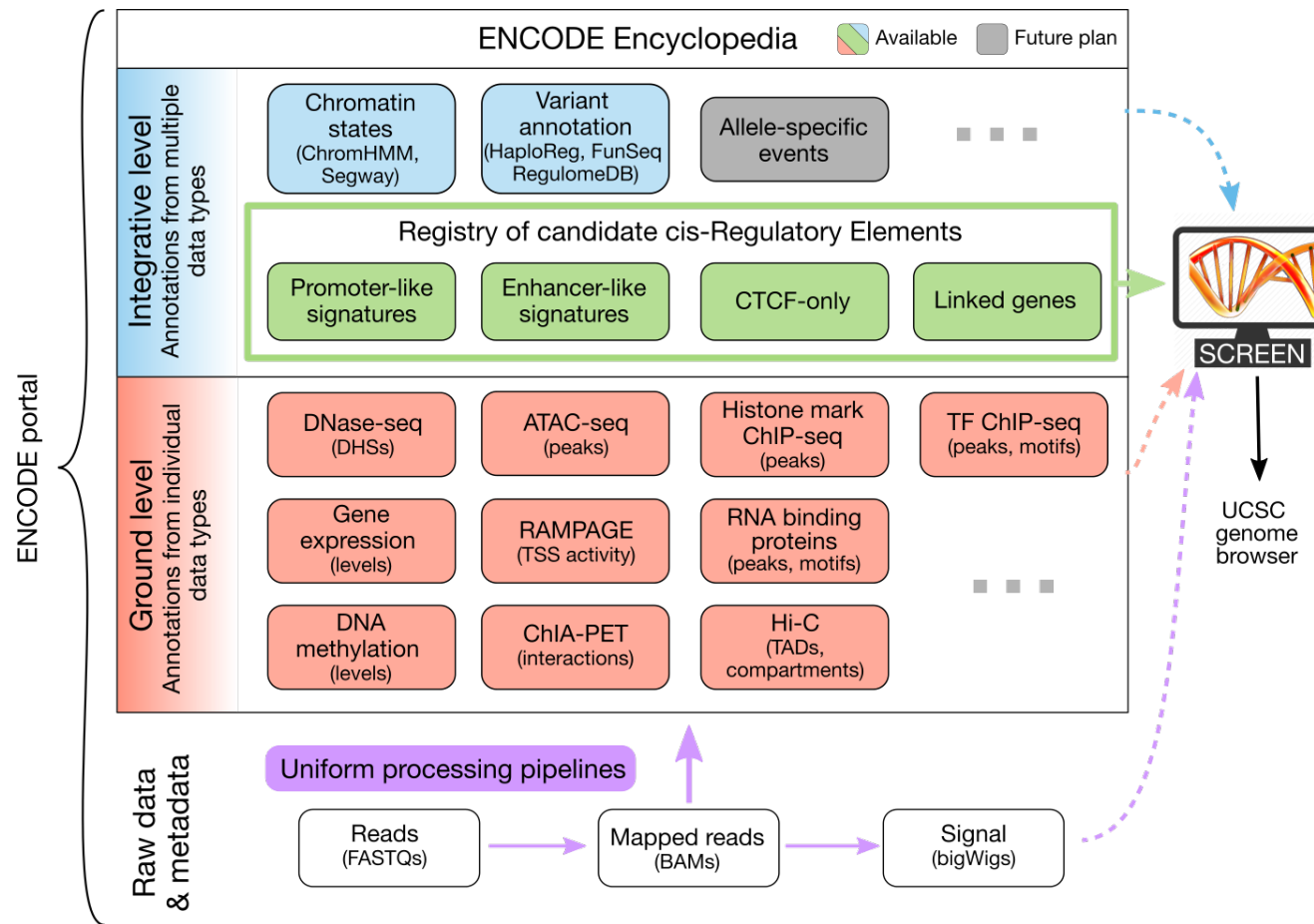
✉️ morris2@mail.nih.gov

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Encyclopedia Versions

- **Ground level** annotations include all released experiments to-date.
- **Integrative level** annotations are included in V3 & V4, as indicated in their metadata.
 - **V4** (based on ENCODE3, ENCODE2 and ROADMAP human data)
 - **V3** (based on ENCODE3, ENCODE2 and ROADMAP human data)
 - **V2** (based on ENCODE2 and ROADMAP human data)
 - V1 (prototype)

ENCODE Encyclopedia Version 5: Genomic and Transcriptomic Annotations



- The ENCODE Consortium not only **produces high-quality data**, but also **analyzes the data** in an integrative fashion. The ENCODE Encyclopedia organizes the most salient analysis products into annotations and **provides tools to search and visualize them**. The Encyclopedia has two levels of annotations:
 - Integrative-level annotations** integrate multiple types of experimental data and ground level annotations.
 - Ground-level annotations** are derived directly from the experimental data, typically produced by uniform processing pipelines.

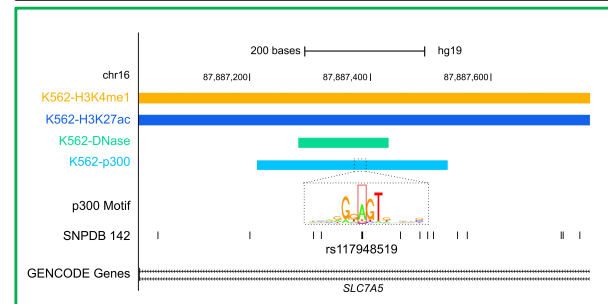
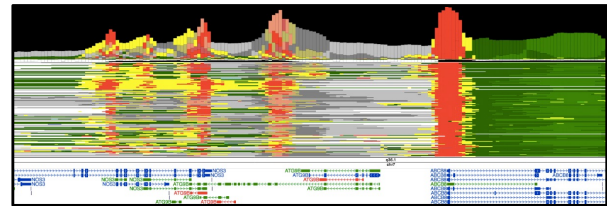
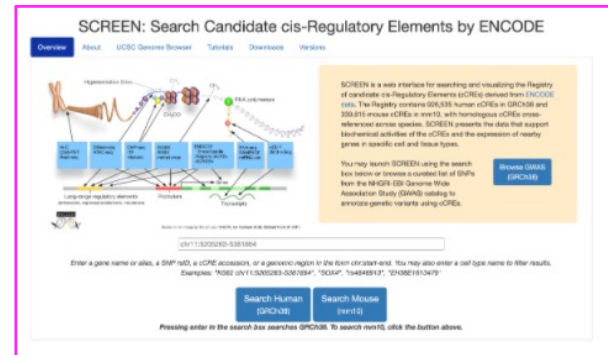
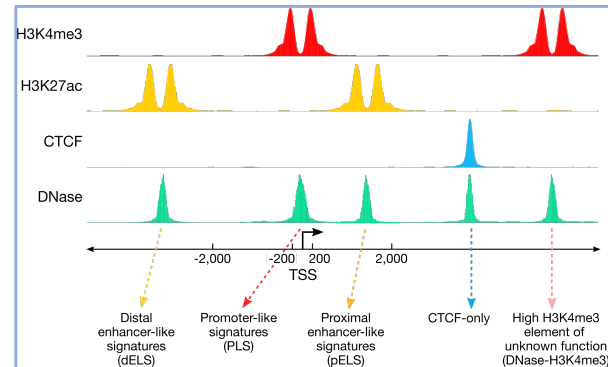
Integrative Level Annotations

The Registry of Candidate
cis-Regulatory Elements

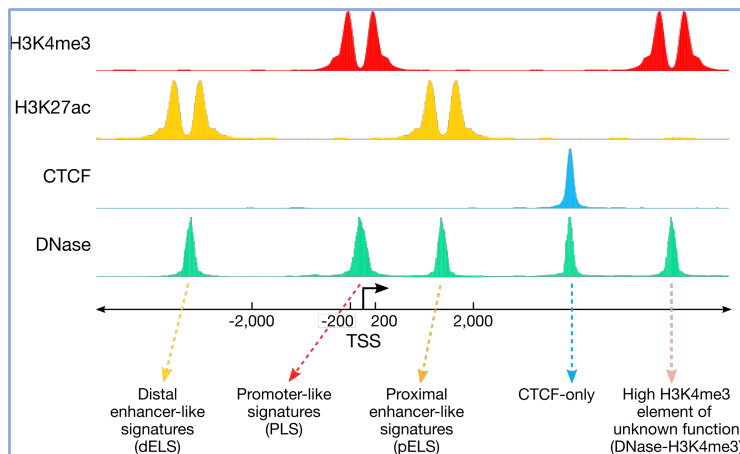
SCREEN

Chromatin states

Variant Annotation

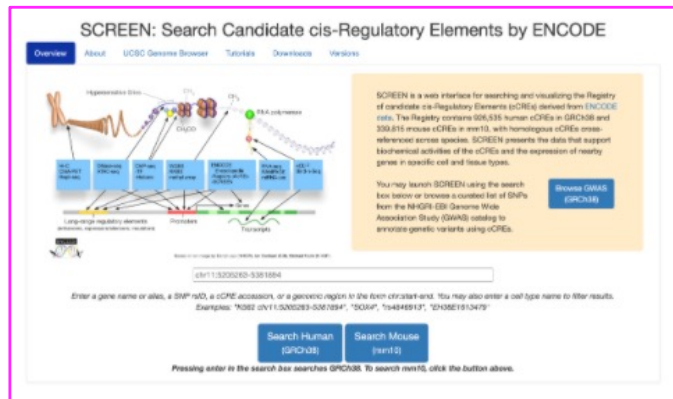


Integrative Level Annotations : The Registry of Candidate cis-Regulatory Elements



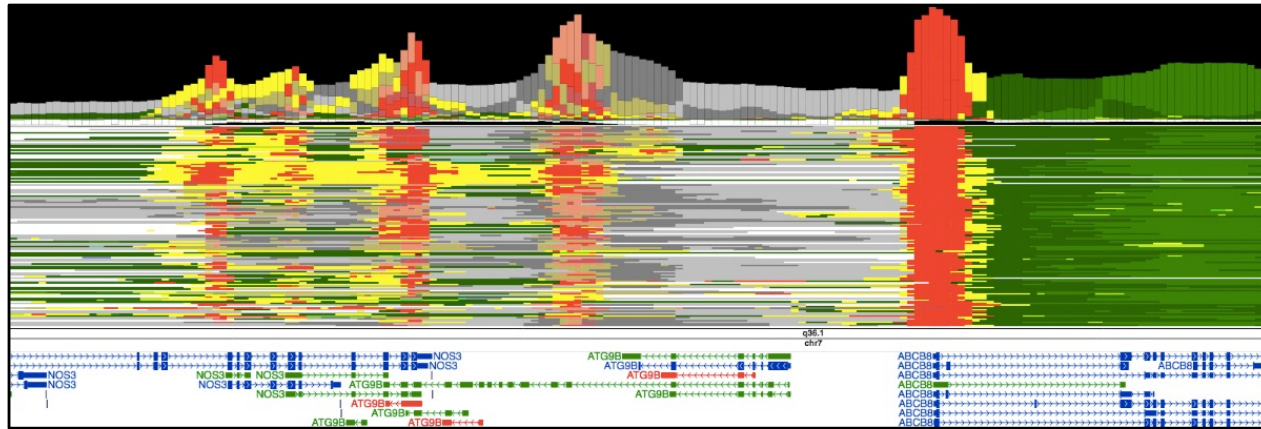
- The **core** of the integrative level of the ENCODE Encyclopedia is **the Registry of candidate cis-Regulatory Elements (cCREs)**.
- Which integrates all high-quality **DNase-seq** and **H3K4me3, H3K27ac, and CTCF ChIP-seq data**.
- Currently the Registry (version 2) comprises **926,535 human cREs** and **339,815 mouse cCREs**.
- Classified cCREs into promoter-like, enhancer-like, DNase-H3K4me3, and CTCF-only groups in a cell-type agnostic manner.
- Currently **25 human (15 mouse)** cell types have complete cell-type-specific cCRE classifications and 839 human (157 mouse) cell types have partial cCRE classifications.

Integrative Level Annotations : SCREEN



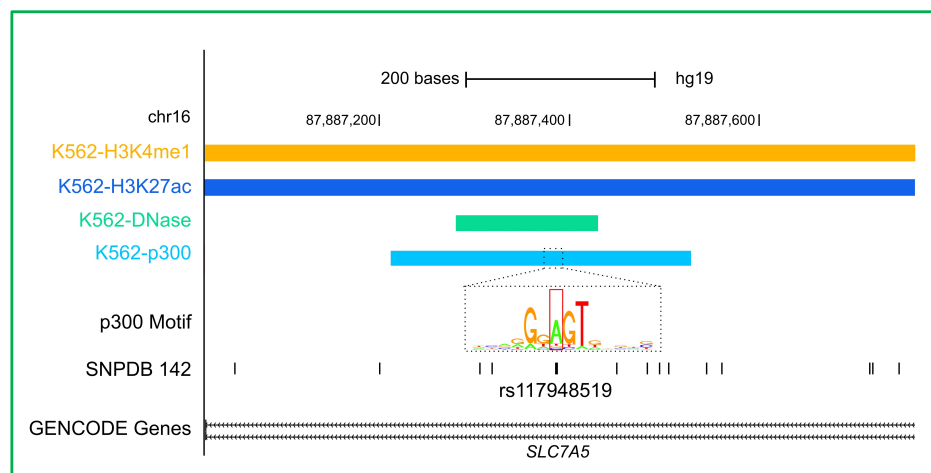
- SCREEN is a **web-based search and visualization** engine specifically designed for the Registry of cCREs.
- SCREEN allows users to explore **cCREs** and investigate how they **connect with other annotations** in the Encyclopedia in a cell-type-specific manner
- As well as the underlying **raw ENCODE data** whenever available.
- SCREEN also presents the results of using **cCREs to interpret the variants** uncovered by Genome-wide Association Studies (GWAS).

Integrative Level Annotations : Chromatin states



- **Semi-automated genomic annotation methods** such as ChromHMM and Segway take as input a panel of epigenomic data (including histone mark ChIP-seq and DNase-seq) in a particular cell type and use machine learning methods to simultaneously partition the genome into segments and assign chromatin states to these segments;
- The states are assigned such that two segments with the same state exhibit similar epigenomic patterns.
- The procedure is "semi-automated" because states are then **manually compared with known biological information** in order to designate each state as an enhancer-like, promoter-like, gene body, etc.
- The chromatin states of **164 human cell types** have been annotated using this strategy by integrating 1,615 genomics datasets (Libbrecht et al., (2019) Genome Biology).
- The chromatin states for **mouse epigenomes of 12 tissue types at 8 different developmental timepoints**, constituting 66 epigenomes of 8 histone marks each, have been annotated using a model integrating 1,056 genomic datasets and their respective controls.

Integrative Level Annotations: Variant Annotation



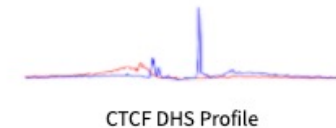
- Genome Wide Association Studies (**GWAS**) have provided insights into how genetic variations contribute to human diseases.
- Over **80%** of the variants reported by GWAS are **in noncoding regions** of the genome.
- By integrating data from the ENCODE project and other public sources, **RegulomeDB** and **HaploReg** are two resources developed by ENCODE labs to aid the research community in annotating GWAS variants.
- **FunSeq** is another ENCODE resource for annotating **both germline and somatic variants**, particularly in the **noncoding regions of cancer genomes**.

Ground Level Annotations

Open chromatin (DNase-seq, ATAC-seq)

DNase I hypersensitive sites (DHSs) computed from DNase-seq experiments, and ATAC-seq peaks (enriched genomic regions).

[\[Open chromatin regions\]](#)



Histone mark enrichment (ChIP-seq)

Peaks (enriched genomic regions) of a variety of histone marks computed from ChIP-seq experiments.

[\[Histone mark peaks\]](#)



H3K27ac from mouse e11.5 hindbrain

Transcription factor binding (TF ChIP-seq)

Peaks (enriched genomic regions) of TFs computed from ChIP-seq experiments.

Visualize sequence motifs and other information on Factorbook.

[\[TF peaks \]](#) [Factorbook](#) [↗](#)



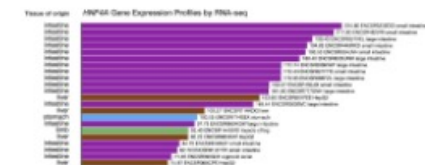
CTCF Motif from [Factorbook](#) [↗](#)

Ground Level Annotations

Gene expression (RNA-seq)

Expression levels of genes and transcripts annotated by GENCODE, which can be visualized on SCREEN.

[[Expression levels](#) | [SCREEN](#)]

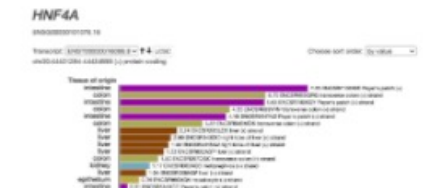


HNF4A Gene Expression

Transcription start site (TSS) activity profiling (RAMPAGE)

Identification of transcription start sites (TSSs) and quantification of transcript expression, which can be visualized on SCREEN.

[[RAMPAGE peaks](#) | [SCREEN](#)]

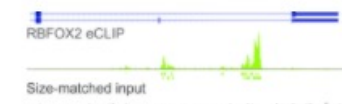


HNF4A Transcript Expression

RNA binding protein occupancy (eCLIP-seq)

Peaks (enriched genomic regions) computed from eCLIP-seq data in human cell lines K562 and HepG2 for RNA Binding Proteins (RBPs).

[[RBP peaks](#)]



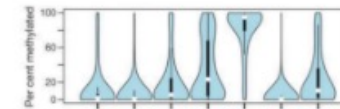
RBFox2 read density

Ground Level Annotations

DNA methylation (RRBS, WGBS)

Genome-wide methylation state of CpG, CHH, and CHG dinucleotides.

[[Methylation levels](#)]

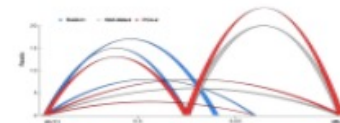


RRBS analysis in GM12878 [↗](#)

Three dimensional chromatin interactions (ChIA-PET)

3D interactions between genomic loci such as promoters and distal enhancers computed from ChIA-PET experiments.

[[Interactions](#)]



ChIA-PET interactions [↗](#)

Topologically associating domains (TADs) (Hi-C)

TADs and A and B compartments computed from Hi-C experiments.

[[TADS](#) | [Compartments](#)]



K562 Interaction Matrix

ENCODE data

Data availability

Data availability

All data is available, from raw data to final processed data

For end-level users:

- Tracks in the UCSC browser with desired level of detail
→ Visualize tracks and explore genomic context

For end-level users and bio-IT:

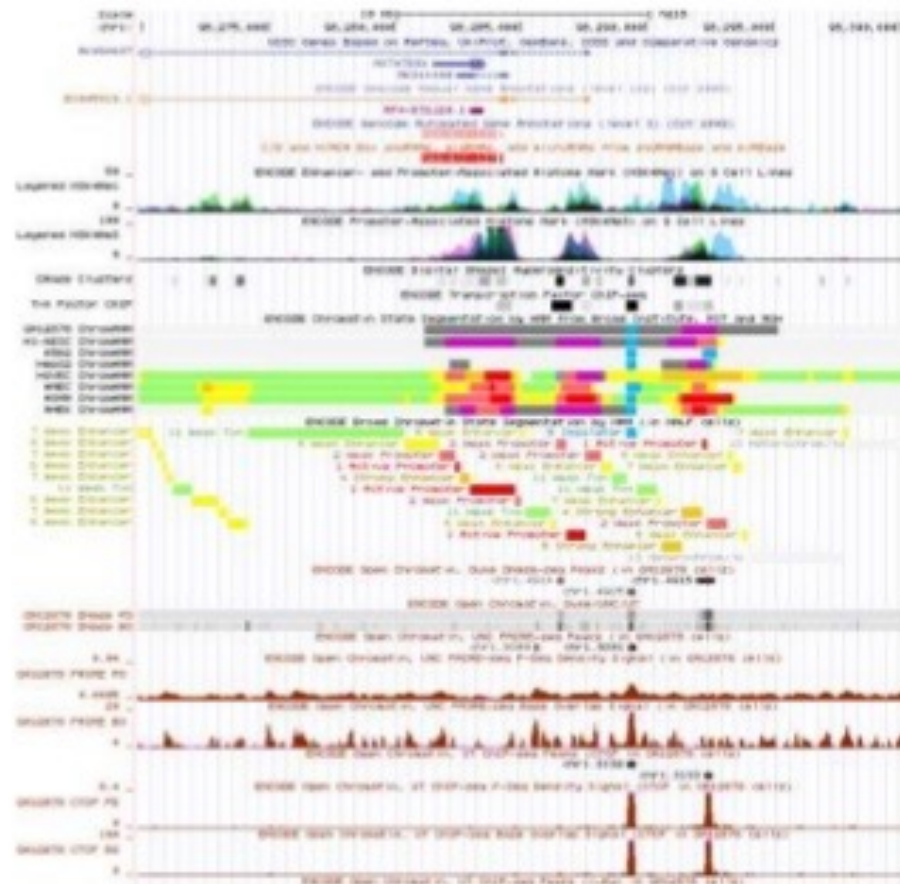
- In UCSC “Table browser” and other UCSC tools
→ Export genomic information, including processed data

For high end-level users and Bio-IT:

- Raw data and semi-processed data in GEO and others

Data availability

Tracks in the UCSC browser with desired level of detail



ENCODE data
Data availability

Raw data

Assays

[illegible]

ENCODE data

Data availability

Production ENCODE data

Experiment	Type	No. Samples in experiment	Lab, Organization	Release date
ENC0001: Whole Genome Bouffard Sequencing by ENCODE/HMB	WGBS	2	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-12
ENC0002: DNase/FAIRE/ChIP Synthesis from ENCODE/OpenChrom/Duke/UNC/UTR	DNase-Seq, FAIRE-Seq, ChIP-Seq	17	Gregory Crawford, Duke University	2012-09-12
ENC0003: CpG Methylation by Illumina Methylation Bead Arrays from ENCODE/HMB	CpG Methylation	63	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-07
ENC0004: CpG Methylation by Methylation Bead Arrays from ENCODE/HMB	CpG Methylation	62	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-07
ENC0005: Genotype (SNP and WGS) by Illumina HiSeq and BG from ENCODE/HudsonAlpha	Genotyping	64	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-07
ENC0006: Chromatin Interactions by 3C from ENCODE/Duke/UNC/UTR	3C	4	Job Dekker, University of Massachusetts Medical School	2012-07-19
ENC0007: Chromatin Interactions by 3C from ENCODE/University of Washington	3C	9	John Stamatoeannopoulos, University of Washington, Seattle	2012-07-19
ENC0008: Chromatin Interaction Analysis Pooled End Tags (ChIP-PET) from ENCODE/UTR	ChIP-PET	6	Michael Snyder, Yale University	2012-07-19
ENC0009: RNA-seq from ENCODE/UTR	RNA-Seq	24	Various	2012-07-12
ENC0010: Targeted RT-PCR arrays spanning unannotated splice junctions sequenced by Roche 454	RNA-Seq	3	Thomas Gingeras, Cold Spring Harbor Lab	2012-05-28
ENC0011: Open Chromatin by DNase-Seq from ENCODE/OpenChrom/Duke University	DNase-Seq	48	Gregory Crawford, Duke University	2012-10-14
ENC0012: Histone Modifications by ChIP-seq from ENCODE/University of Washington	ChIP-Seq	108	John Stamatoeannopoulos, University of Washington, Seattle	2012-09-07
ENC0013: Chromatin State Segmentation by HMM from ENCODE/Broad	ChIP-Seq	9	Bradley Bernstein, Broad Institute of MIT and Harvard	2012-05-24
ENC0014: RNA-seq from ENCODE/HudsonAlpha	RNA-Seq	20	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-17
ENC0015: Replication Timing by RepH-seq from ENCODE/University of Washington	RepH-Seq	13	John Stamatoeannopoulos, University of Washington, Seattle	2012-04-27
ENC0016: [6-ATAC-Seq] GENCODE PCR-Seq Batch VI	RNA-Seq	6	Teahty Hubbard, Wellcome Trust Sanger Institute	2012-09-15
ENC0017: RNA-seq from ENCODE/Stanford/Yale/UC/Harvard	RNA-Seq	4	Various	2012-09-15
ENC0018: Nucleosome Position by DNase-seq from ENCODE/Stanford/UTR	DNase-Seq	2	Michael Snyder, Yale University	2012-09-15
ENC0019: [6-ATAC-Seq] GENCODE PCR-Seq Batch VI	RNA-Seq	6	Teahty Hubbard, Wellcome Trust Sanger Institute	2012-09-15
ENC0020: [6-ATAC-Seq] GENCODE PCR-Seq Batch V	RNA-Seq	6	Teahty Hubbard, Wellcome Trust Sanger Institute	2012-09-15
ENC0021: [6-ATAC-Seq] GENCODE PCR-Seq Batch IV	RNA-Seq	10	Teahty Hubbard, Wellcome Trust Sanger Institute	2012-09-15
ENC0022: RNA Subcellular Compartment Localization from ENCODE/HMB	RNA-Seq	78	Thomas Gingeras, Cold Spring Harbor Lab	2012-12-11
ENC0023: RNA-seq analysis of K562 cells (K562-98) with stable integration of a tel-indeletable cDNA targeting a candidate (X) transcription factor	RNA-Seq	4	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-12-11
ENC0024: RNA Subcellular Localization by Pooled End Tag Sequencing from ENCODE/UTR	RNA-Seq	11	Youn Ryan, Genome Institute of Singapore	2012-12-11
ENC0025: Genetic Framework for GATA Factor Function in Vascular Biology	ChIP-Seq	5	Michael Snyder, Yale University	2012-09-11
ENC0026: Open Chromatin (H3K4) by ChIP-seq from ENCODE/OpenChrom/UTR	ChIP-Seq	68	Various	2012-10-27
ENC0027: ENCODE Cold Spring Harbor Lab Long RNA-seq (Seq18)	RNA-Seq	2	Thomas Gingeras, Cold Spring Harbor Lab	2012-10-27
ENC0028: Histone Modifications by ChIP-seq from ENCODE/Broad Institute	ChIP-Seq	296	Bradley Bernstein, Broad Institute of MIT and Harvard	2012-09-02
ENC0029: Transcription Factor Binding Sites by ChIP-seq from ENCODE/HMB	ChIP-Seq	218	Richard Myers, HudsonAlpha Institute for Biotechnology	2012-09-29
ENC0030: ENCODE SUNY RNA Binding Protein Genes Data for various cell lines	Expression	72	Scott Tenenbaum, University at Albany State University of New York	2012-09-29
ENC0031: Histone Modifications by ChIP-seq from ENCODE/Stanford/Yale/UC/Harvard	ChIP-Seq	32	Various	2012-09-11
ENC0032: ENCODE Transcription Factor Binding Sites by ChIP-seq from Stanford/Yale/UC/Harvard	ChIP-Seq	426	Various	2012-09-11
ENC0033: Transcription Factor Binding Sites by ReproTag ChIP-seq from ENCODE/University of Chicago	ChIP-Seq	12	Kevin White, University of Chicago	2012-09-11
ENC0034: [6-ATAC-Seq] GENCODE PCR-Seq Batch I	RNA-Seq	7	Teahty Hubbard, Wellcome Trust Sanger Institute	2012-07-14

Regulome, Haploreg and Genebore

<http://goo.gl/jhBvS>

HaploReg v2



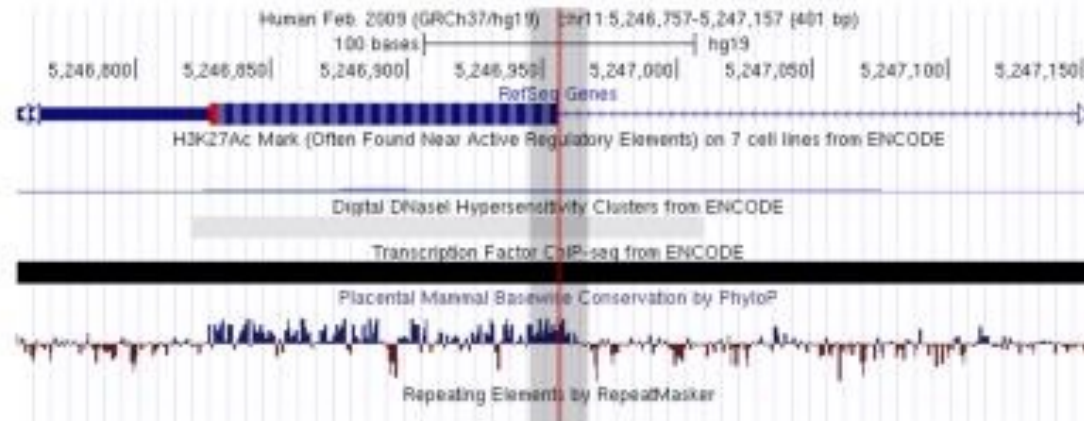
<http://goo.gl/oP5gj>



Regulome

Score: 2a

Likely to affect binding



Protein Binding

Filter:

Method	Location	Bound Protein	Cell Type	Additional Info	Reference
ChIP-seq	chr11:5244812..5248771	POLR2A	PBDE		ENCODE
ChIP-seq	chr11:5244812..5248771	POLR2A	PBDE		ENCODE
ChIP-seq	chr11:5244812..5248771	POLR2A	PBDE		ENCODE
ChIP-seq	chr11:5244812..5248771	POLR2A	PBDE		ENCODE
ChIP-seq	chr11:5244812..5248771	POLR2A	PBDE		ENCODE
ChIP-seq	chr11:5246719..5247202	GATA1	CD36	shbrg1	21795385
ChIP-seq	chr11:5246723..5247183	GATA1	CD36	shLuc	21795385

HaploReg

HaploReg v2



HaploReg is a tool for exploring annotations of the noncoding genome at variants on haplotype blocks, such as candidate regulatory SNPs at disease-associated loci. Using LD information from the 1000 Genomes Project, linked SNPs and small indels can be visualized along with their predicted chromatin state, their sequence conservation across mammals, and their effect on regulatory motifs. HaploReg is designed for researchers developing mechanistic hypotheses of the impact of non-coding variants on clinical phenotypes and normal variation.

Update 2013.02.14: Version 2 now includes an expanded library of SNPs (based on dbSNP 137), motif instances (based on PWMs discovered from ENCODE experiments), enhancer annotations (adding 90 cell types from the Roadmap Epigenome Mapping Consortium), and eQTLs (from the GTex eQTL browser). In addition, LD calculations are provided based on the 1000 Genomes Phase 1 individuals, and r^2 and D' measurements are available down to an r^2 threshold of 0.2. Display improvements include improved cell metadata, gene metadata, and PWM display on the detail pages and the option for text output. Version 1 is available [here](#).

[Build Query](#) [Set Options](#) [Documentation](#)

Use one of the three methods below to enter a set of variants. If an r^2 threshold is specified (see the Set Options tab), results for each variant will be shown in a separate table along with other variants in LD. If r^2 is set to NA, only queried variants will be shown, together in one table.

Query (comma-delimited list of rsIDs OR a single region as chr1: start-end):

or, upload a text file (one refSNP ID per line): No file chosen

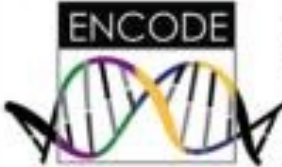
or, select a GWAS:

Query SNP: **rs2853669** and variants with $r^2 \geq 0.8$

chr	pos (hg19)	LD (r ²)	LD (D')	variant	Ref	Alt	AFR freq	AMR freq	ASN freq	EUR freq	SAF freq	Promoter histone marks	Enhancer histone marks	DNase	Protein bound	eQTL tissues	Motifs changed	GENCODE genes	dbSNP func annot
5	1295348	1	1	rs2853669	A	G	0.88	0.23	0.31	0.29		enhancer	enhancer	GM12878/Huh-7/Ishikawa	POU2F1		7 altered motifs	154kb 5' of TERT	
5	1297075	0.8	0.97	rs201976870	CC4	C	0.97	0.20	0.23	0.24							21 altered motifs	1 kb 5' of TERT	
5	1297077	0.8	0.97	rs145544123	ACC	A	0.97	0.21	0.24	0.27							19 altered motifs	1 kb 5' of TERT	
5	1297488	0.8	0.97	rs2738108	C	T	0.97	0.23	0.26	0.28				10 cell type	EDF1		HEN1,2p161	2 kb 5' of TERT	

ENCODE Tutorials

ENCODE Tutorial May 2013, Biology of Genomes, CSHL



On May 8, 2013, an ENCODE tutorial was held at the Biology of Genomes Meeting at Cold Spring Harbor Laboratory (CSHL), hosted jointly by the ENCODE (NIH/NHGRI) project.

The tutorial was prepared for biologists using human and/or mouse genetic data to study disease and basic biology and explain what data are available, what they mean, how they can be displayed and downloaded, and how they can be used in genetic research on human disease.

Agenda and Resources

All files are in  format.

Introduction to ENCODE datatypes and their interpretation

Mike Snyder, Stanford University

- Presentation: [ENCODE: Understanding the Genome](#)

How to display and download ENCODE data

Mike Pazin, NHGRI

- Presentation: [How to display and download ENCODE data](#)
- [ENCODE User's Guide](#)
- [Downloading and Displaying ENCODE Data](#)
- [Using HaploReg and RegulomeDB to Mine ENCODE Data](#)
- [Using sessions to save and share ENCODE browser views and settings](#)
- [A standard session for viewing ENCODE data](#)
To use this session file, which is in .txt format, download it to your computer, then upload the file to the UCSC browser as described in the handout on 'Using sessions ...'

Using ENCODE and Roadmap data to illuminate disease-associated variation

John Stamatoyannopoulos, University of Washington

- Presentation: [Using ENCODE and Roadmap data to illuminate disease-associated variation](#)

<http://www.genome.gov/27553901>

NCBI Roadmap Epigenomics Page

OVERVIEW

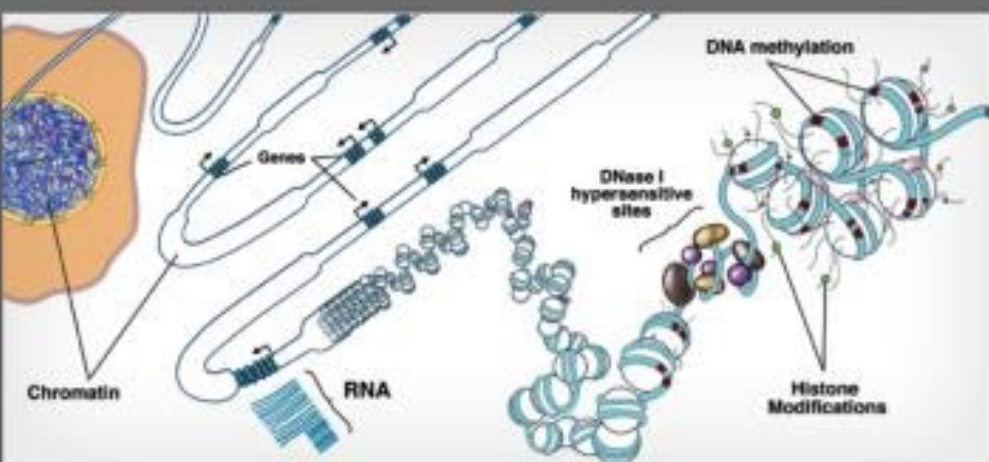
PROJECT DATA

MAPPING CENTERS

PROTOCOLS & STANDARDS

PUBLICATIONS

NEWS



NIH Roadmap Epigenomics Mapping Consortium

The NIH Roadmap Epigenomics Mapping Consortium was launched with the goal of producing a public resource of human epigenomic data to catalyze basic biology and disease-oriented research. The Consortium leverages experimental pipelines built around next-generation sequencing technologies to map DNA methylation, histone modifications, chromatin accessibility and small RNA transcripts in stem cells and primary ex vivo tissues selected to represent the normal counterparts of tissues and organ systems frequently involved in human disease. The Consortium expects to deliver a collection of normal epigenomes that will provide a framework or reference for comparison and integration within a broad array of future studies. The Consortium also aims to close the gap between data generation and its public dissemination by rapid release of raw sequence data, profiles of epigenomics features and higher-level integrated maps to the scientific community. The Consortium is also committed to the development, standardization and dissemination of protocols, reagents and analytical tools to enable the research community to utilize, integrate and expand upon this body of data.

VIEW/DOWNLOAD QUICK LINKS

Genome Browsers

- <http://www.epigenomebrowser.org>
- <http://igenomebrowser.wustl.edu/>
- <http://epigenomegateway.wustl.edu/>

Data Repositories

- [NCBI Epigenomics Gateway](#)
- [Epigenome Atlas](#)

NEWS

13 ENCODE/Roadmap Epigenomics Tutorial
DEC

27 Disease Risk Links to Gene Regulation
MAY

Archives ▶

The ENCODE Phase I Project: The Pilot Project

2003-2007

Introduction_Pilot Phase

- The pilot phase tested and compared existing methods to rigorously analyze a defined portion of the human genome sequence.
- It was organized as an open consortium and brought together investigators with diverse backgrounds and expertise to evaluate the relative merits of each of a diverse set of techniques, technologies and strategies.
- The concurrent technology development phase of the project aimed to develop new high throughput methods to identify functional elements.

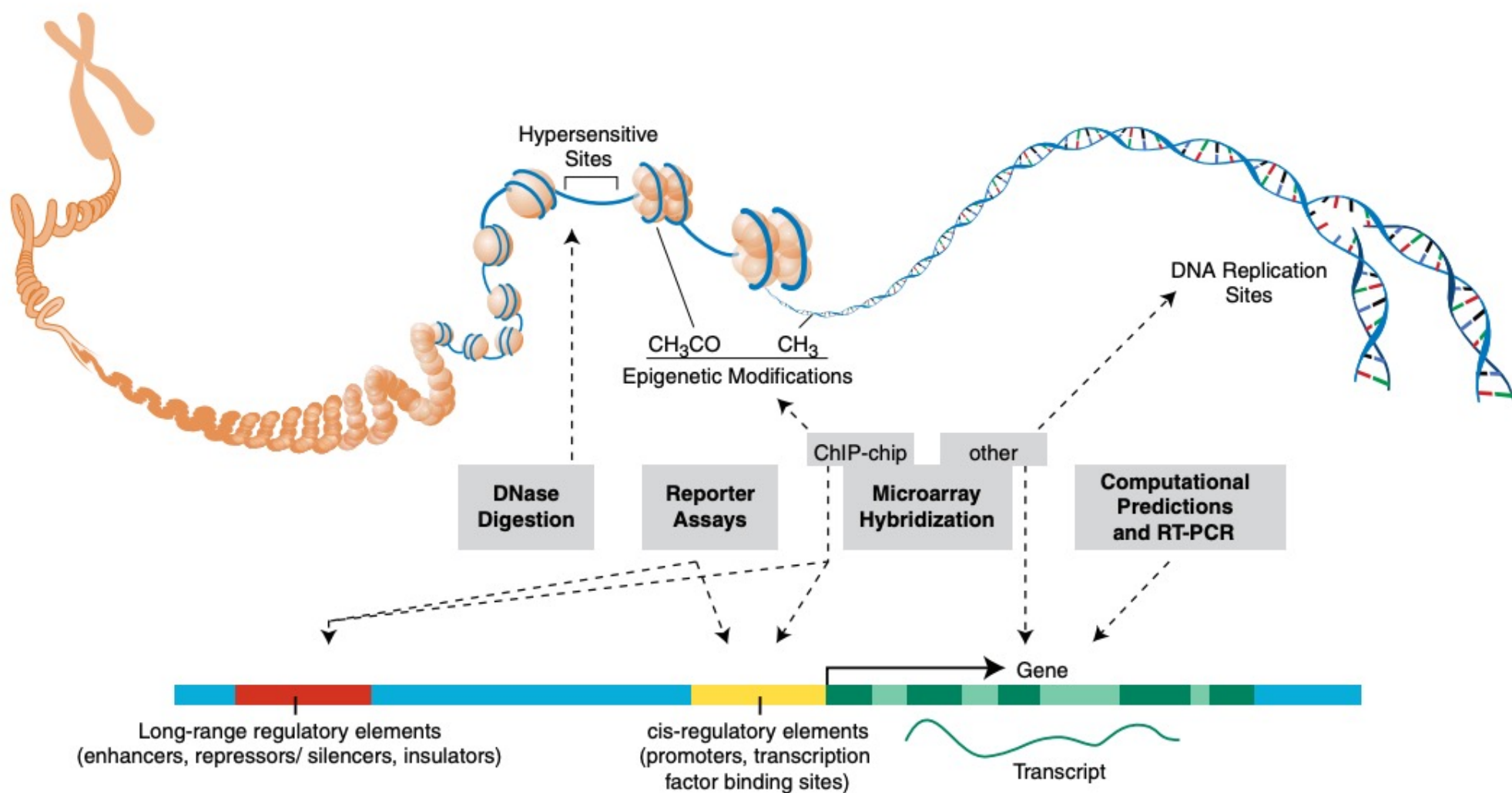


Fig. 1. Functional genomic elements being identified by the ENCODE pilot phase. The indicated methods are being used to identify different types of functional elements in the human genome.

Introduction_Pilot Phase

- The goal of these efforts was to identify a suite of approaches that would allow the comprehensive identification of all the functional elements in the human genome.
- Through the ENCODE pilot project, [National Human Genome Research Institute](#) (NHGRI) assessed the abilities of different approaches to be scaled up for an effort to analyse the entire human genome and to find gaps in the ability to identify functional elements in genomic sequence.

Pilot Phase_goal

- Evaluate a variety of different methods for use in later stages.
- Using a number of existing techniques to analyse a portion of the genome equal to about 1% (30Mb).
- 35 groups provided more than 200 experimental computational data.

The ENCODE (ENCyclopedia Of DNA Elements) Project

The ENCODE Project Consortium*†

The ENCyclopedia Of DNA Elements (ENCODE) Project aims to identify all functional elements in the human genome sequence. The pilot phase of the Project is focused on a specified 30 megabases (~1%) of the human genome sequence and is organized as an international consortium of computational and laboratory-based scientists working to develop and apply high-throughput approaches for detecting all sequence elements that confer biological function. The results of this pilot phase will guide future efforts to analyze the entire human genome.

approaches, such as cDNA-cloning efforts (4, 5) and chip-based transcriptome analyses (6, 7), have revealed the existence of many transcribed sequences of unknown function. As a reflection of this complexity, about 5% of the human genome is evolutionarily conserved with respect to rodent genomic sequences, and therefore is inferred to be

*Affiliations for all members of the ENCODE Consortium can be found on *Science* Online at www.sciencemag.org/cgi/content/full/306/5696/636/DC1.

†To whom correspondence should be addressed.
E-mail: elise_feingold@nih.gov

ENCODE Project Scientific Management:

National Human Genome Research Institute (E. A. Feingold, P. J. Good, M. S. Guyer, S. Kamholz, L. Liefer, K. Wetterstrand, F. S. Collins).

Initial ENCODE Pilot Phase Participants:

Affymetrix, Inc. (T. R. Gingeras, D. Kampa, E. A. Sekinger, J. Cheng, H. Hirsch, S. Ghosh, Z. Zhu, S. Patel, A. Piccolboni, A. Yang, H. Tammana, S. Bekiranov, P. Kapranov, R. Harrison, G. Church, K. Struhl); Ludwig Institute for Cancer Research (B. Ren, T. H. Kim, L. O. Barrera, C. Qu, S. Van Calcar, R. Luna, C. K. Glass, M. G. Rosenfeld); Municipal Institute of Medical Research (R. Guigo, S. E. Antonarakis, E. Birney, M. Brent, L. Pachter, A. Reymond, E. T. Dermitzakis, C. Dewey, D. Keefe, F. Denoeud, J. Lagarde, J. Ashurst, T. Hubbard, J. J. Wesselink, R. Castelo, E. Eyras); Stanford University (R. M. Myers, A. Sidow, S. Batzoglou, N. D. Trinklein, S. J. Hartman, S. F. Aldred, E. Anton, D. I. Schroeder, S. S. Marticke, L. Nguyen, J. Schmutz, J. Grimwood, M. Dickson, G. M. Cooper, E. A. Stone, G. Asimenos, M. Brudno); University of Virginia (A. Dutta, N. Karnani, C. M. Taylor, H. K. Kim, G. Robins); University of Washington (G. Stamatoyannopoulos, J. A. Stamatoyannopoulos, M. Dorschner, P. Sabo, M. Hawrylycz, R. Humbert, J. Wallace, M. Yu, P. A. Navas, M. McArthur, W. S. Noble); Wellcome Trust Sanger Institute (I. Dunham, C. M. Koch, R. M. Andrews, G. K. Clelland, S. Wilcox, J. C. Fowler, K. D. James, P. Groth, O. M. Dovey, P. D. Ellis, V. L. Wraight, A. J. Mungall, P. Dhami, H. Fiegler, C. F. Langford, N. P. Carter, D. Vetrie); Yale University (M. Snyder, G. Euskirchen, A. E. Urban, U. Nagalakshmi, J. Rinn, G. Popescu, P. Bertone, S.

Hartman, J. Rozowsky, O. Emanuelsson, T. Royce, S. Chung, M. Gerstein, Z. Lian, J. Lian, Y. Nakayama, S. Weissman, V. Stolc, W. Tongprasit, H. Sethi).

Additional ENCODE Pilot Phase Participants:

British Columbia Cancer Agency Genome Sciences Centre (S. Jones, M. Marra, H. Shin, J. Schein); Broad Institute (M. Clamp, K. Lindblad-Toh, J. Chang, D. B. Jaffe, M. Kamal, E. S. Lander, T. S. Mikkelsen, J. Vinson, M. C. Zody); Children's Hospital Oakland Research Institute (P. J. de Jong, K. Osoegawa, M. Nefedov, B. Zhu); National Human Genome Research Institute/Computational Genomics Unit (A. D. Baxevas, T. G. Wolfsberg); National Human Genome Research Institute/Molecular Genetics Section (F. S. Collins, G. E. Crawford, J. Whittle, I. E. Holt, T. J. Vasicek, D. Zhou, S. Luo); NIH Intramural Sequencing Center/National Human Genome Research Institute (E. D. Green, G. G. Bouffard, E. H. Margulies, M. E. Portnoy, N. F. Hansen, P. J. Thomas, J. C. McDowell, B. Maskeri, A. C. Young, J. R. Idol, R. W. Blakesley); National Library of Medicine (G. Schuler); Pennsylvania State University (W. Miller, R. Hardison, L. Elnitski, P. Shah); The Institute for Genomic Research (S. L. Salzberg, M. Pertea, W. H. Majoros); University of California, Santa Cruz (D. Haussler, D. Thomas, K. R. Rosenbloom, H. Clawson, A. Siepel, W. J. Kent).

ENCODE Technology Development Phase Participants:

Boston University (Z. Weng, S. Jin, A. Halees, H. Burden, U. Karaoz, Y. Fu, Y. Yu, C. Ding, C. R. Cantor); Massachusetts General Hospital (R. E. Kingston, J. Dennis); NimbleGen Systems, Inc. (R. D. Green, M. A. Singer, T. A. Richmond, J. E. Norton, P. J. Farnham, M. J. Oberley, D. R. Inman); NimbleGen Systems, Inc. (M. R. McCormick, H. Kim, C. L. Middle, M. C. Pirrung); University of California, San Diego (X. D. Fu, Y. S. Kwon, Z. Ye); University of Massachusetts Medical School (J. Dekker, T. M. Tabuchi, N. Gheldof, J. Dostie, S. C. Harvey).

Pilot Phase_Target Seletion

- 50% were selected **manually**
50% were selected **randomly**
 - The two main criteria for manually :
 - The presence of **well-studied genes** or other known sequence elements
 - The existence of a substantial amount of comparative sequence data
- 

Pilot Phase_Target Selection

The randomly selected sequences

- composed of 500kb regions
- selected according to a stratified random-sampling strategy based on
 - gene density –
 $\frac{\text{\#bases in genes}}{\text{\#other bases}}$
 - level of non-exonic conservation
 - 125 bases windows, base alignment with mouse 75%+, score (prediction), took the low score

ARTICLES

Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project

The ENCODE Project Consortium*

We report the generation and analysis of functional data from multiple, diverse experiments performed on a targeted 1% of the human genome as part of the pilot phase of the ENCODE Project. These data have been further integrated and augmented by a number of evolutionary and computational analyses. Together, our results advance the collective knowledge about human genome function in several major areas. First, our studies provide convincing evidence that the genome is pervasively transcribed, such that the majority of its bases can be found in primary transcripts, including non-protein-coding transcripts, and those that extensively overlap one another. Second, systematic examination of transcriptional regulation has yielded new understanding about transcription start sites, including their relationship to specific regulatory sequences and features of chromatin accessibility and histone modification. Third, a more sophisticated view of chromatin structure has emerged, including its inter-relationship with DNA replication and transcriptional regulation. Finally, integration of these new sources of information, in particular with respect to mammalian evolution based on inter- and intra-species sequence comparisons, has yielded new mechanistic and evolutionary insights concerning the functional landscape of the human genome. Together, these studies are defining a path for pursuit of a more comprehensive characterization of human genome function.

Phase 2

2008-2012

Overview of Results – I [pilot]

- The human genome is pervasively transcribed;
- Many novel non-protein-coding transcripts have been identified;
- Numerous novel TSSs, many of which show chromatin structure and sequence-specific protein-binding properties;
- DNA replication timing is correlated with chromatin structure;
- A total of 5% of the bases in the genome can be confidently identified as being under evolutionary constraint in mammals, of which ~60% showed convinced function;
- Different functional elements vary greatly in their context;
- Many functional elements are seemingly unconstrained across mammalian evolution;
-

Overview of Results – II [production]

- The majority (80.4%) of the human genome participates in at least one biochemical events in at least one cell type;
- Primate-specific elements as well as elements without detectable mammalian constraint show evidence of negative selection (functional);
- The genome can be classified into different chromatin states with distinct functional properties;
- RNA expression could be predictable with both chromatin marks and transcription factor binding at promoters;
- Many non-coding variants in individual genome sequences lie in ENCODE-annotated functional regions;
- SNPs associated with disease by GWAS are enriched within non-coding functional elements;
-

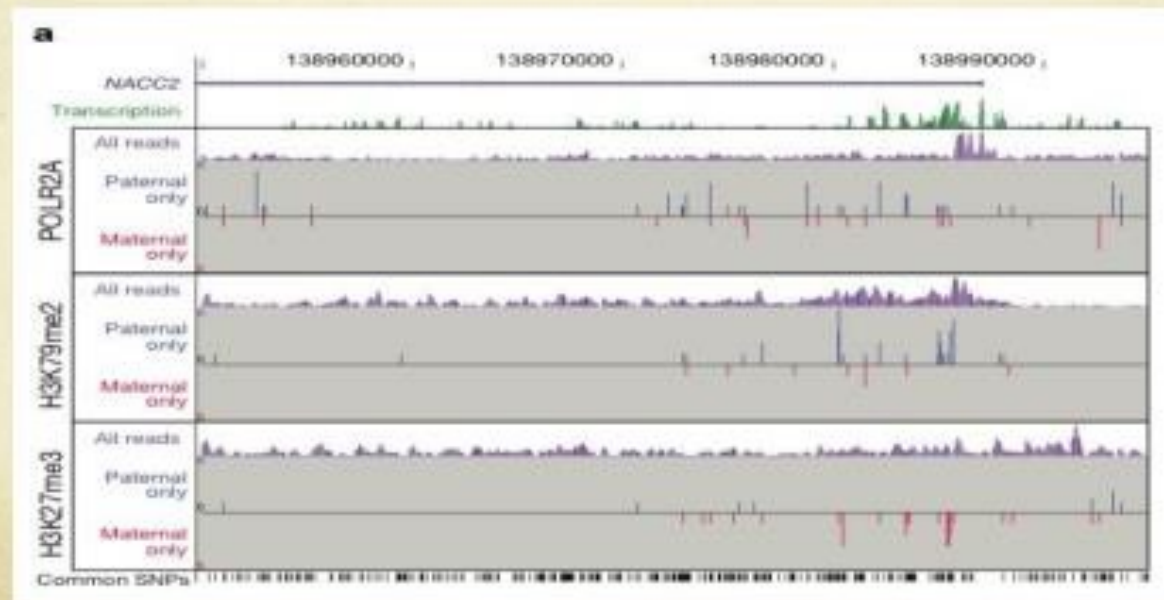
Decode the Genome

- 80.4%, covered by at least one ENCODE-annotated element
- 62%, different RNA types
- 56.1%, regions highly enriched for histone modifications
- 44.2%, covered excluding RNAs and histone elements
- 19.4%, at least one DHS or TF CHIP-seq peak
- 15.2%, regions of open chromatin
- 8.5%, either a TF-binding-site motif (4.6%) or a DHS footprint (5.7%)
- 8.1%, sites of TF binding
- could be underestimated ...



Insights into human genomic variation

- examined the allele-specific variation (NA12878, along with parents)
- found instances of preferential binding towards each parental allele.



Common variants associated with disease

- GWAS outputs a series of SNPs associated with a phenotype (un-necessarily the functional variants)
- 88% of these SNPs are either intronic or intergenic
- examined 4860 SNP-phenotype associations for 4492 SNPs
- 12% overlap TF-occupied regions; 34% overlap DHSs
- GWAS SNPs were consistently enriched beyond all the genotyping SNPs in function-rich partitions of the genome, and depleted in function-poor partitions; GWAS SNPs are particularly enriched in the segmentation classes associated with enhancers and TSSs
- Considering the LD, up to 71% of GWAS SNPs have a potential causative SNP overlapping a DNase I site, and 31% of loci have a candidate SNP that overlaps a binding site occupied by a transcription factor.

Introduction - I

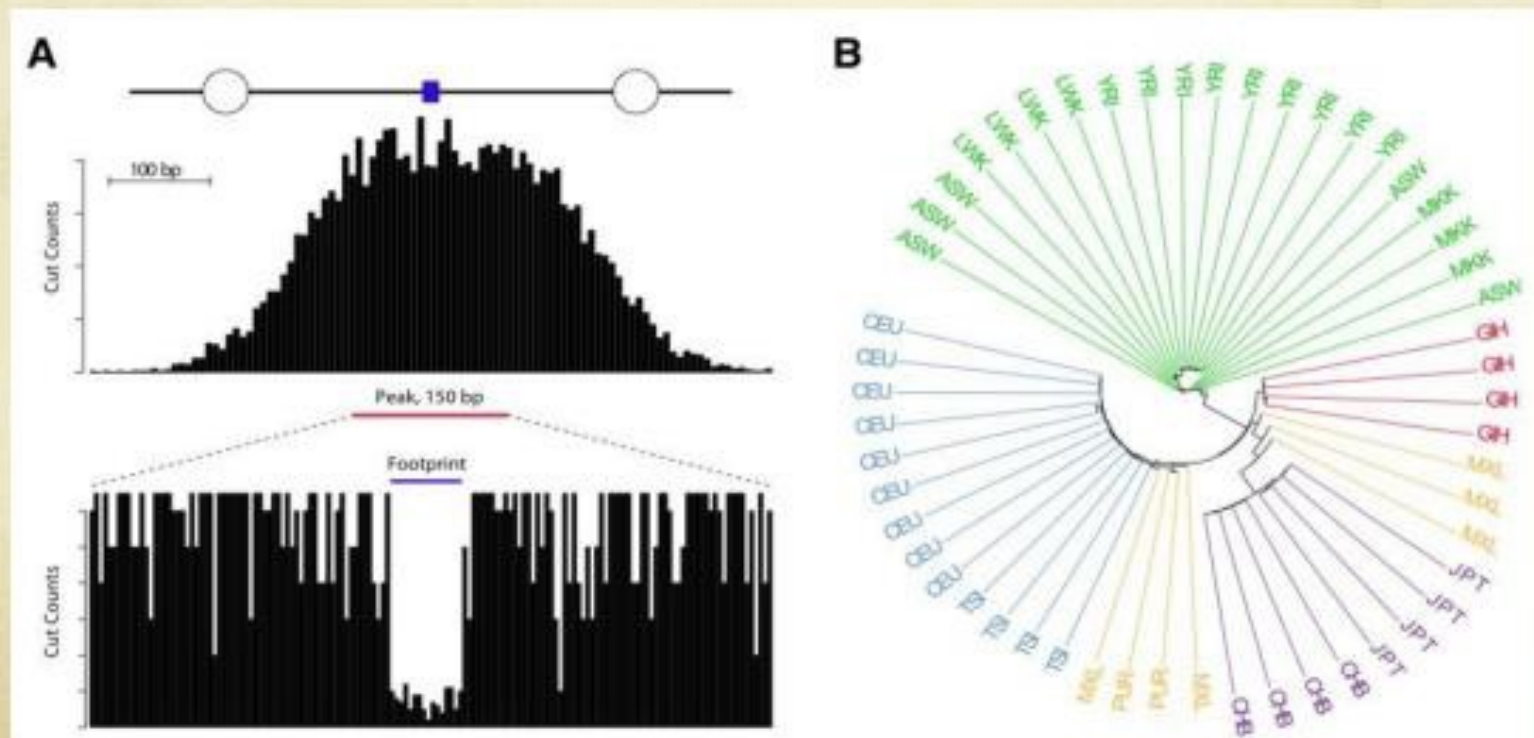
- Protein-coding DNA constitute $\sim 1.5\%$ of the human genome, but $\sim 2.5\%$ - 15% is estimated to be functionally constrained.
- A number of examples in humans have been described of positive selection that are due to adaptive evolution of non-coding DNA.

Introduction - II

- Hypersensitivity to the nonspecific endonuclease DNase I has been used for over 30 yr as a probe for regulatory DNA
- The binding of sequence-specific transcriptional regulators in place of canonical nucleosomes creates DNase I cleavage patterns allows identification of the “footprints” of DNA-bound regulators
- the nonspecificity of DNase I is a powerful feature that allows all DNA-protein interactions to be queried in a single experiment

Overview of Data

- the ENCODE project enables to create a genome-scale map of diverse functional non-coding elements marked by DHSs.
- 53unrelated individuals that encompass five geographically diverse populations (avg. $\sim 40\times$)



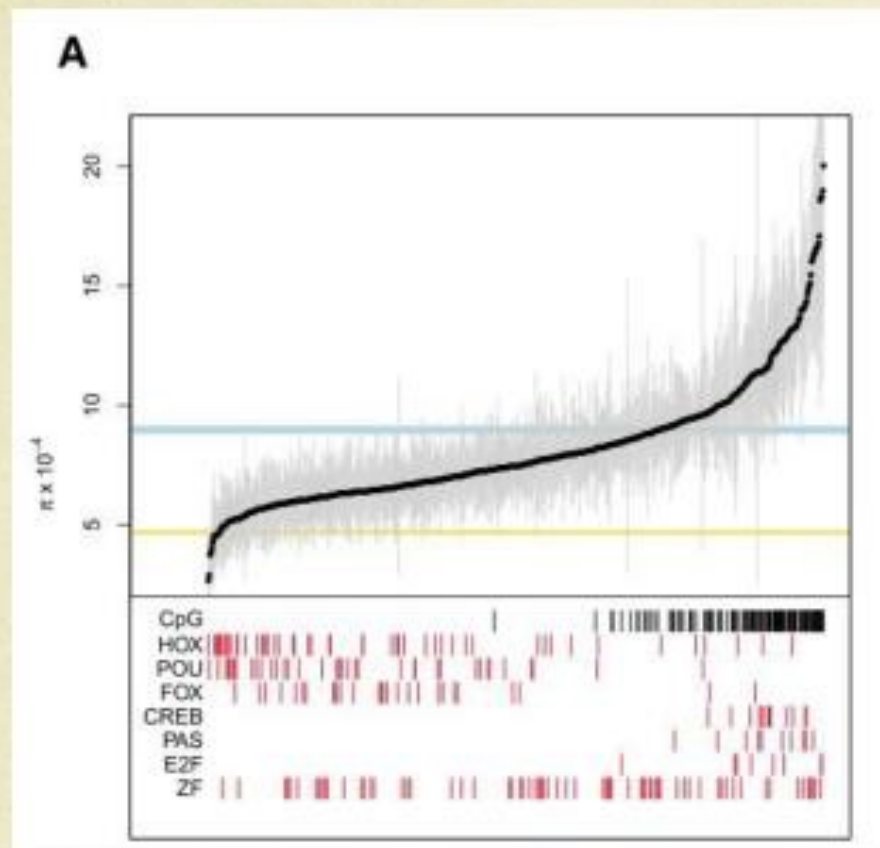
Results - I

- Pervasive regulatory variation across the human genome
 - 2.9 M DNase I peaks, 8.4 M DNase I footprints spanning 577 M and 156 M of the genome, respectively
 - for DNase I peaks, DNase I footprints, and exome, 3.85 M, 1.01 M, and 0.15 M variants were observed (avg., 6.7, 6.5, and 4.2 variants per kb)
 - GERP score, a measure of evolutionary constraint.

Results - II

- Patterns of nucleotide diversity in regulatory DNA sequence motifs
 - scanned DNase I footprints for 732 known motifs
 - for each motif, calculating nucleotide diversity, π
 - also calculated π for fourfold synonymous sites, a proxy for neutrally evolving DNA, and protein-coding sequences
 - Approximately 60% of motifs have average diversities significantly lower than fourfold synonymous sites (blue line), indicative of purifying selection.

Results – II _[cont.]

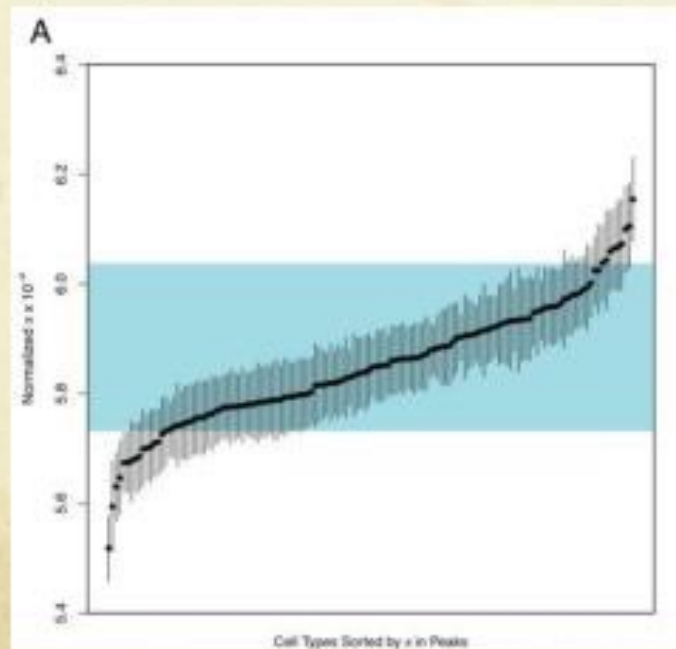


- highlighting motif diversity for several important classes of transcriptional regulators
- the ubiquitous presence of CpG sites is a common characteristic of motifs with high levels of diversity

○ Heterogeneity in both selective constraint and mutation rate likely contribute to the differences in diversity observed among motifs.

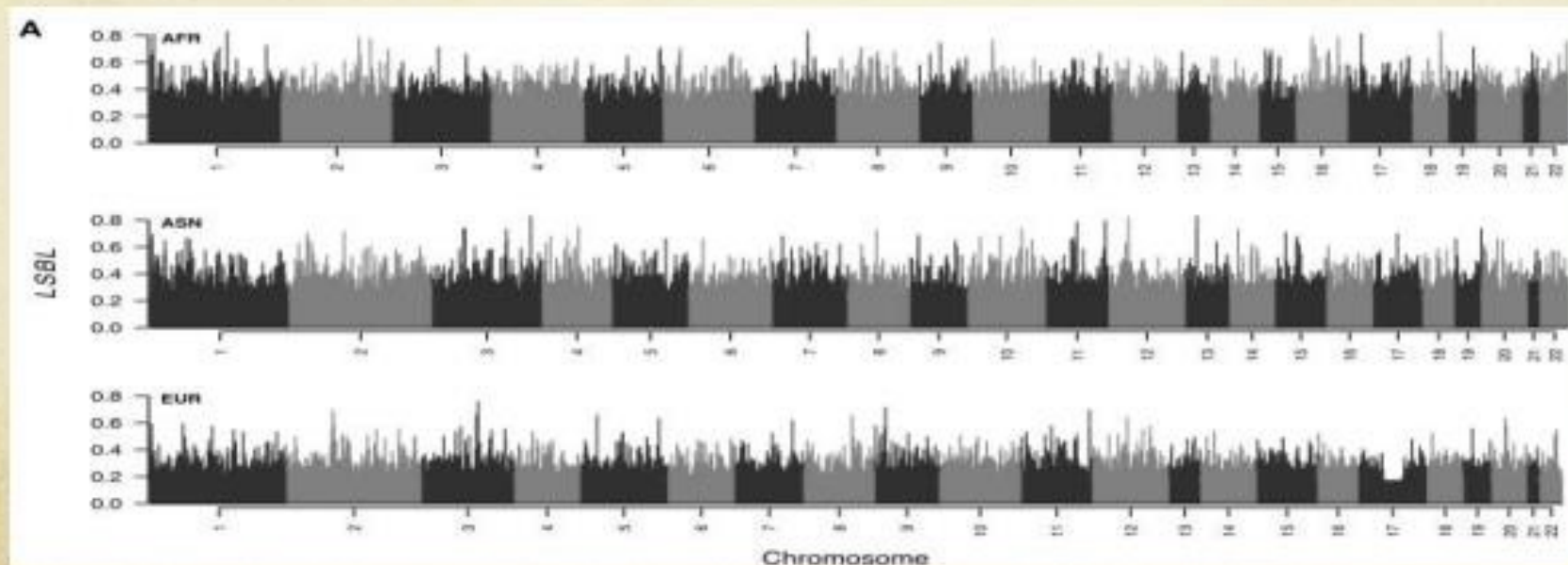
Results - III

- Heterogeneity of functional constraint across cell types
 - calculated the normalized π averaged across all DNase I peaks for each of the 138 cell lines
 - marked differences were shown between cell lines
 - the majority of cell types exhibited average levels of normalized diversity that are within the range of fourfold degenerate sites



Results - IV

- Signatures of positive selection
 - calculated Locus-Specific Branch Lengths (LSBLs) for variants in DNase I peaks in Africans, Asians, and Europeans.
 - signals: 1% tail of the empirical distribution
 - genes within 50 kb
 - enrichment of KEGG



Results – IV [cont.]

Table 1. Enriched KEGG pathways for genes within 50 kb of highly differentiated DNase I peaks

Population	KEGG pathway (Identification)	No. of genes	P-value
European	Melanogenesis (04916)	31	0.0001
	Arrhythmogenic right ventricular cardiomyopathy (05412)	28	0.0006
	ECM-receptor interaction (04512)	30	0.0007
	Pathways in cancer (05200)	80	0.0033
	Dilated cardiomyopathy (05414)	29	0.0052
	Hypertrophic cardiomyopathy (05410)	27	0.0066
	Axon guidance (04360)	37	0.0075
	Focal adhesion (04510)	51	0.0087
	Wnt signaling pathway (04310)	41	0.0087
	Calcium signaling pathway (04020)	46	0.0095
African	Vascular smooth muscle contraction (04270)	40	0.0008
	Chemokine signaling pathway (04062)	57	0.0008
	Adherins junction (04520)	29	0.0009
	Pathways in cancer (05200)	88	0.0012
	Adipocytokine signaling pathway (04920)	25	0.0027
	Wnt signaling pathway (04310)	45	0.0045
	Axon guidance (04360)	39	0.0097
	Arrhythmogenic right ventricular cardiomyopathy (05412)	31	0.00009
Asian	Vascular smooth muscle contraction (04270)	36	0.0049
	Pathways in cancer (05200)	83	0.0049
	Hypertrophic cardiomyopathy (05410)	28	0.0059
	Dilated cardiomyopathy (05414)	30	0.0059

P-values shown are adjusted for multiple testing using FDR (see Methods). Pathways in bold denote those shared between two or more populations.

Guideline to ENCODE

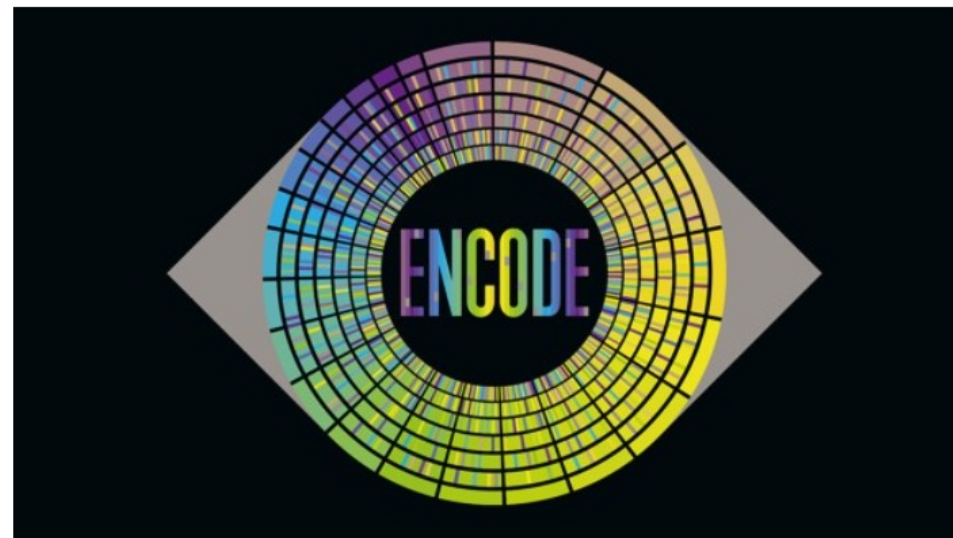


nature > collection

COLLECTION | 05 SEPTEMBER 2012

Encode

Access the collected papers by exploring the thematic threads that run through them, with topics such as DNA methylation, RNA or machine learning.



<https://www.nature.com/collections/aghcdefffg/#/>

What and who is ENCODE

Main aims, funding and the institutions/labs behind the 200 M \$

Who?

International consortium

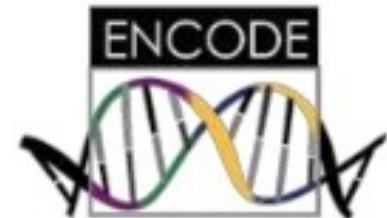
Funded by NHGRI – National Human Genome Research Institute
200 million dollar

Main collaborators (for human data)

Broad Institute (ChIP-seq)
HudsonAlpha Institute for Biotechnology (methylation)
Sanger Institute (RNA-seq)
Duke University (DNase)
Yale University (Pol II)
EBI (data analysis)

Main aims

“Build a comprehensive parts list of functional elements in the human genome, including elements that act at the protein and RNA levels, and regulatory elements that control cells and circumstances in which a gene is active”



What and who is ENCODE

Main aims, funding and the institutions/labs behind the 200 M \$

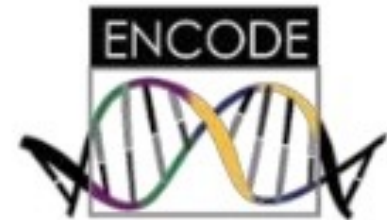
What's so hot... It has been running for years?

Started in 2007 – pilot project
1% of the genome

2007-2012

Since then, introduction of new technologies

- Higher throughput
- Genome-wide
- Much more samples and different tissues (different 'tiers' – see later)
- Better data analysis and integration

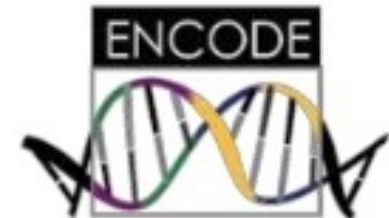


What and who is ENCODE

Main aims, funding and the institutions/labs behind the 200 M \$

What's so hot... It has been running for years?

World wide press attention



The New York Times **Science**

WORLD | U.S. | N.Y. | REGION | BUSINESS | TECHNOLOGY | SCIENCE | HEALTH | SPORTS | OPEDION
ENVIRONMENT | SPACE & COSMOS

Bits of Mystery DNA, Far From 'Junk,' Play Crucial Role

By DNA KOLATA
Published: September 5, 2012 | 574 Comments

Among the many mysteries of human biology is why complex diseases like diabetes, high blood pressure and psychiatric disorders are so difficult to predict and, often, to treat. An equally perplexing puzzle is why one individual gets a disease like cancer or depression, while an identical twin remains perfectly healthy.

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voor abonnees • infolijn • Vrijwilligers

Schakelbord regeert de genen

Menselijk genoom is veel meer dan enkel genen

Woensdag 10 september 2012, 05u42 | Rikie Van den Eynde

AANRADER 3
BRUSSEL - In ons DNA ligt verscholen tussen de genen een immens controleplan dat hun werking aanstuurt. Het oude idee dat ons erfelijk materiaal vooral uit versniet bestaat, moet nu wel.

the guardian

News | Sport | Comment | Culture | Business | Money | Life & style

News > Science > Genetics

Breakthrough study overturns theory of 'junk DNA' in genome

The international Encode project has found that about a fifth of the human genome regulates the 2% that makes proteins

What and who is ENCODE

Main aims, funding and the institutions/labs behind the 200 M \$

What's so hot... It has been running for years?

World wide press attention... and criticisms

"Popular" media focus on the "junk DNA aspect"

The authors also claim in their press-release that > 80% of the genome is 'biologically active' (<> may be involved in regulation in one way or another <> junk DNA)

ENCODE reveals for the first time a lot of factors of the **very complex switching board** controlling expression / ...

This 100,000 word post on the ENCODE media bonanza will cure cancer

By Michael Eisen | September 6, 2012

It is oddly fitting that the papers describing the results of the NIH's massive \$200m ENCODE project were published in the midst of political convention season. For this was no typical scientific publication, but a carefully orchestrated spectacle, meant to justify a massive, expensive undertaking, and to convince us that we are better off now than we were five years ago.

I'll touch more on details of the science, and the way it was carried out, in another, longer, post. But I want to try to explain to people who were asking on twitter why I found today's media blitz to promote the ENCODE publications so off-putting. Because, as cynical as I am about this kind of thing, I still found myself incredibly disheartened by the degree to which the ENCODE press release and many of the interviews published today push a narrative about their results that is, at best, misleading.

The issues all stem, ultimately, from the press releases issued by the ENCODE team, [one of which](#) begins:

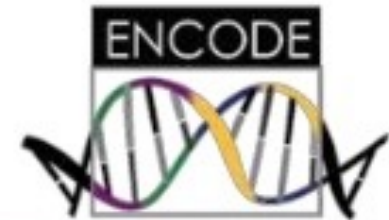
The hundreds of researchers working on the ENCODE project have revealed that much of what has been called 'junk DNA' in the human genome is actually a massive control panel with millions of switches regulating the activity of our genes. Without these switches, genes would not work – and mutations in these regions might lead to human disease. The new information delivered by ENCODE is so comprehensive and complex that it has given rise to a new publishing model in which electronic documents and datasets are interconnected.

What and who is ENCODE

Main aims, funding and the institutions/labs behind the 200 M \$

What's so hot... It has been running for years?

30 (!) research papers published in three journals at the same time



ENCODE

Encyclopedia of DNA Elements



Outline

What and who is ENCODE

Key ENCODE topics and most important papers for our research

ENCODE data – make use of the encyclopedia...

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Key topics

Transcription factor binding motifs
Chromatin patterns at transcription factor binding sites
Characterization of intergenic regions and gene definitions
RNA and chromatin modification patterns around promoters
Epigenetic regulation of RNA processing
Non-coding RNA characterisation
DNA-methylation
Enhancer discovery and characterization
3D connections across the genome
Characterisation of network topology
Machine learning approaches to genomics
Impact of functional information on understanding variation
Impact of evolutionary selection on functional regions

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Main paper

95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction

Classifying the genome into seven chromatin states indicates an initial set of 399,124 regions with enhancer-like features and 70,292 regions with promoter-like features

It is possible to correlate quantitatively RNA sequence production and processing with both chromatin marks and transcription factor binding at promoters, indicating that promoter functionality can explain most of the variation in RNA expression

Single nucleotide polymorphisms (SNPs) associated with disease by GWAS are enriched within non-coding functional elements, with a majority residing in or near ENCODE-defined regions that are outside of protein-coding genes. In many cases, the disease phenotypes can be associated with a specific cell type or transcription factor

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Main paper

Techniques used:

RNA-seq

ChIP-seq

DNase-seq

DNA-methylation arrays and bisulfite seq

FAIRE-seq

Tier 1: three cell lines (K652 – GM12878 – H1 hESC)

Tier 2: cell line panel (HeLa-S3 – HepG2 – HUVECs)

Tier 3 (all other cell types)

Total: 1640 datasets / 147 different cell types

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Main paper

Table 1 | Summary of transcription factor classes analysed in ENCODE

Acronym	Description	Factors analysed
ChromRem	ATP-dependent chromatin complexes	5
DNARep	DNA repair	3
HISase	Histone acetylation, deacetylation or methylation complexes	8
Other	Cyclin kinase associated with transcription	1
Pol2	Pol II subunit	1 (2 forms)
Pol3	Pol III-associated	6
TFNS	General Pol II-associated factor, not site-specific	8
TFSS	Pol II transcription factor with sequence-specific DNA binding	87

Table 2 | Summary of ENCODE histone modifications and variants

Histone modification or variant	Signal characteristics	Putative functions
H2A.Z	Peak	Histone protein variant (H2A.Z) associated with regulatory elements with dynamic chromatin
H3K4me1	Peak/region	Mark of regulatory elements associated with enhancers and other distal elements, but also enriched downstream of transcription starts
H3K4me2	Peak	Mark of regulatory elements associated with promoters and enhancers
H3K4me3	Peak	Mark of regulatory elements primarily associated with promoters/transcription starts
H3K9ac	Peak	Mark of active regulatory elements with preference for promoters
H3K9me1	Region	Preference for the 5' end of genes
H3K9me3	Peak/region	Repressive mark associated with constitutive heterochromatin and repetitive elements
H3K27ac	Peak	Mark of active regulatory elements; may distinguish active enhancers and promoters from their inactive counterparts
H3K27me3	Region	Repressive mark established by polycomb complex activity associated with repressive domains and silent developmental genes
H3K36me3	Region	Elongation mark associated with transcribed portions of genes, with preference for 3' regions after intron 1
H3K79me2	Region	Transcription-associated mark, with preference for 5' end of genes
H4K20me1	Region	Preference for 5' end of genes

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Main paper

95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction

Classifying the genome into **seven chromatin states** indicates an initial set of 399,124 regions with enhancer-like features and 70,292 regions with promoter-like features

It is possible to **correlate quantitatively RNA sequence production and processing with both chromatin marks and transcription factor binding at promoters**, indicating that promoter functionality can explain most of the variation in RNA expression

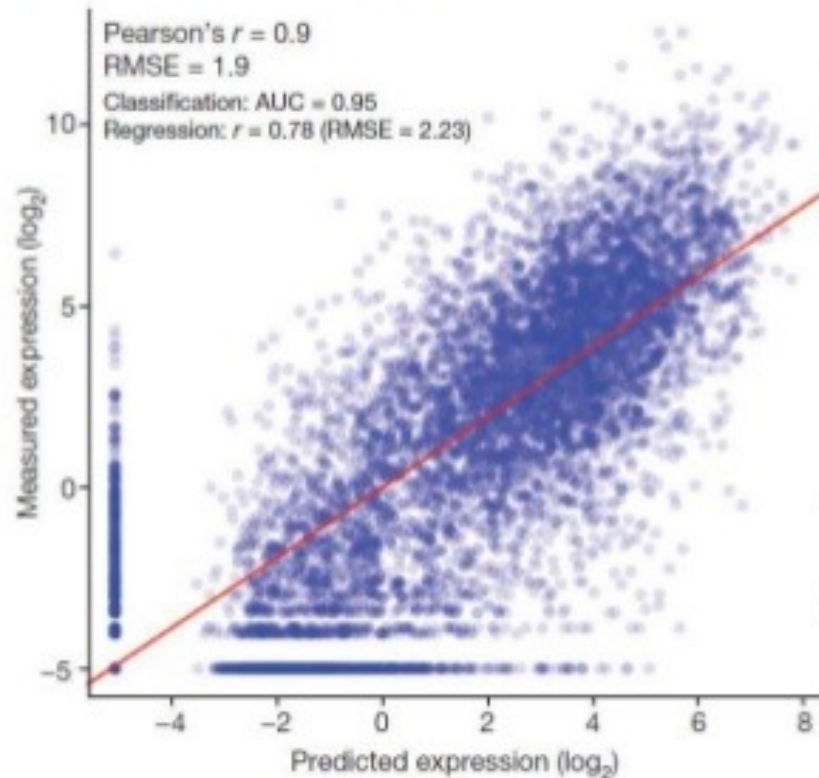
Single nucleotide polymorphisms (SNPs) associated with disease by GWAS are enriched within non-coding functional elements, with a majority residing in or near ENCODE-defined regions that are outside of protein-coding genes. In many cases, the disease phenotypes can be associated with a specific cell type or transcription factor

Key ENCODE topics

Main ENCODE topics and selection of most important papers

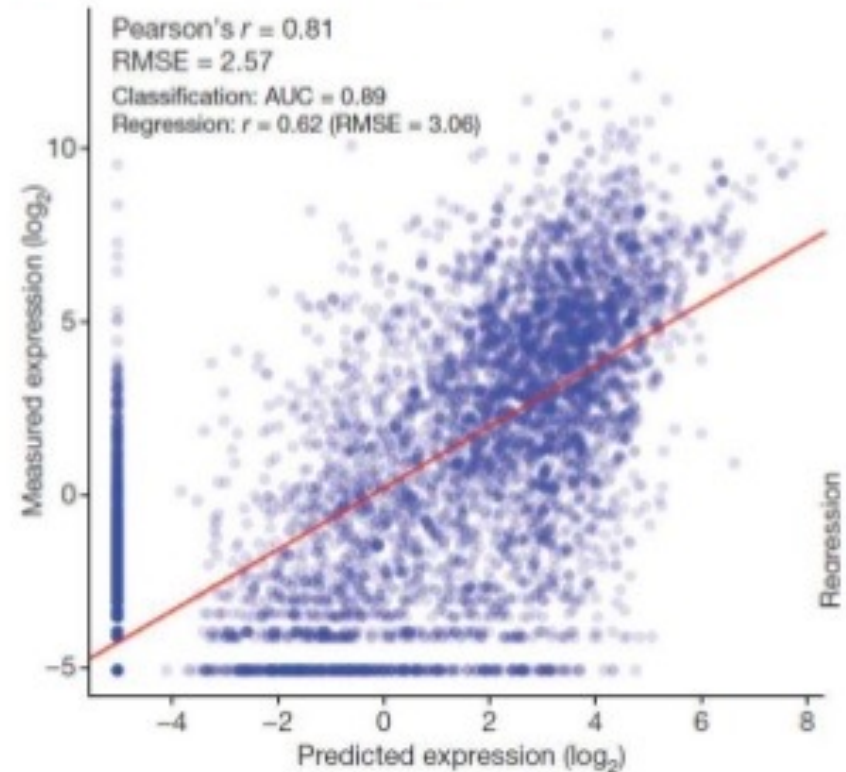
Expression – chromatin state

a CAGE poly(A)⁺ K562 whole cell



Expression – transcription factors

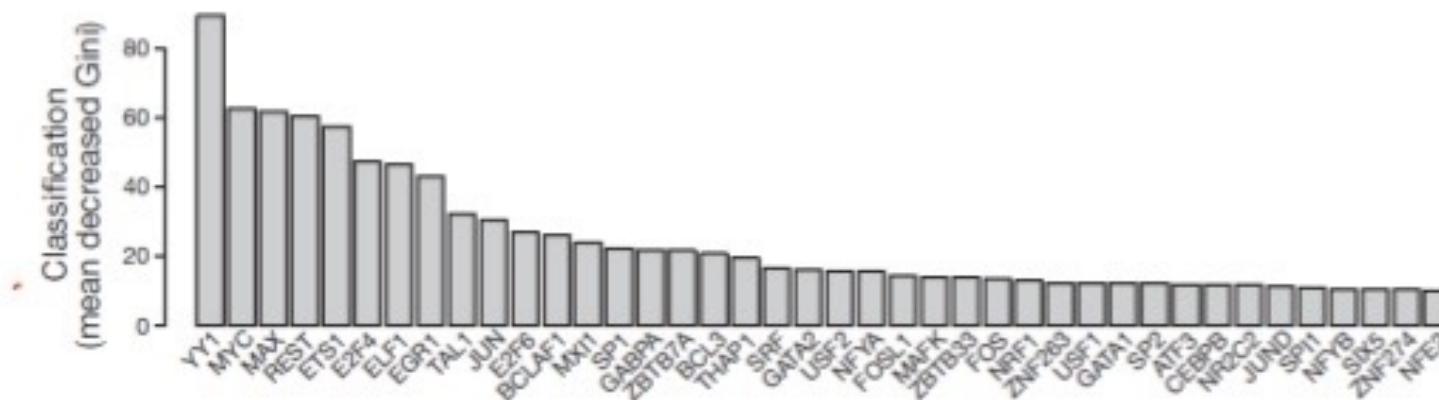
b CAGE poly(A)⁺ K562 whole cell



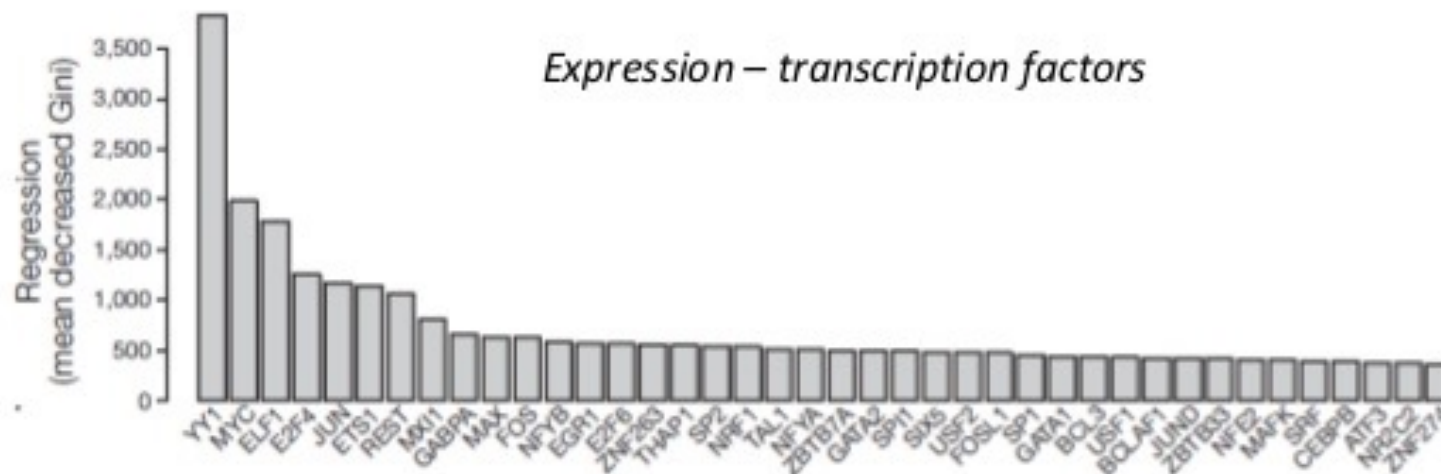
Key ENCODE topics

Main ENCODE topics and selection of most important papers

Relative importance of variables

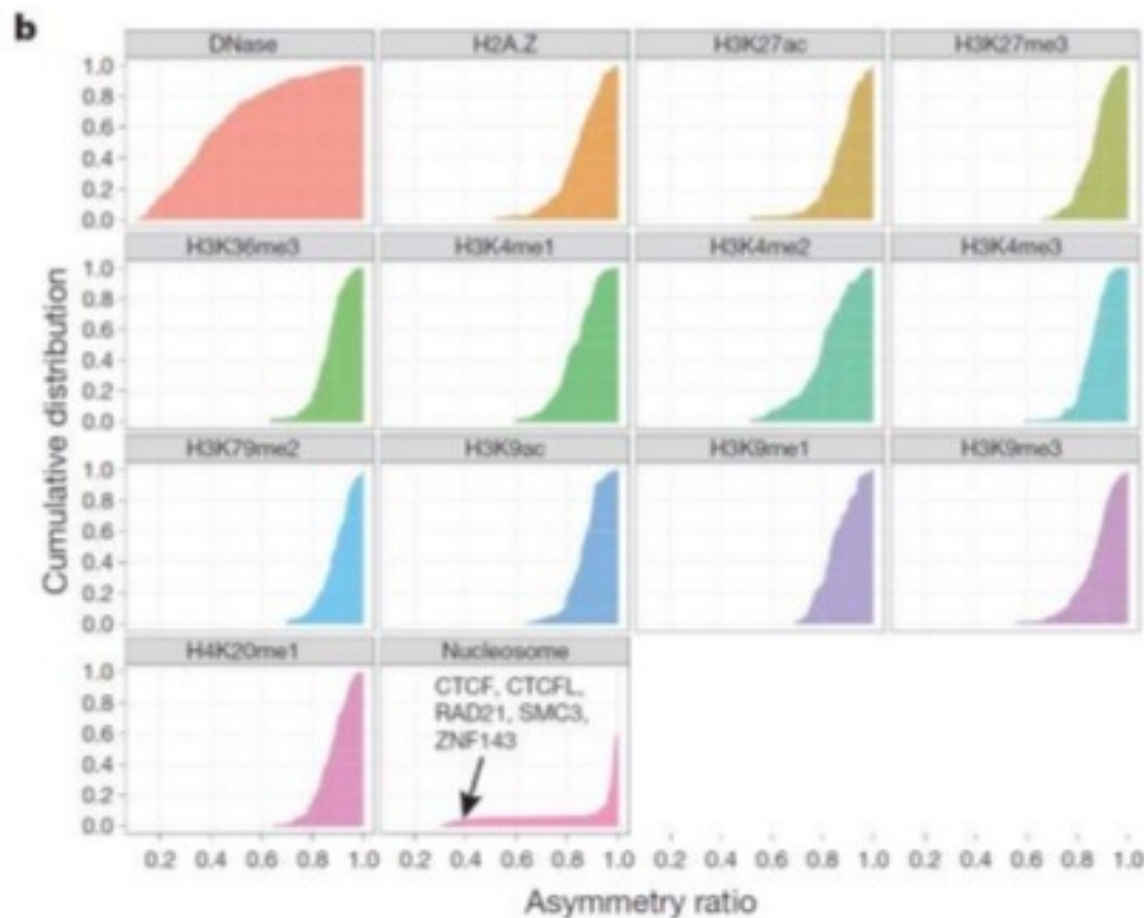


Expression – transcription factors



Key ENCODE topics

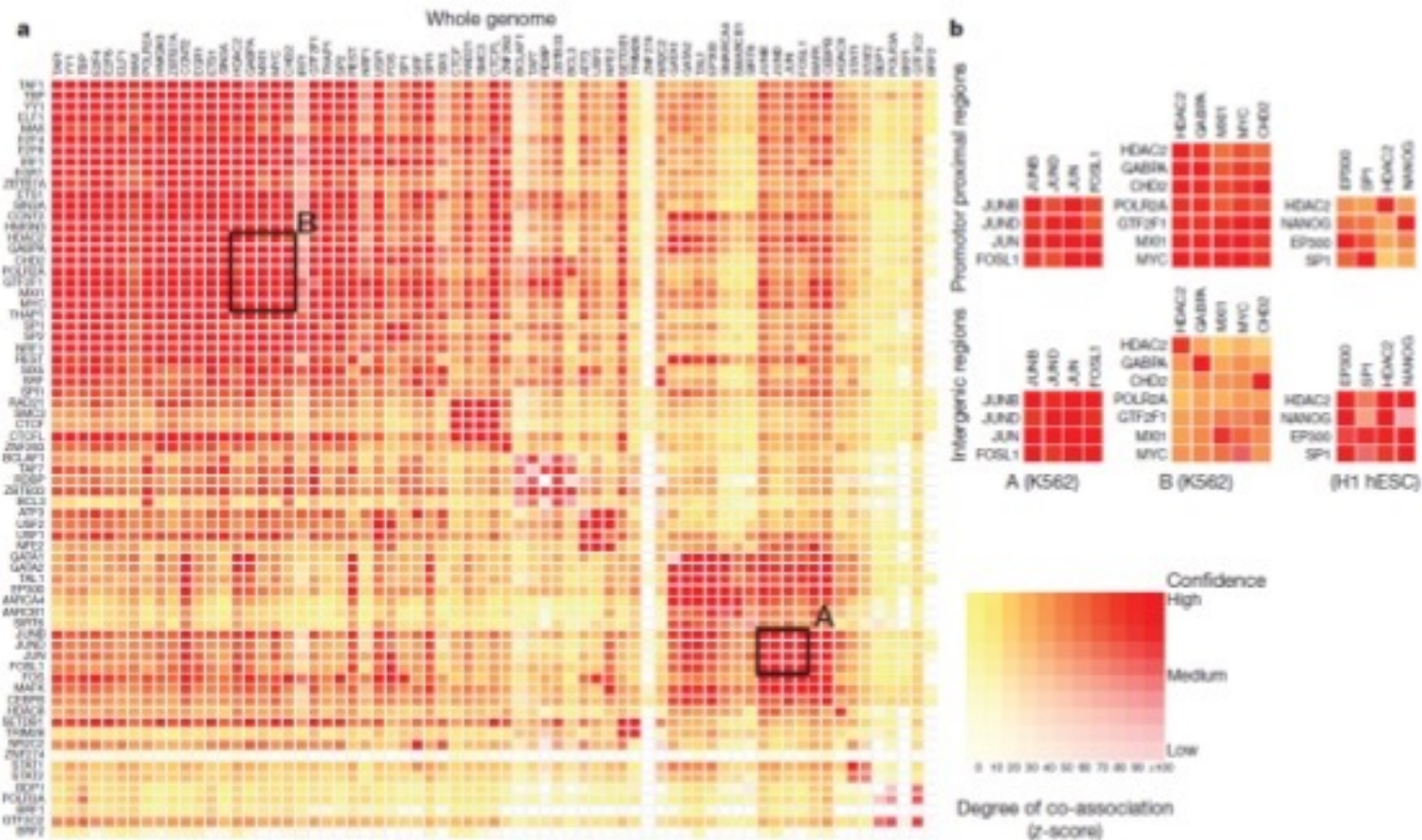
Main ENCODE topics and selection of most important papers



*Chromatin state patterns at
transcription-factor binding
sites*

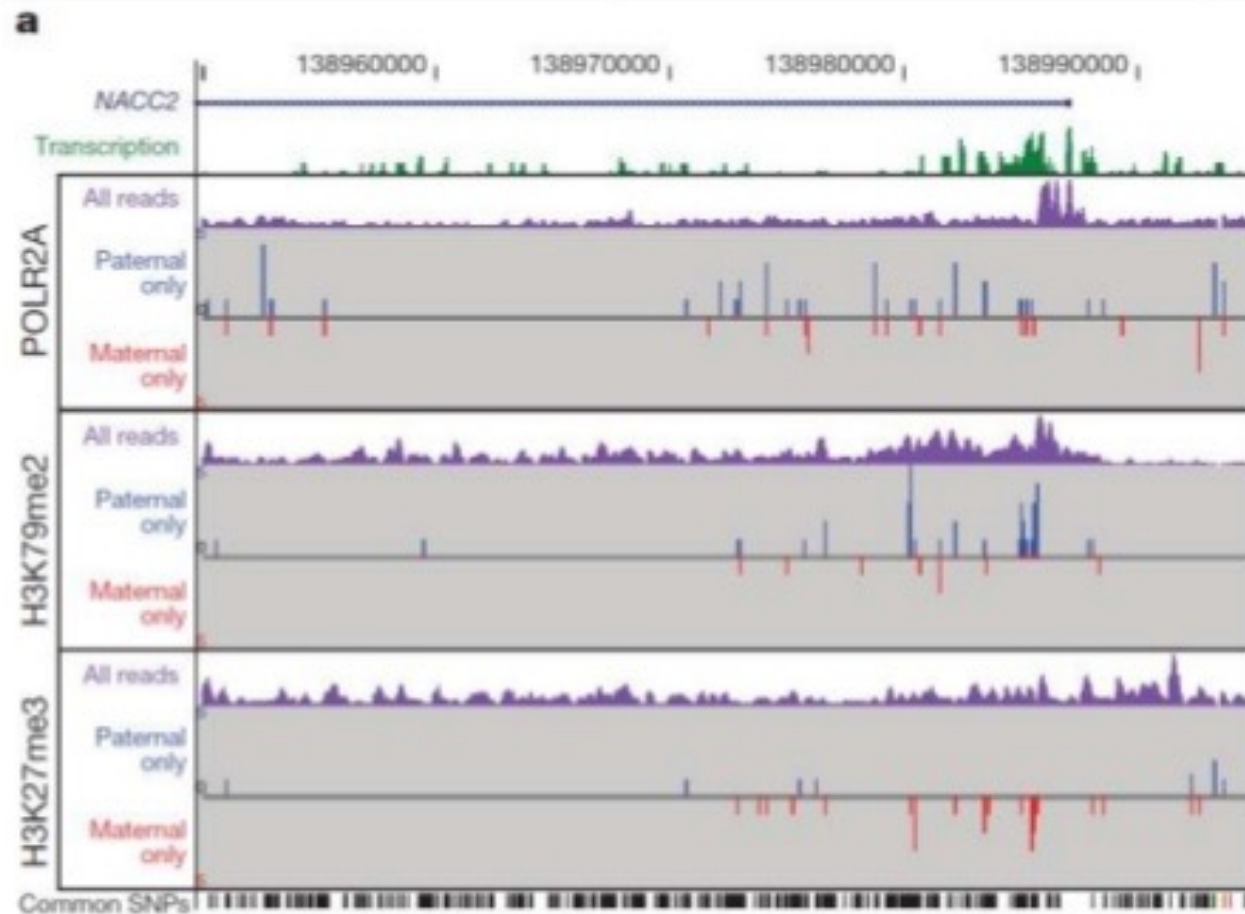
Key ENCODE topics

Main ENCODE topics and selection of most important papers



Key ENCODE topics

Main ENCODE topics and selection of most important papers



Insight in genomic variation – allele specific variation

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Main paper

95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction

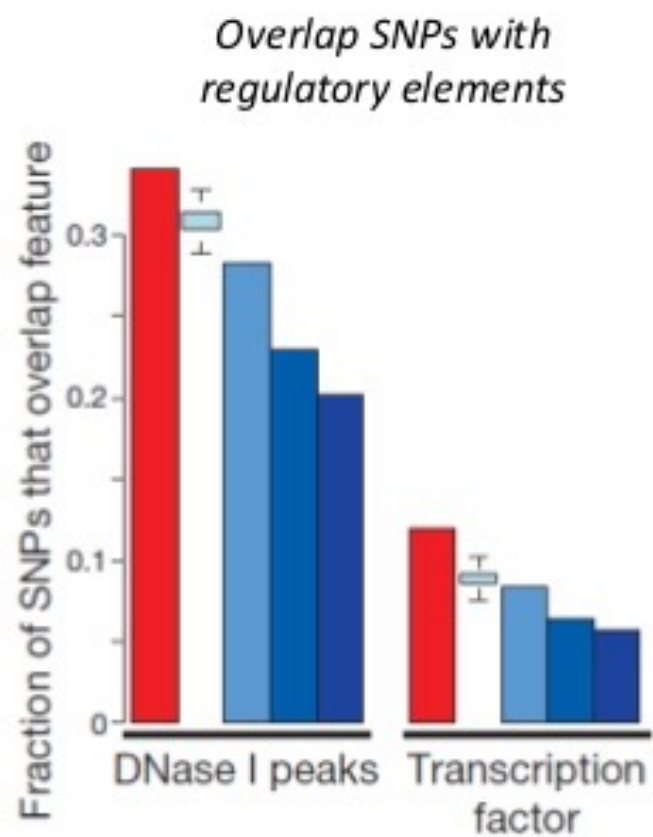
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It is possible to correlate quantitatively RNA sequence production and processing with both chromatin marks and transcription factor binding at promoters, indicating that promoter functionality can explain most of the variation in RNA expression

Single nucleotide polymorphisms (**SNPs**) **associated with disease by GWAS are enriched within non-coding functional elements**, with a majority residing in or near ENCODE-defined regions that are outside of protein-coding genes. In many cases, the disease phenotypes can be associated with a specific cell type or transcription factor

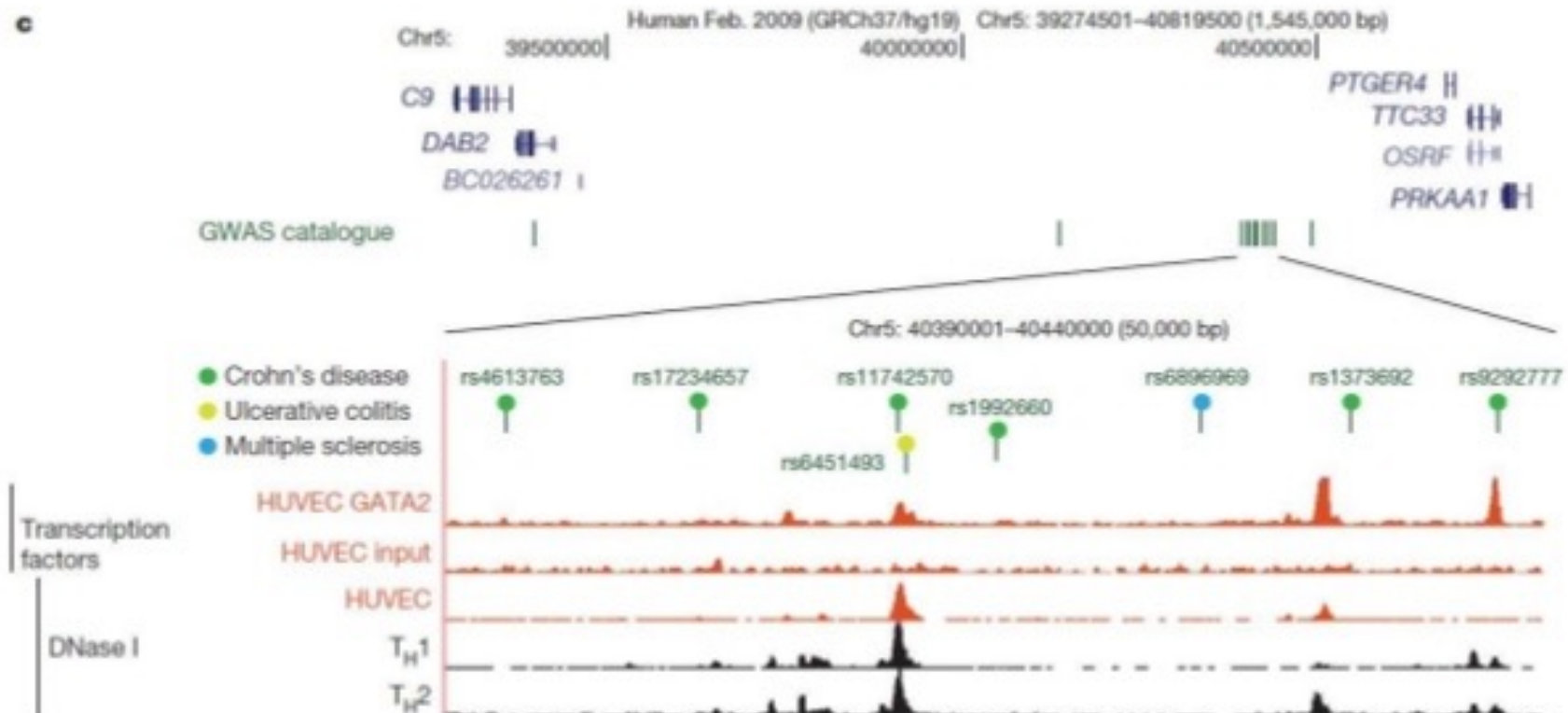
Key ENCODE topics

Main ENCODE topics and selection of most important papers



Key ENCODE topics

Main ENCODE topics and selection of most important papers



Overlap SNPs with regulatory elements and 'open' chromatin

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

ARTICLE

doi:10.1038/nature11232

The accessible chromatin landscape of the human genome

Robert E. Thurman^{1*}, Eric Rynes^{1*}, Richard Humbert^{1*}, Jeff Vierstra¹, Matthew T. Maurano¹, Eric Haugen¹, Nathan C. Sheffield², Andrew B. Stergachis¹, Hao Wang¹, Benjamin Vernot¹, Kavita Garg³, Sam John¹, Richard Sandstrom¹, Daniel Bates¹, Lisa Boatman⁴, Theresa K. Canfield¹, Morgan Diegel¹, Douglas Dunn¹, Abigail K. Ebersol⁴, Tristan Frum⁴, Erika Giste¹, Audra K. Johnson¹, Ericka M. Johnson⁴, Tanya Kutvavin¹, Bryan Lajoie⁵, Bum-Kyu Lee⁶, Kristen Lee¹, Darin London², Dimitra Lotakis⁴, Shane Neph¹, Fidencio Neri¹, Eric D. Nguyen⁴, Hongzhu Qu^{1,7}, Alex P. Reynolds¹, Vaughn Roach¹, Alexias Safi², Minerva E. Sanchez⁴, Amartya Sanyal⁵, Anthony Shafer¹, Jeremy M. Simon⁸, Lingyun Song², Shinny Vong¹, Molly Weaver¹, Yongqi Yan⁴, Yancheng Zhang⁸, Zhuzhu Zhang⁸, Boris Lenhard^{9†}, Muneesh Tewari³, Michael O. Dorschner¹⁰, R. Scott Hansen⁴, Nick A. Navas⁴, George Stamatoyannopoulos⁴, Vishwanath R. Iyer⁶, Jason D. Lieb⁸, Shamil R. Sunyaev¹¹, Joshua M. Akey¹, Sabo¹, Rajinder Kaul⁴, Terrence S. Furey⁸, Job Dekker⁵, Gregory E. Crawford² & John A. Stamatoyannopoulos^{1,12}

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Accessible chromatin landscape

DNAseI treatment

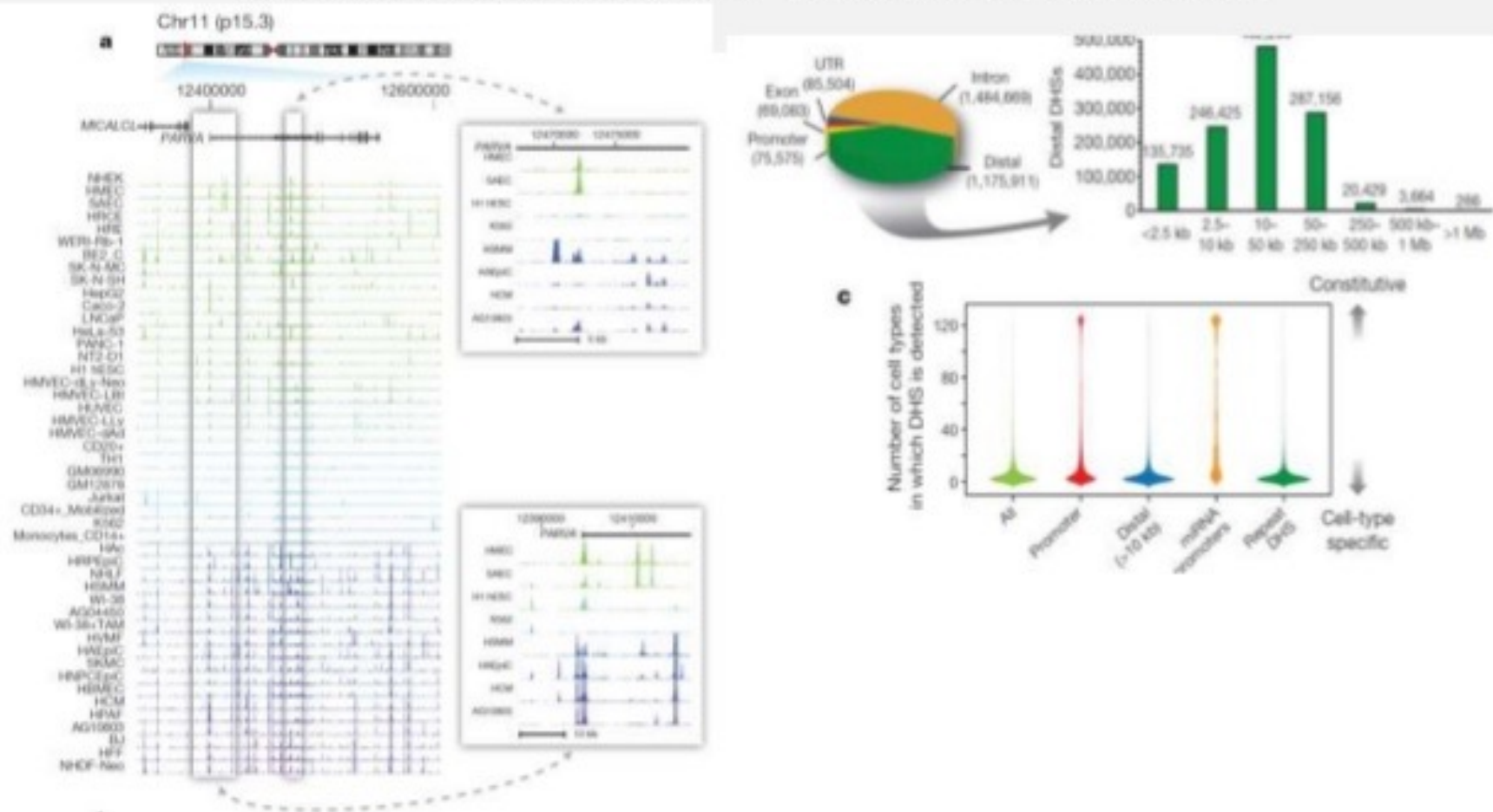
Combined analysis with TFs and H3K4me3

→ Identification of “accessible” chromatin regions

Key ENCODE topics

Main ENCODE topics and selection of most important papers

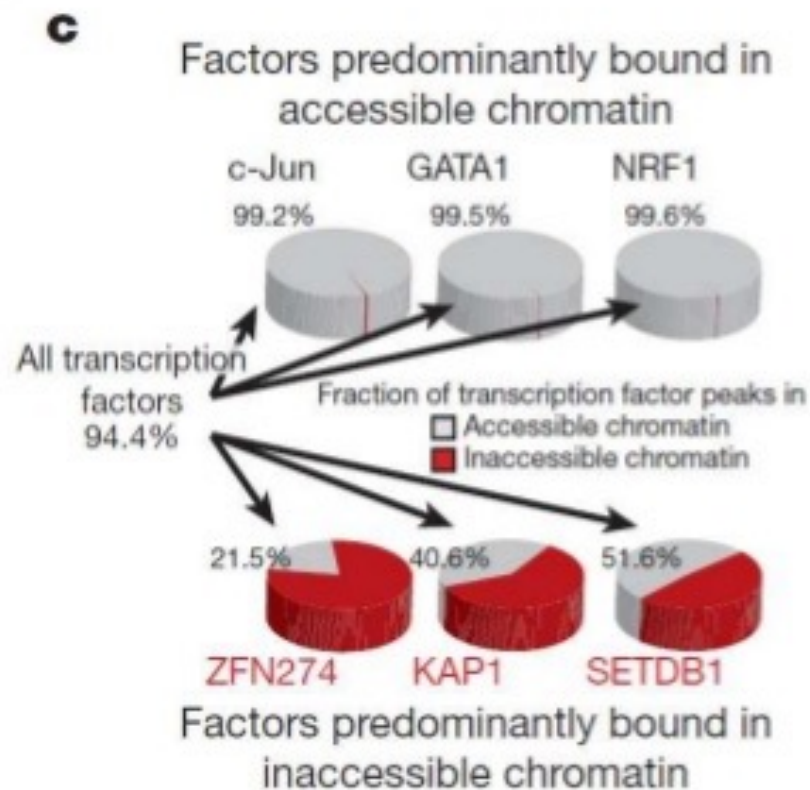
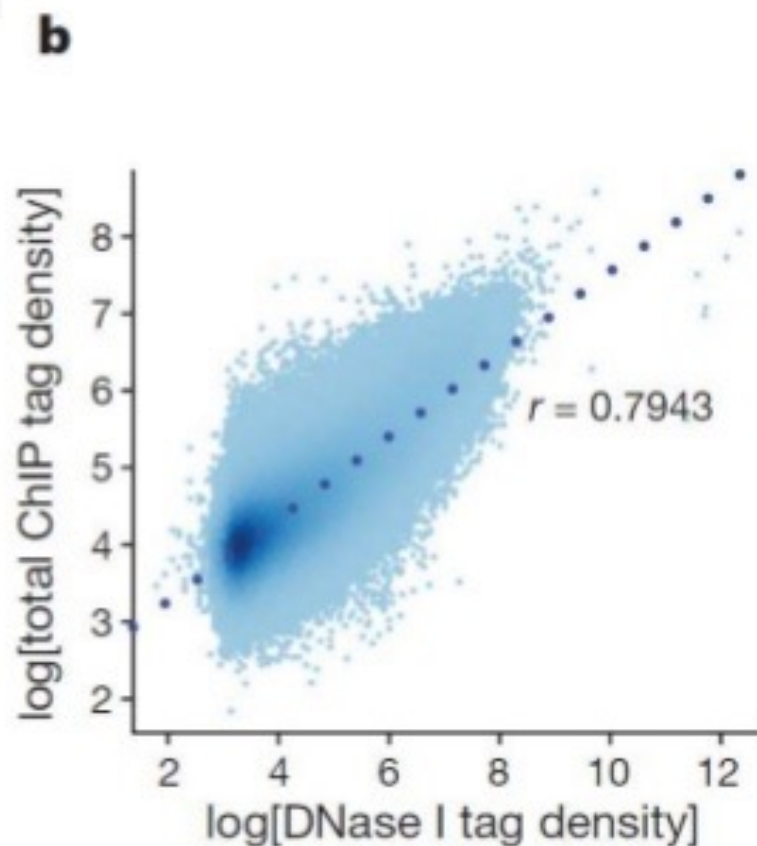
Accessible chromatin landscape – location of accessible regions



Key ENCODE topics

Main ENCODE topics and selection of most important papers

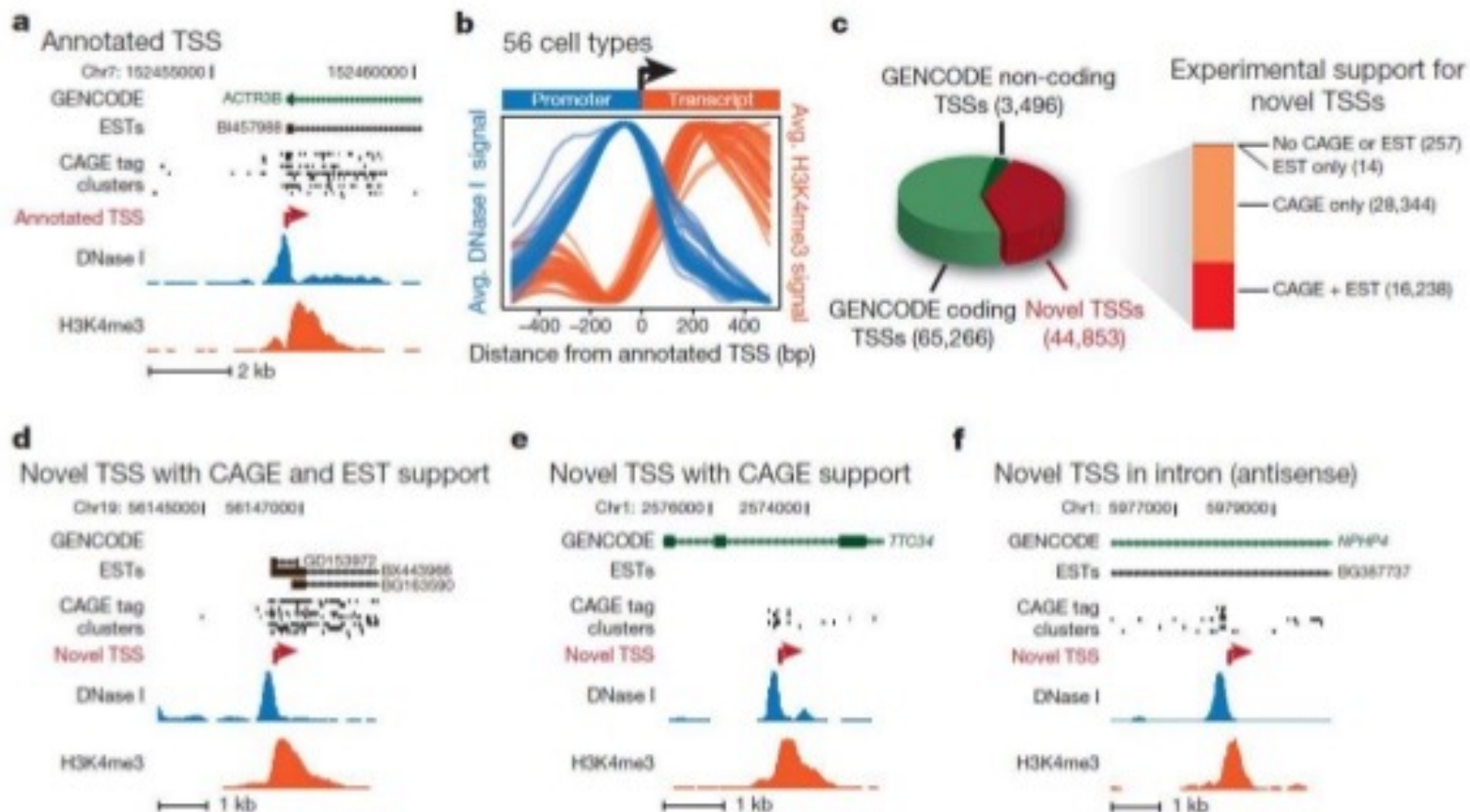
Accessible chromatin landscape – association with ChIP-seq and TFs



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Accessible chromatin landscape – novel transcripts



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

ARTICLE

doi:10.1038/nature11233

Landscape of transcription in human cells

Sarah Djebali^{1*}, Carrie A. Davis^{2*}, Angelika Merkel¹, Alex Dobin², Timo Lassmann³, Ali Mortazavi^{4,5}, Andrea Tanzer¹, Julien Lagarde¹, Wei Lin², Felix Schlesinger², Chenghai Xue², Georgi K. Marinov⁴, Jainab Khatun⁶, Brian A. Williams⁴, Chris Zaleski², Joel Rozowsky^{7,8}, Maik Röder¹, Felix Kokocinski⁹, Rehab F. Abdelhamid³, Tyler Alioto^{1,10}, Igor Antoshechkin⁴, Michael T. Baer², Nadav S. Bar¹¹, Philippe Batut², Kimberly Bell², Ian Bell¹², Sudipto Chakraborty², Xian Chen¹³, Jacqueline Chrast¹⁴, Joao Curado¹, Thomas Derrien¹, Jorg Drenkow², Erica Dumais¹², Jacqueline Dumais¹², Radha Dutttagupta¹², Emilie Falconnet¹⁵, Meagan Fastuca², Kata Fejes-Toth², Pedro Ferreira¹, Sylvain Foissac¹², Melissa J. Fullwood¹⁶, Hui Gao¹², David Gonzalez¹, Assaf Gordon², Harsha Gunawardena¹³, Cedric Howald¹⁴, Sonali Jha², Rory Johnson¹, Philipp Kapranov^{12,17}, Brandon King⁴, Colin Kingswood^{1,10}, Oscar J. Luo¹⁶, Eddie Park⁵, Kimberly Persaud², Jonathan B. Preall², Paolo Ribeca^{1,10}, Brian Risk⁶, Daniel Robyr¹⁵, Michael Sammeth^{1,10}, Lorian Schaffer⁴, Lei-Hoon See², Atif Shahab¹⁶, Jorgen Skancke^{1,11}, Ana Maria Suzuki³, Hazuki Takahashi³, Hagen Tilgner¹⁴, Diane Trout⁴, Nathalie Walters¹⁴, Huaiwen Wang², John Wrobel⁶, Yibao Yu¹³, Xiaolan Ruan¹⁶, Yoshihide Hayashizaki³, Jennifer Harrow⁹, Mark Gerstein^{7,8,18}, Tim Hubbard⁹, Andre Reymond¹⁴, Stylianos E. Antonarakis¹⁵, Gregory Hannon², Morgan C. Giddings^{6,13}, Yijun Ruan¹⁶, Barbara Wold⁴, Arnaldi³, Roderic Guigo^{1,19} & Thomas R. Gingeras^{2,12}

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Landscape of transcription

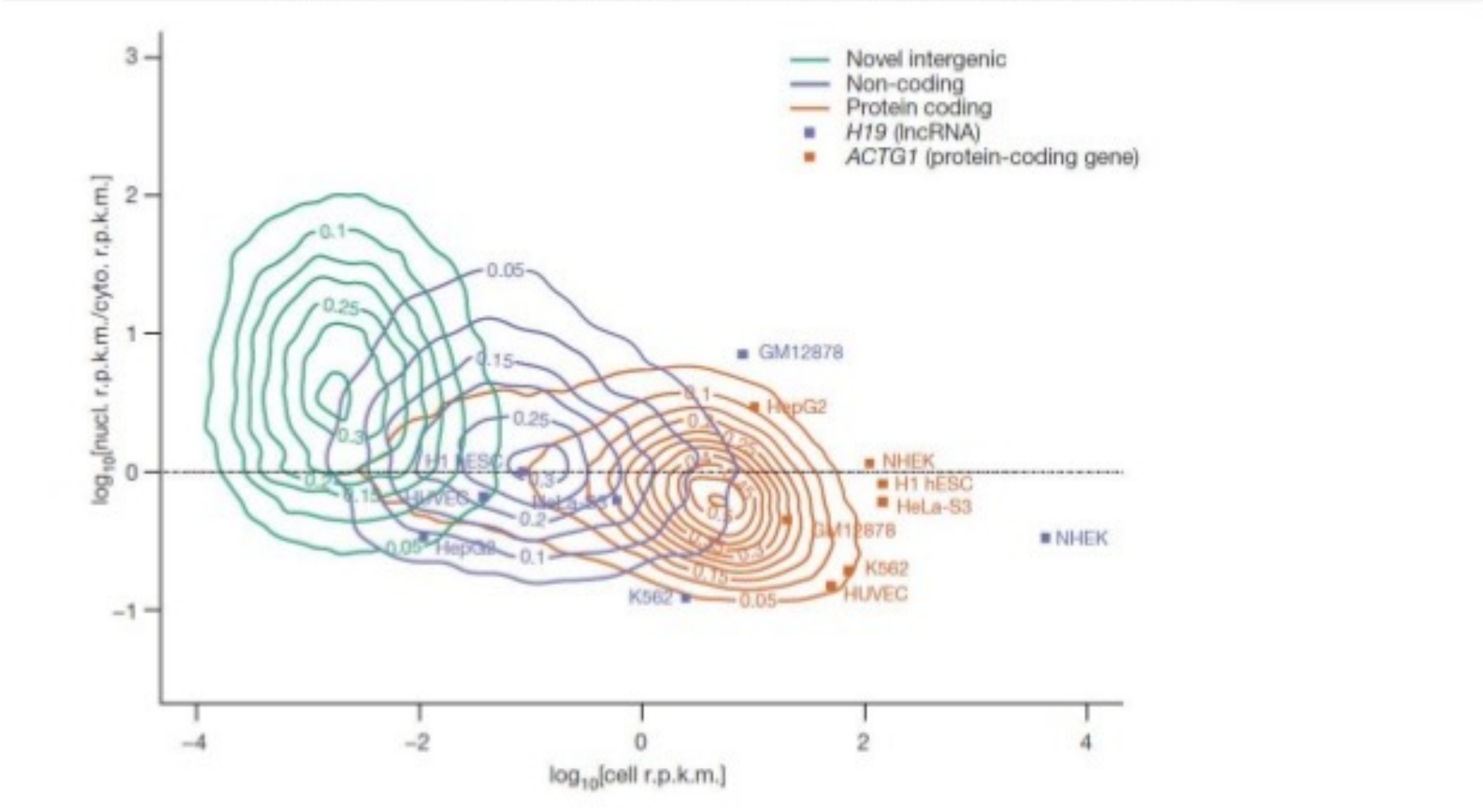
RNA-seq

→ Get a grip on what is transcribed, including novel transcripts and RNAs

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Landscape of transcription – nucleolar fraction vs. whole cell



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Landscape of transcription

Table 1 | Long polyadenylated and non-polyadenylated RNAs

Expression of GENCODE (v7) annotated elements (a)

Gene type	Detected exons† (annotation no.)	Detected splice junctions† (annotation no.)	Detected transcripts† (annotation no.)	Detected genes† (annotation no.)	Exon nucleotide coverage‡ (%)	Number of genes expressed in at least one cell line	Number of genes expressed in only one cell line	Proportion over genes expressed§ (%)	Number of genes expressed in 14 cell lines	Proportion over genes expressed (%)
Long non-coding	22,381 (41,467)	8,017 (26,872)	6,521 (14,880)	5,906 (9,277)	87.5	5,906	1,386	23.5	631	10.7
Protein coding	288,322 (318,514)	194,752 (244,158)	59,822 (76,006)	18,939 (20,679)	98.1	18,939	1,082	5.7	10,571	55.8
Other*	102,000 (133,937)	19,277 (47,663)	45,410 (71,113)	10,649 (21,750)	95.2	10,649	2,453	23.0	1,896	17.8
Total annotated	412,703 (493,918)	222,046 (318,693)	111,753 (161,999)	35,494 (51,706)	96.7	35,394	4,921	13.9	13,098	37.0

Table 2 | Short RNAs

Expression of GENCODE (v7) annotated small RNA genes (a)

Gene type*	GENCODE total	Detected genes (% detected)	No. genes expressed in only one cell line (% detected)	No. genes expressed in 12 cell lines (% detected)	miRNA guide fragment‡	miRNA passenger fragment§	Internal fragments of annotated small RNA (average per detected gene)
miRNA	1,756	497 (28)	59 (12)	147 (30)	454 (454)	175 (175)	18
snoRNA	1,521	458 (30)	73 (16)	223 (49)	NA	NA	60
snRNA	1,944	378 (19)	123 (33)	41 (11)	NA	NA	36
IRNA	624	465 (75)	29 (6)	197 (42)	NA	NA	52
Other†	1,209	191 (16)	69 (36)	24 (13)	NA	NA	32
Total GENCODE	7,054	1,989 (28)	353 (18)	632 (32)	NA	NA	40

Expression of unannotated short RNAs (b)

Cell compartment	Unannotated short RNAs	Exonic	Intronic	Exon-intron boundaries	Genic	Gene-intergene boundaries	Intergenic
Cell	57,393	14,116	13,773	1,818	29,707	13,048	25,906
Nucleus	82,297	19,334	40,136	5,248	64,718	7,417	16,289
Cytosol	25,455	6,183	5,605	665	12,453	6,631	12,447
Three compartments	150,165	38,969	55,061	7,552	101,582	23,185	45,081

NA = not available

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

LETTER

doi:10.1038/nature11279

The long-range interaction landscape of gene promoters

Amartya Sanyal^{1*}, Bryan R. Lajoie^{1*}, Gaurav Jain¹ & Job Dekker¹

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Long-range interaction of promoters

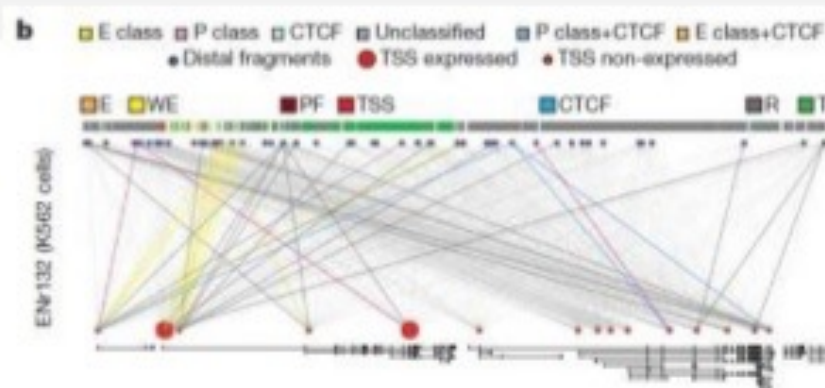
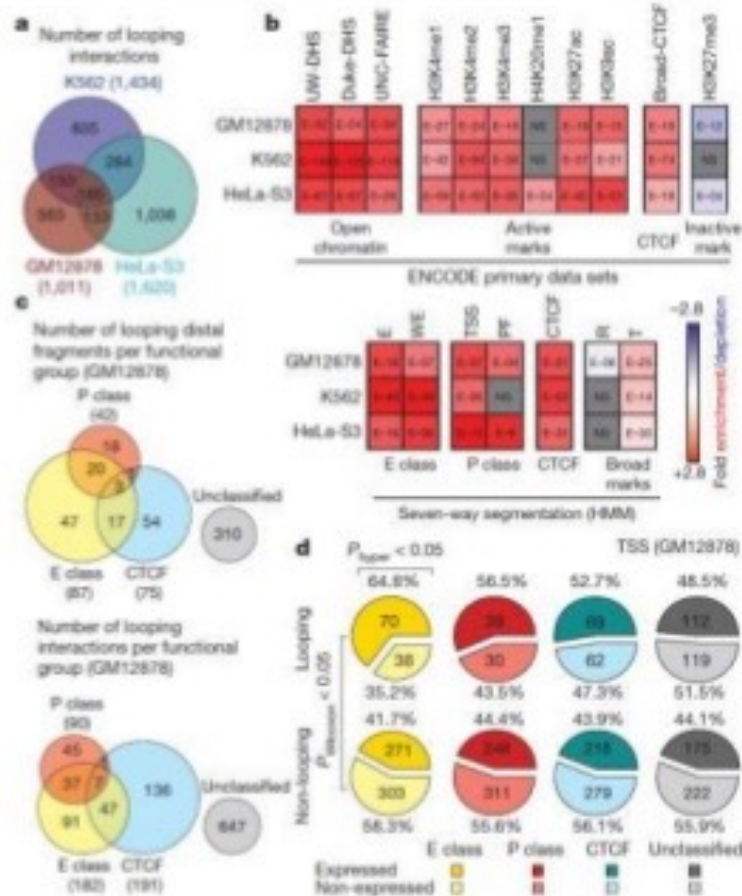
5C mapping (chromatin interaction mapping technology)

→ Long-range interactions of promoter regions

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Long-range interaction of promoters



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

RNA editing in the human ENCODE RNA-seq data

Eddie Park,^{1,2} Brian Williams,^{3,4} Barbara J. Wold,^{3,4} and Ali Mortazavi^{1,2,5}

¹Department of Developmental and Cell Biology, University of California Irvine, Irvine, California 92697, USA; ²Center for Complex Biological Systems, University of California Irvine, Irvine, California 92697, USA; ³Division of Biology, California Institute of Technology, Pasadena, California 91125, USA; ⁴Beckman Institute, California Institute of Technology, Pasadena, California 91125, USA

Long noncoding RNAs are rarely translated in two human cell lines

Balázs Bánfai,^{1,7} Hui Jia,^{2,7} Jainab Khatun,^{3,7} Emily Wood,² Brian Risk,³ William E. Gundling Jr.,² Anshul Kundaje,⁴ Harsha P. Gunawardena,⁵ Yanbao Yu,⁵ Ling Xie,⁵ Krzysztof Krajewski,⁵ Brian D. Strahl,⁵ Xian Chen,⁵ Peter Bickel,¹ Morgan C. Giddings,⁶ James B. Brown,^{1,7,8} and Leonard Lipovich^{2,7,8}

¹Department of Statistics, University of California, Berkeley, California 94720, USA; ²Center for Molecular Medicine and Genetics, School of Medicine, Wayne State University, Detroit, Michigan 48201, USA; ³Biomolecular Research Center, Boise State University, Boise, Idaho 83725, USA; ⁴Department of Computer Science, Stanford University, Palo Alto, California 94305, USA; ⁵University of North Carolina School of Medicine, Chapel Hill, North Carolina 27599, USA; ⁶College of Arts and Sciences, Boise State University, Boise, Idaho 83725, USA; ⁷83725, USA

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Understanding transcriptional regulation by integrative analysis of transcription factor binding data

Chao Cheng,^{1,2} Roger Alexander,^{1,2} Renqiang Min,^{1,2} Jing Leng,² Kevin Y. Yip,^{1,2,3}
Joel Rozowsky,^{1,2} Koon-Kiu Yan,^{1,2} Xianjun Dong,⁴ Sarah Djebali,⁵ Yijun Ruan,⁶
Carrie A. Davis,⁷ Piero Carninci,⁸ Timo Lassman,⁸ Thomas R. Gingeras,⁷ Roderic Guigó,⁵
van Birney,⁹ Zhiping Weng,⁴ Michael Snyder,¹⁰ and Mark Gerstein^{1,2,11,12}

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Transcriptional regulation

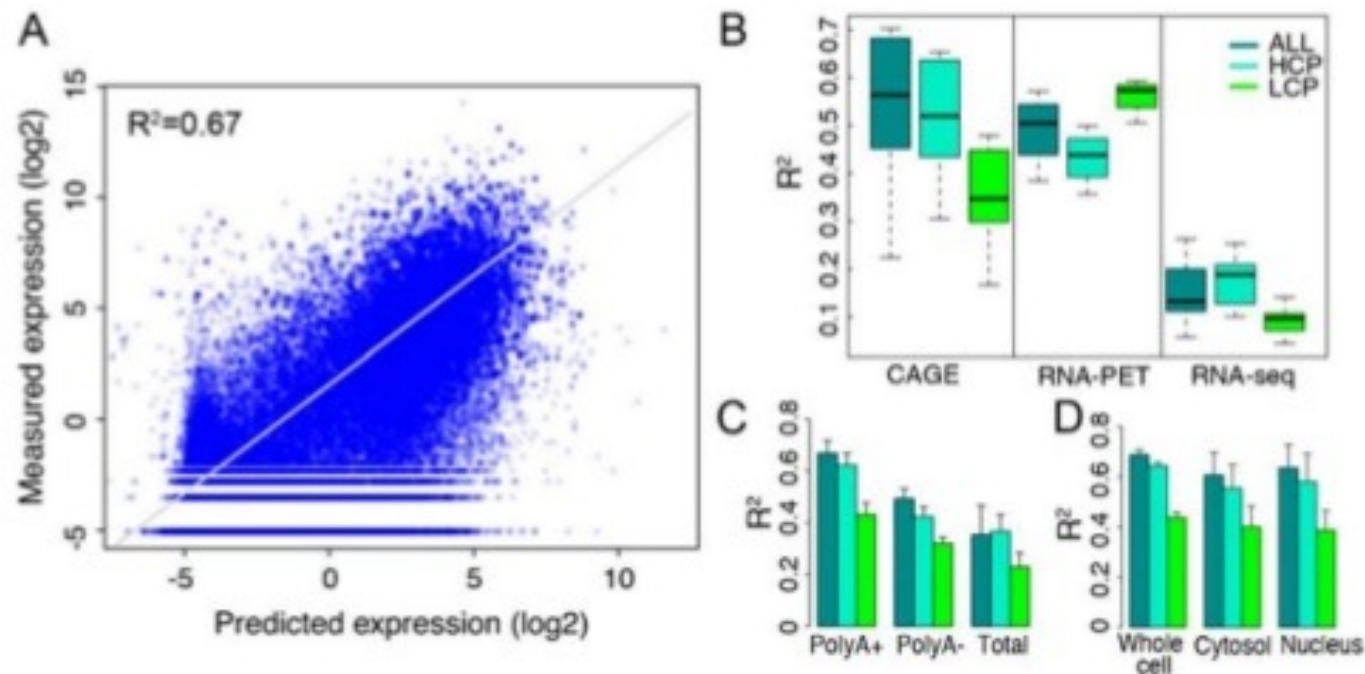
ChIP-seq <> expression detection

→ Predict transcriptional regulation

Key ENCODE topics

Main ENCODE topics and selection of most important papers

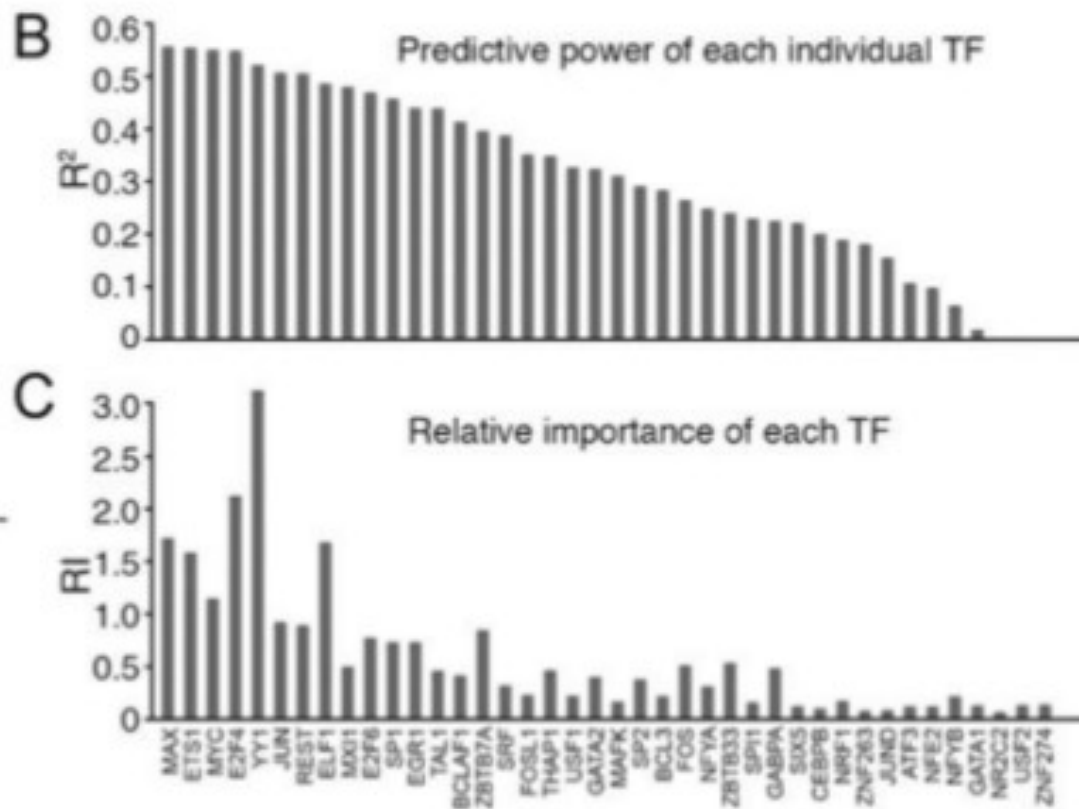
Transcriptional regulation – predict transcription



Key ENCODE topics

Main ENCODE topics and selection of most important papers

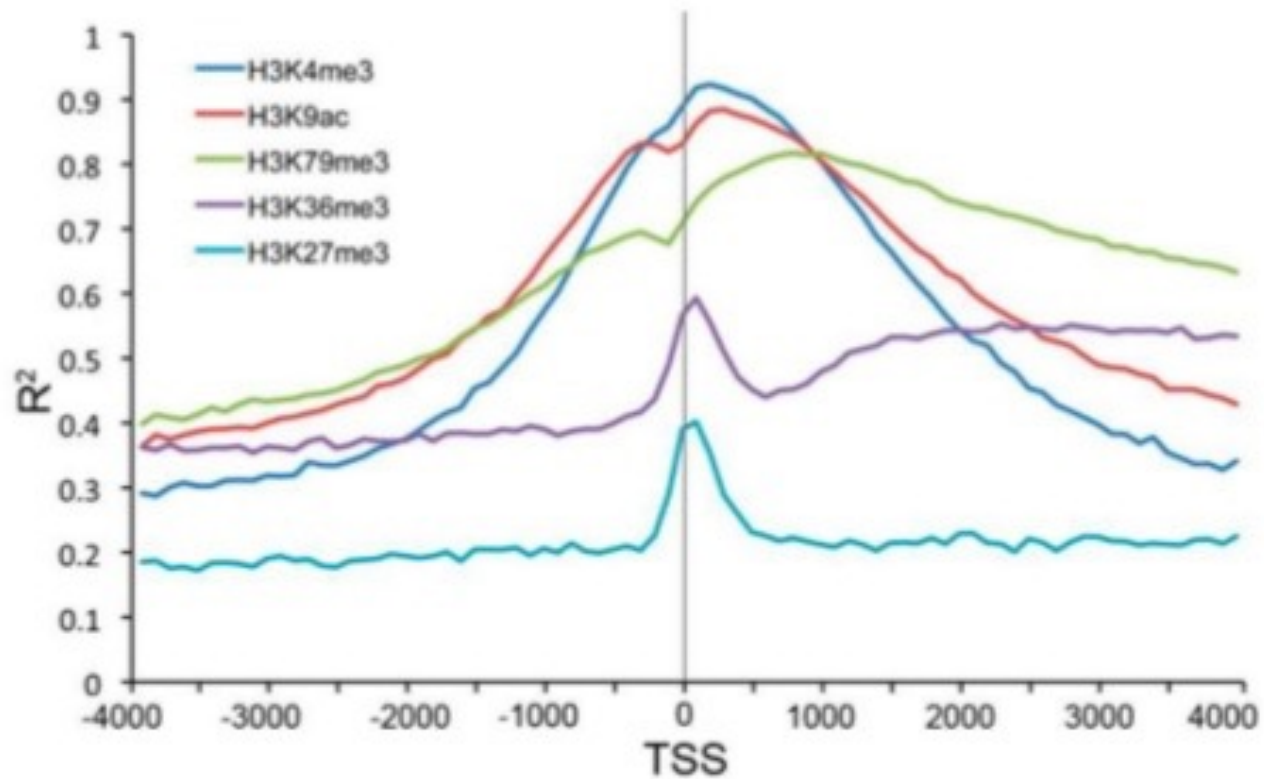
Transcriptional regulation – expression prediction



Key ENCODE topics

Main ENCODE topics and selection of most important papers

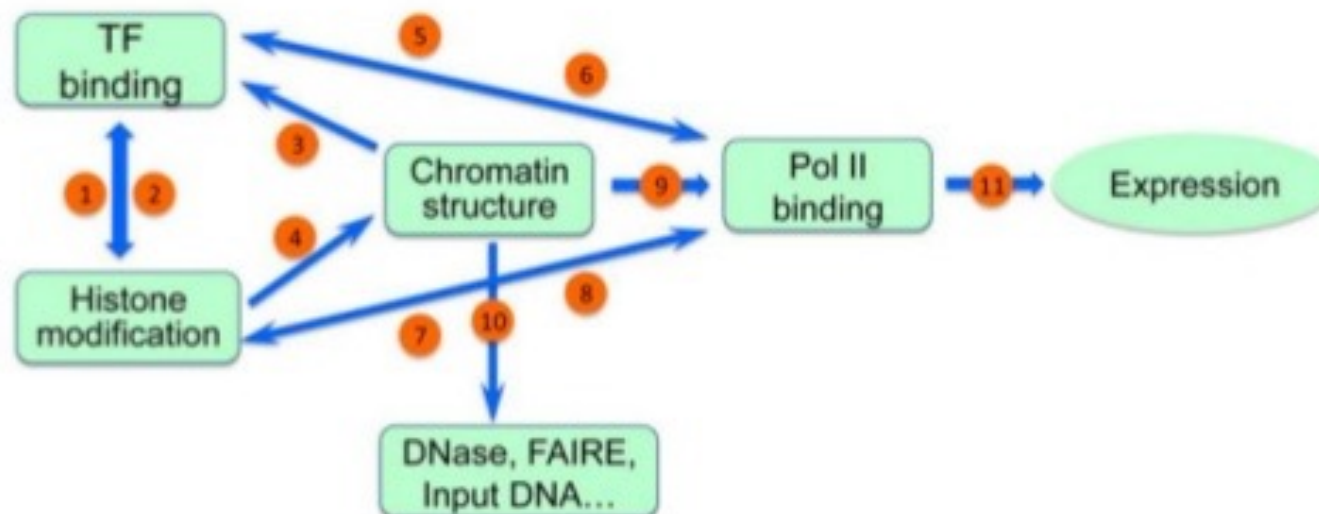
Transcriptional regulation – TFs predict location of histone modifications



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Transcriptional regulation – model



- (1) Recruiting histone modifiers
- (2) Recruiting TFs
- (3) Accessibility
- (4) Remodeling
- (5) Recruiting general TFs
- (6) Interacting with TFs

- (7) Recruit general TFs
- (8) Interacting with histone modifiers
- (9) Accessibility
- (10) Accessibility
- (11) Transcription

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Predicting cell-type-specific gene expression from regions of open chromatin

Anirudh Natarajan,¹ Galip Gürkan Yardımcı,¹ Nathan C. Sheffield,¹
Gregory E. Crawford,^{2,3,5} and Uwe Ohler^{2,4,5}

¹Program in Computational Biology and Bioinformatics, Duke University, Durham, North Carolina 27708, USA; ²Institute for Genome Sciences & Policy, Duke University, Durham, North Carolina 27708, USA; ³Department of Pediatrics, Division of Medical Genetics, Duke University, Durham, North Carolina 27708, USA; ⁴Department of Biostatistics & Bioinformatics, Duke University, Durham, North Carolina 27708, USA

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Cell-type specific gene expression from open chromatin regions

Table 1. Candidate TFs identified by the classifier for each cell line using Split DHS from the top 10 highest absolute Z-score of expression and nonredundant TFs

Cell type	UR-UR Other genes			UR-DR genes		
	AuROC	TFs Positive coefficient	TFs Negative coefficient	AuROC	TFs Positive coefficient	TFs Negative coefficient
Chorion	0.55		OSR2, ELK1	0.83	ZFP161	E2F3, ELK1
Medulloblastoma	0.77	CRX, REST		0.8	CRX, REST, NR2F2, SOX11	
FB0167P	0.72	BACH2, ZIC1, AIRE		0.72	STAT1, ZBTB12, BACH2, HOXC11, ZIC1, AIRE	
GM12878	0.75	EGR2, SPIB		0.64	SPIB	ARID5A
H1_E5	0.67	ZIC3, OTX2, NR2F2		0.69	NR2F2, ZIC3, OTX2	MEIS1, NR2F2
Glioblastoma	0.81	ZIC1, IRF3, BAPX1		0.74	ZIC1, HOXD10	
HeLa S3	0.84	PAX6, E2F2, FOXF2, ELK1, ARNT, ESRRA		0.84	MEOX1, ARNT, FOXF2, PAX6, ELK1, ESRRA	
Hepatocytes	0.70	FOXJ3, HNF4A, RXRA, STAT3	RXRA, RFX7, HOXA6, FOXJ3, E2F3, STAT3	0.78		E2F3
HepG2	0.77	GFI1, HNF4A		0.67	SOX9, FOXA2, HNF4A	
HMEC	0.6	STAT4		0.68	STAT4, IRF6	
HUVEC	0.67	SOX17		0.71	HIC1, SOX17	
K562	0.66	GATA1		0.64	GATA1	
LnCAP	0.64	NKX3-1		0.6	ZBTB78	
MCF7	0.74	GATA3		0.68	GATA1, ESR1	
Melanocyte	0.76	MAF, LEFT1, IRF4, CLUX1, GABPA		0.83	IRF4, GABPA	TBX5, GABPA
HSMMtube	0.61	HLXB9, MYF6, ZBTB3	SOX11, SIX1, ZBTB12, STRA13, ZBTB3	0.67	MYF6	MYF6, STRA13, ZBTB12, SOX11, GATA6, ZBTB6
NHEK	0.72	MTF1, MAF		0.67	MTF1	
Osteoblast	0.63	BACH1, STAT4, GLIS2		0.56	STAT4, BACH1	
AoSMC	0.83	MEIS1, OSR1, HOXC11, SOX8, PITX3, PAX4		0.82	PITX3, MEIS1, OSR1, HOXC11	

UR vs. UR-Other and UR vs. DR classification tasks are shown. TFs with positive and negative coefficients are shown for both tasks. Genes in bold are up-regulated and other genes are down-regulated in the cell line. Several of the same factors help in classifying UR vs. UR-Other genes and UR vs. DR genes.

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Sequence and chromatin determinants of cell-type-specific transcription factor binding

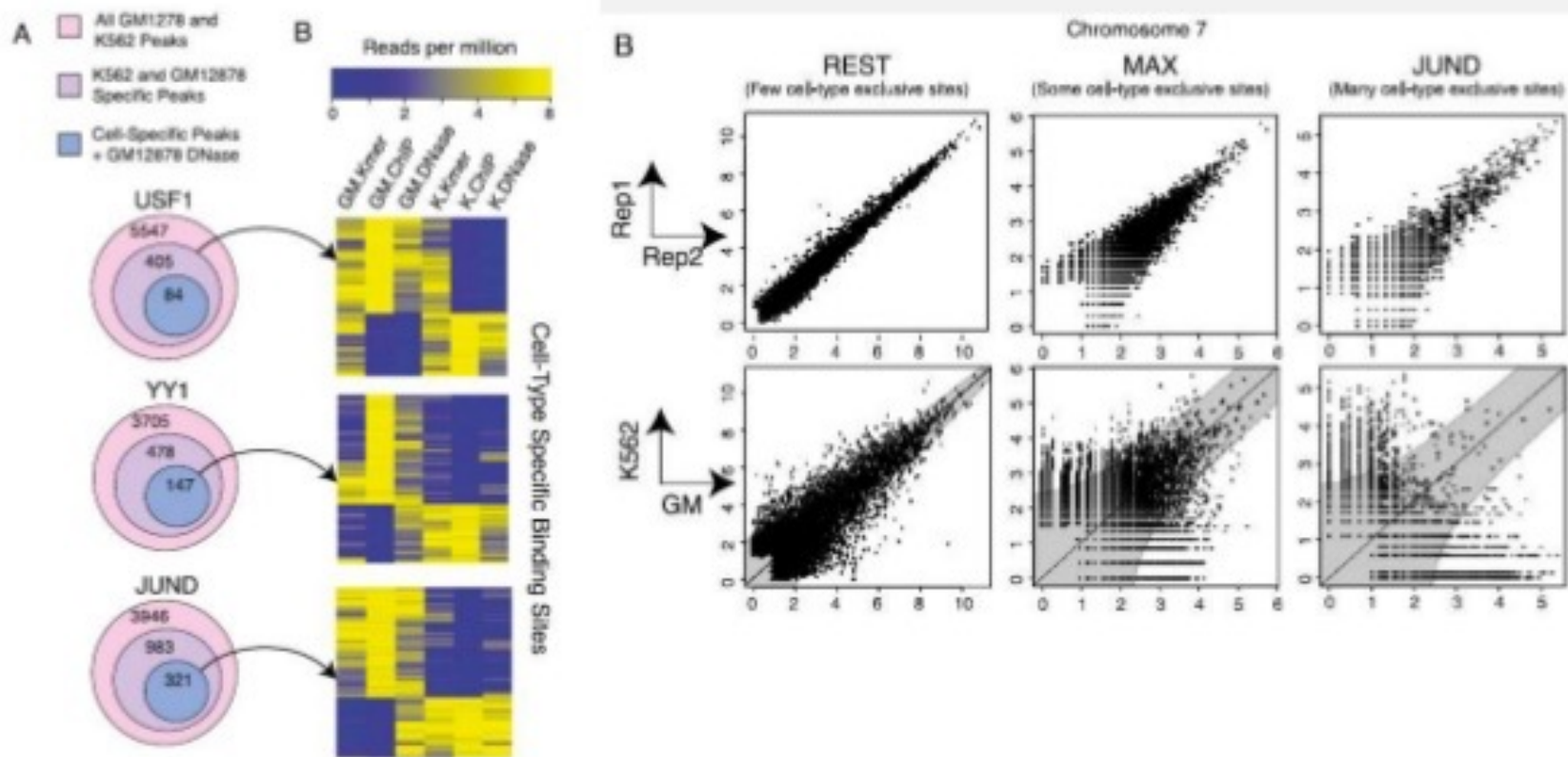
Aaron Arvey,¹ Phaedra Agius,¹ William Stafford Noble,² and Christina Leslie^{1,3}

¹Computational Biology Program, Memorial Sloan-Kettering Cancer Center, New York, New York 10065, USA; ²Department of Genome Sciences, University of Washington, Seattle, Washington 98195, USA

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Cell-type specific TF binding



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Linking disease associations with regulatory information in the human genome

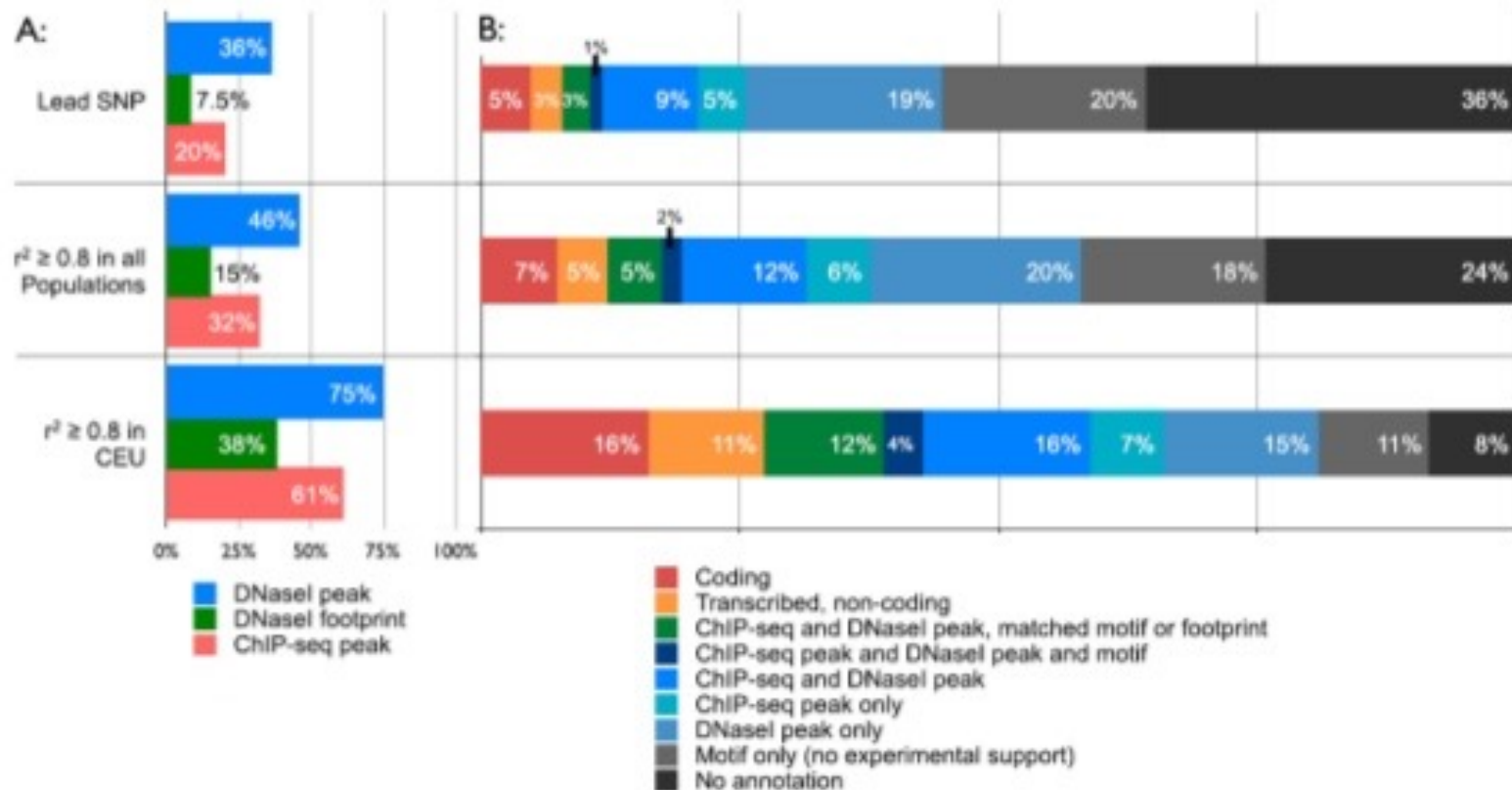
Marc A. Schaub,¹ Alan P. Boyle,² Anshul Kundaje,¹ Serafim Batzoglou,^{1,3} and Michael Snyder^{2,3,4}

¹Department of Computer Science, Stanford University, Stanford, California 94305, USA; ²Department of Genetics, Stanford University School of Medicine, Stanford, California 94305, USA

Key ENCODE topics

Main ENCODE topics and selection of most important papers

SNPs in regulatory regions



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Sequence features and chromatin structure around the genomic regions bound by 119 human transcription factors

Jie Wang,^{1,8} Jiali Zhuang,^{1,8} Sowmya Iyer,^{1,2,8} XinYing Lin,^{1,8} Troy W. Whitfield,¹ Melissa C. Greven,¹ Brian G. Pierce,¹ Xianjun Dong,¹ Anshul Kundaje,³ Yong Cheng,⁴ Oliver J. Rando,¹ Ewan Birney,⁵ Richard M. Myers,⁶ William S. Noble,⁷ Michael Snyder,⁴ and Zhiping Weng^{1,9}

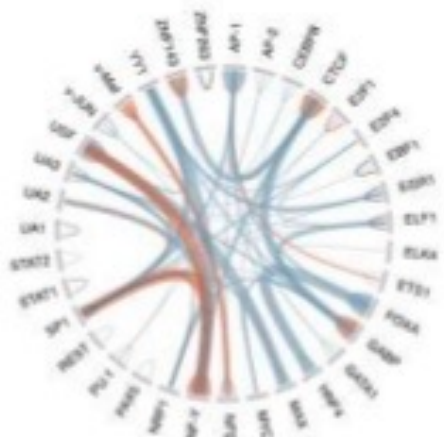
¹Program in Bioinformatics and Integrative Biology, Department of Biochemistry and Molecular Pharmacology, University of Massachusetts Medical School, Worcester, Massachusetts 01605, USA; ²Bioinformatics Program, Boston University, Boston, Massachusetts 02215, USA; ³Department of Computer Science, Stanford University, Stanford, California 94305, USA; ⁴Department of Genetics, Stanford University, Stanford, California 94305, USA; ⁵Vertebrate Genomics Group, European Bioinformatics Institute (EMBL-EBI), Wellcome Trust Genome Campus, Hinxton, Cambridgeshire CB10 1SD, United Kingdom; ⁶HudsonAlpha Institute Biotechnology, Huntsville, Alabama 35806, USA; ⁷Department of Genome Sciences, University of Washington, Seattle, Washington 98195, USA

Key ENCODE topics

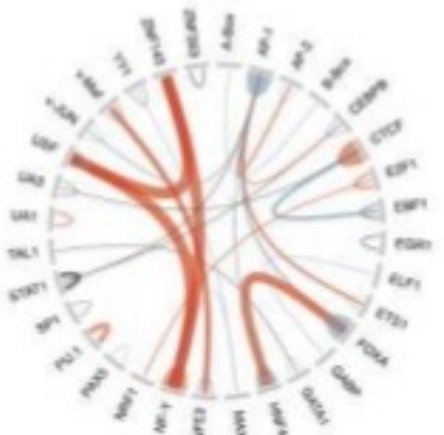
Main ENCODE topics and selection of most important papers

TF binding - interactions

B



C

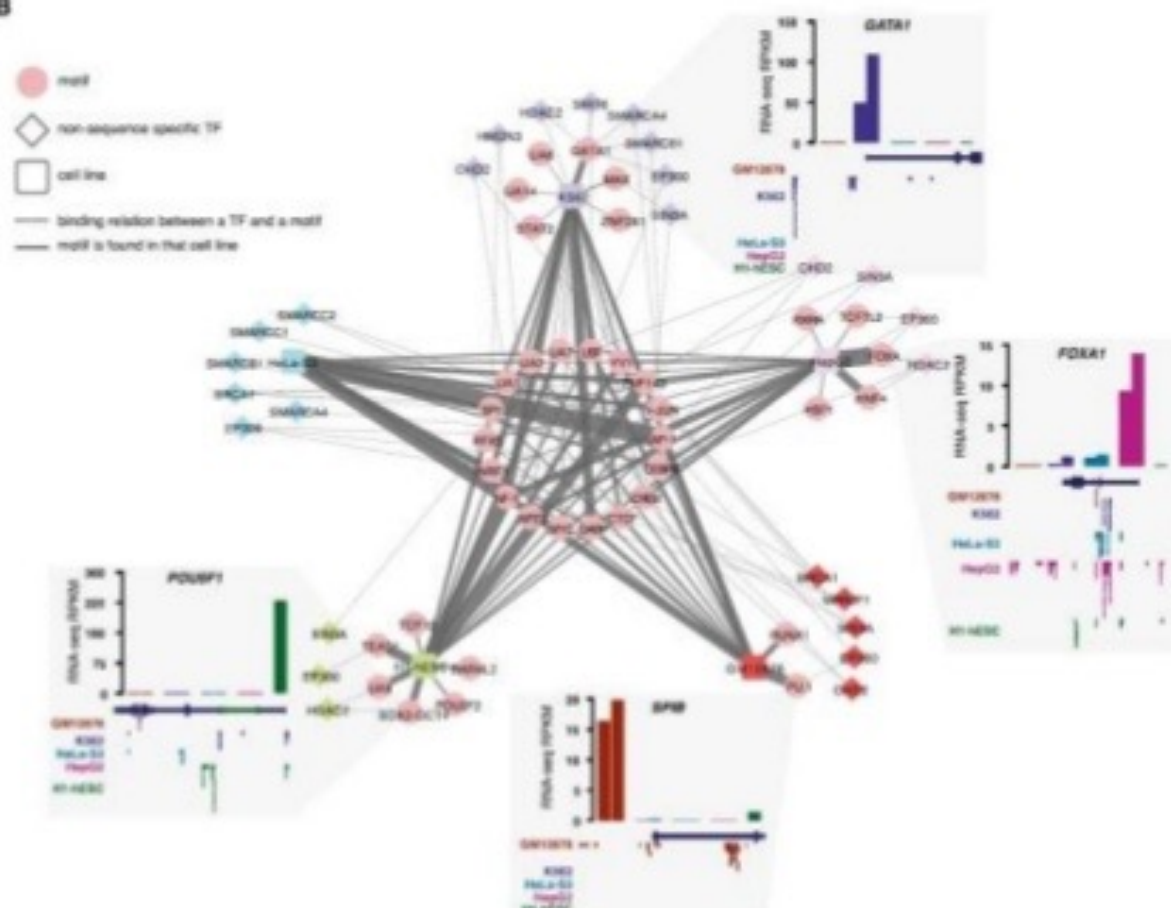


Key ENCODE topics

Main ENCODE topics and selection of most important papers

TF binding – cell-type specificity

B



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Other important papers to us

Yip et al. *Genome Biology* 2012, **13**:R48
<http://genomebiology.com/2012/13/9/R48>



RESEARCH

Open Access

Classification of human genomic regions based on experimentally determined binding sites of more than 100 transcription-related factors

Yip Y Yip^{1,2,3,4,5}, Chao Cheng^{1,2}, Nitin Bhardwaj^{1,2}, James B Brown⁶, Jing Leng¹, Anshul Kundaje⁷,
Rozowsky^{1,2}, Ewan Birney⁸, Peter Bickel⁶, Michael Snyder⁹ and Mark Gerstein^{1,2,10*}

Key ENCODE topics

Main ENCODE topics and selection of most important papers

Classification of genomic regions

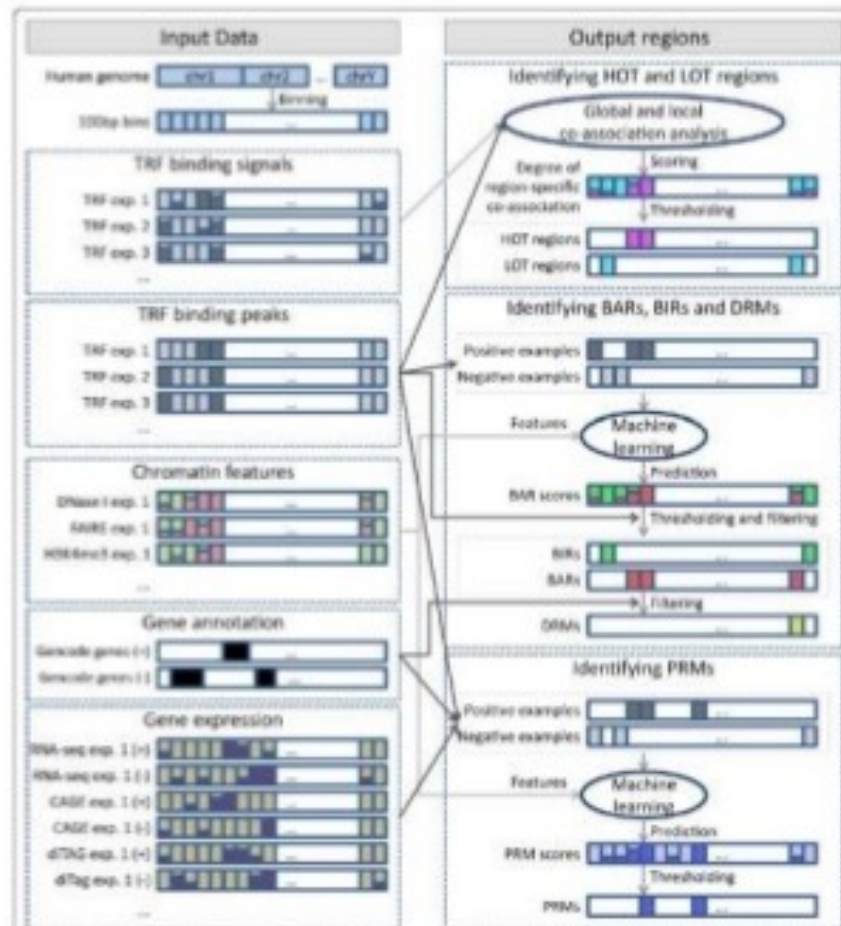
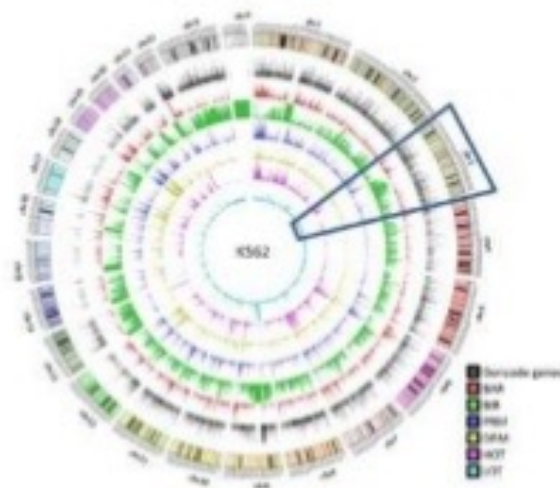


Table 1 Total sizes of the six types of genomic regions derived from transcription factor binding data in the five cell lines

Region type	GM12878	K562	Hep-G2	Hep-G2	K562
Binding active regions (BARs)	109 Mbp (0.33%)	76.8 Mbp (0.23%)	93.8 Mbp (0.28%)	88.4 Mbp (0.27%)	98.7 Mbp (0.29%)
Binding inactive regions (BIRs)	1,000 Mbp (3.03%)	1,200 Mbp (3.59%)	1,000 Mbp (3.03%)	891 Mbp (2.69%)	1,240 Mbp (3.71%)
Promoter proximal regulatory modules (PRMs)	25.4 Mbp (0.076%)	15.3 Mbp (0.045%)	20.7 Mbp (0.062%)	17.1 Mbp (0.051%)	24.1 Mbp (0.072%)
Generalist regulatory modules (GRMs)	24.6 Mbp (0.074%)	18.2 Mbp (0.054%)	25.3 Mbp (0.076%)	21.1 Mbp (0.063%)	22.0 Mbp (0.066%)
High occupancy of TF (HOT) regions (whole genome)	25.9 Mbp (0.078%)	26.4 Mbp (0.078%)	25.6 Mbp (0.077%)	26.1 Mbp (0.078%)	26.4 Mbp (0.079%)
High occupancy of TF (HOT) regions (intergenic regions only)	5.62 Mbp (0.017%)	6.43 Mbp (0.019%)	6.70 Mbp (0.020%)	6.74 Mbp (0.020%)	5.18 Mbp (0.015%)
Low occupancy of TF (LOT) regions (whole genome)	24.7 Mbp (0.074%)	23.1 Mbp (0.069%)	23.8 Mbp (0.072%)	23.6 Mbp (0.071%)	24.2 Mbp (0.073%)
Low occupancy of TF (LOT) regions (intergenic regions only)	10.7 Mbp (0.032%)	11.0 Mbp (0.033%)	10.1 Mbp (0.031%)	10.7 Mbp (0.032%)	12.4 Mbp (0.037%)

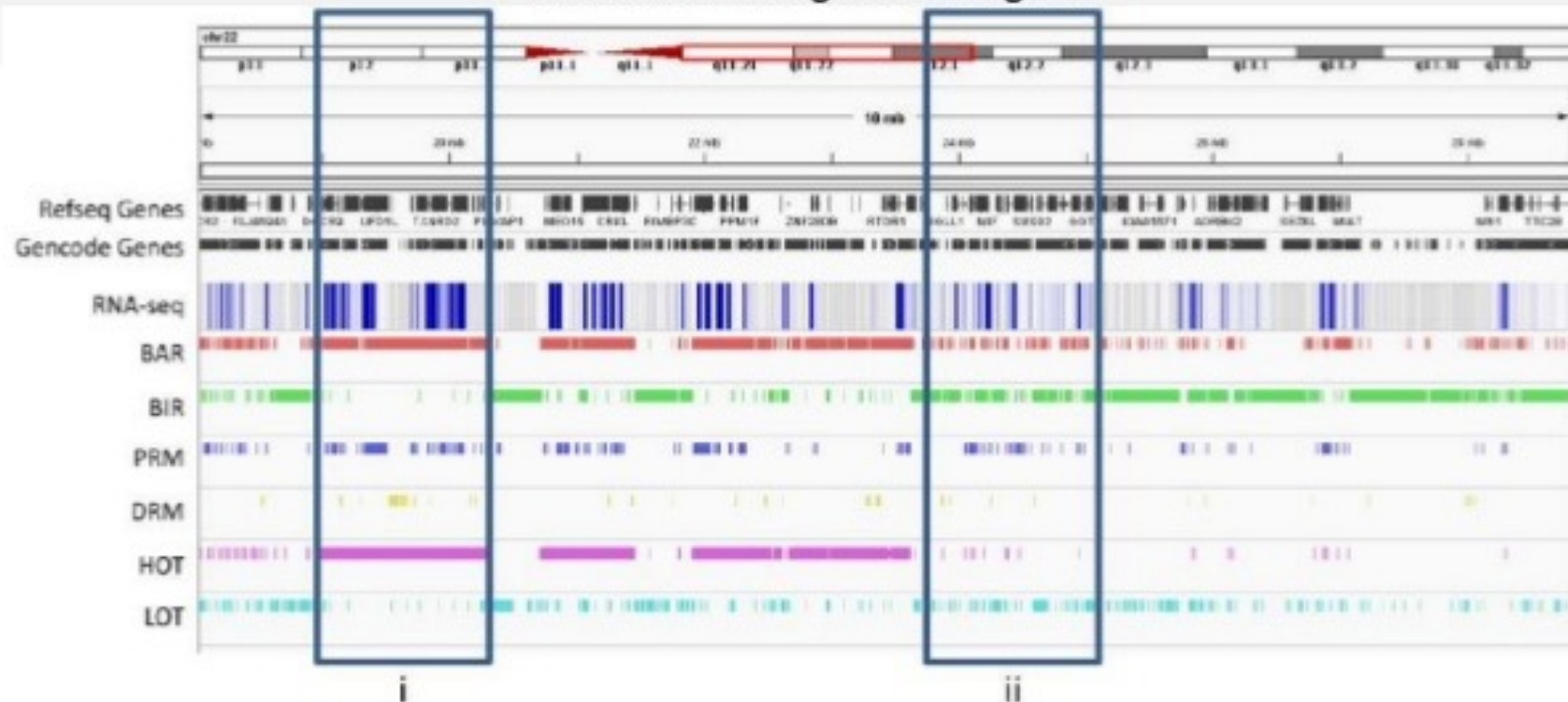
Numbers in parentheses are the percentages of the whole human genome covered by the regions. Intergenic regions are defined as regions of least 10 Mbp from any level 1 or level 2 gene defined in Genecode version 3.



Key ENCODE topics

Main ENCODE topics and selection of most important papers

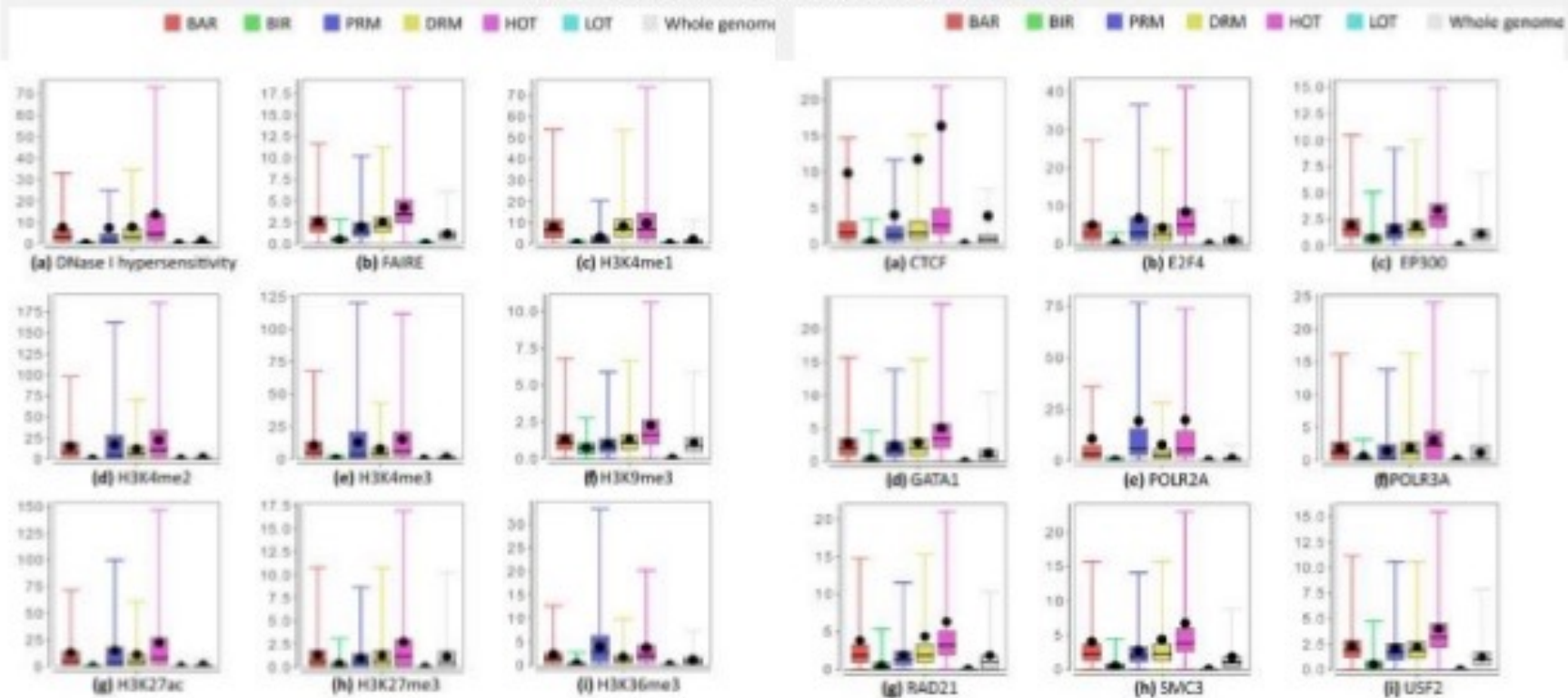
Classification of genomic regions



Key ENCODE topics

Main ENCODE topics and selection of most important papers

Classification of genomic regions



Phase 3

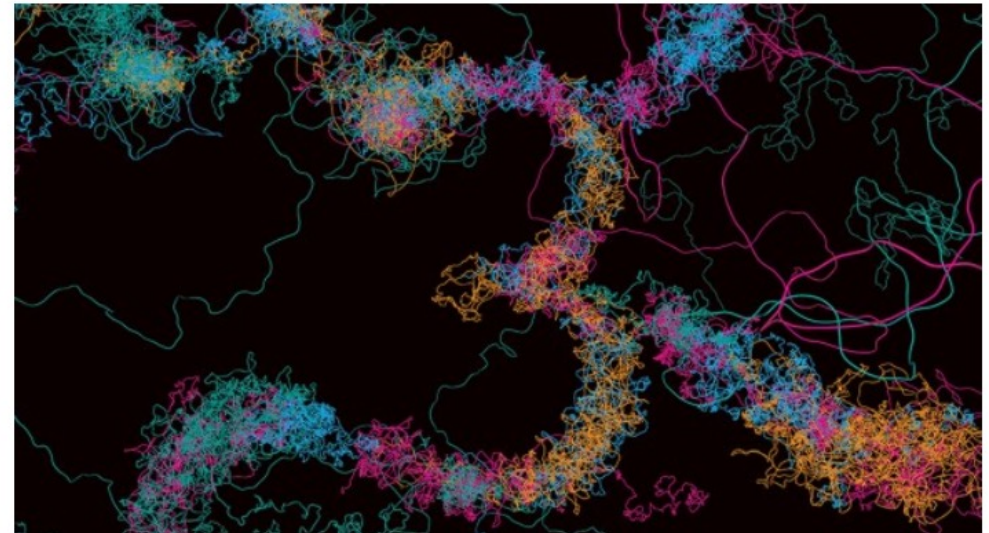
2013-2016

nature > special

SPECIAL | 29 JULY 2020

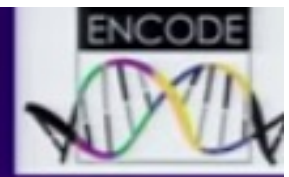
ENCODE 3

How cells, tissues and organisms interpret the information encoded in the genome has vital implications for our understanding of development, health and disease. Launched in 2003, the ENCyclopedia Of DNA Elements (ENCODE) project aims to map the functional elements in the human genome (later expanded to include model organisms).



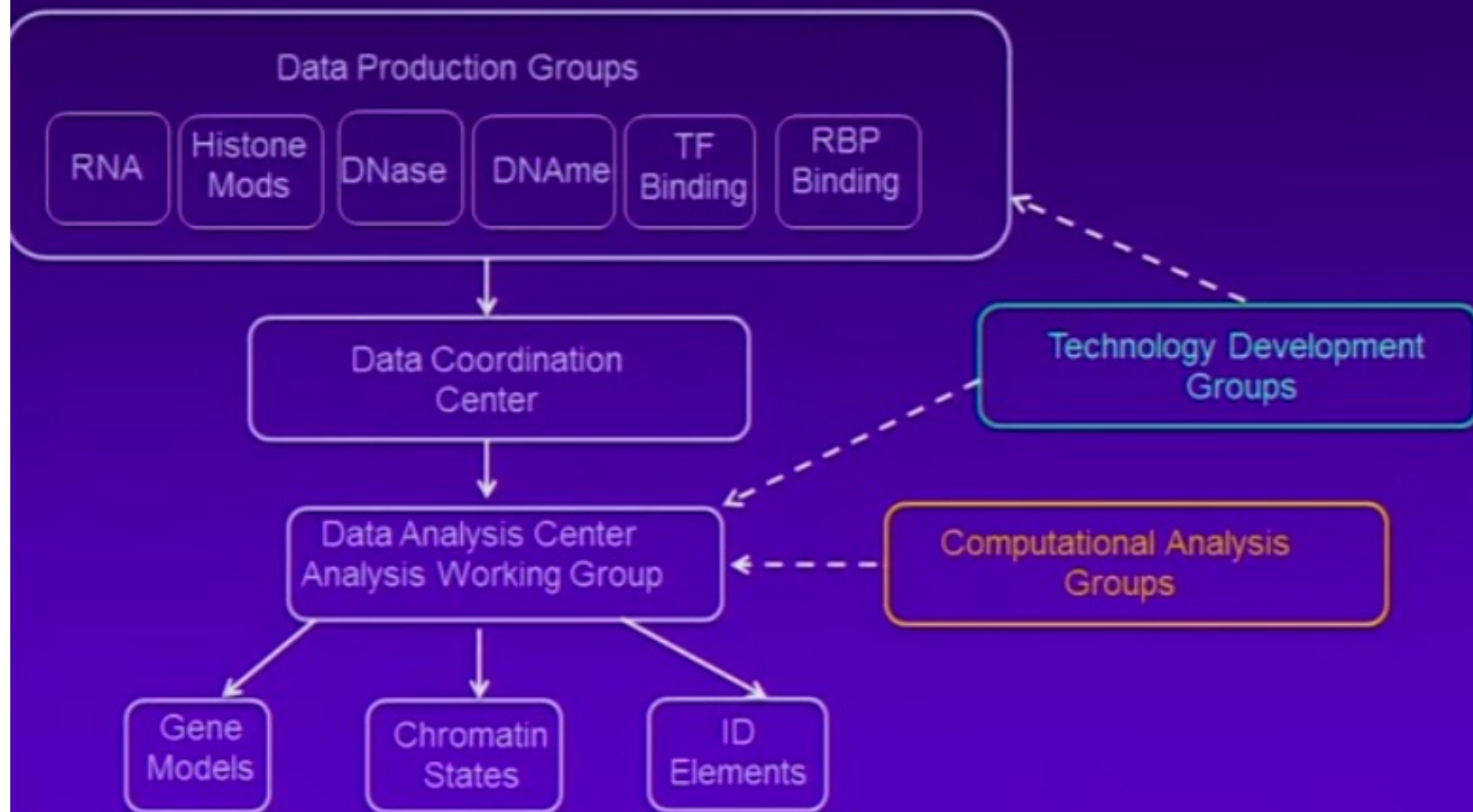
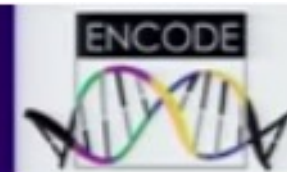
This Collection showcases the main articles and related content resulting from the third phase of ENCODE, during which almost 6,000 new experiments were performed (4,834 involving human samples and 1,158 with mouse samples) and an online registry of more than one million human and mouse candidate cis-regulatory elements (cCREs) was curated. The Encyclopedia paper gives an overview of the various assays and describes the registry of cCREs, and in the companion articles, individual data types are analysed in more detail and the development of novel methodology is reported. This dedicated collection also contains a Perspective, News & Views articles and links to other resources [show less](#)

Goals of ENCODE 3

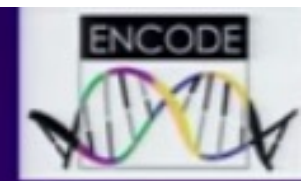


- Expand towards completion the catalog of functional elements in human and mouse genomes
 - Additional cell types, more primary sources
 - Additional data types, e.g., RBPs
- Generate high quality data using high throughput pipelines
- Develop new technologies and analytical tools to generate, analyze and validate data
- Provide data and tools to the community in as useful form as possible
 - Provide easy and rapid access
 - Provide sufficient metadata
 - Describe ways to use it
 - Recognize needs of different types of users

ENCODE 3 Structure



← **The ENCYCLOPEDIA** →

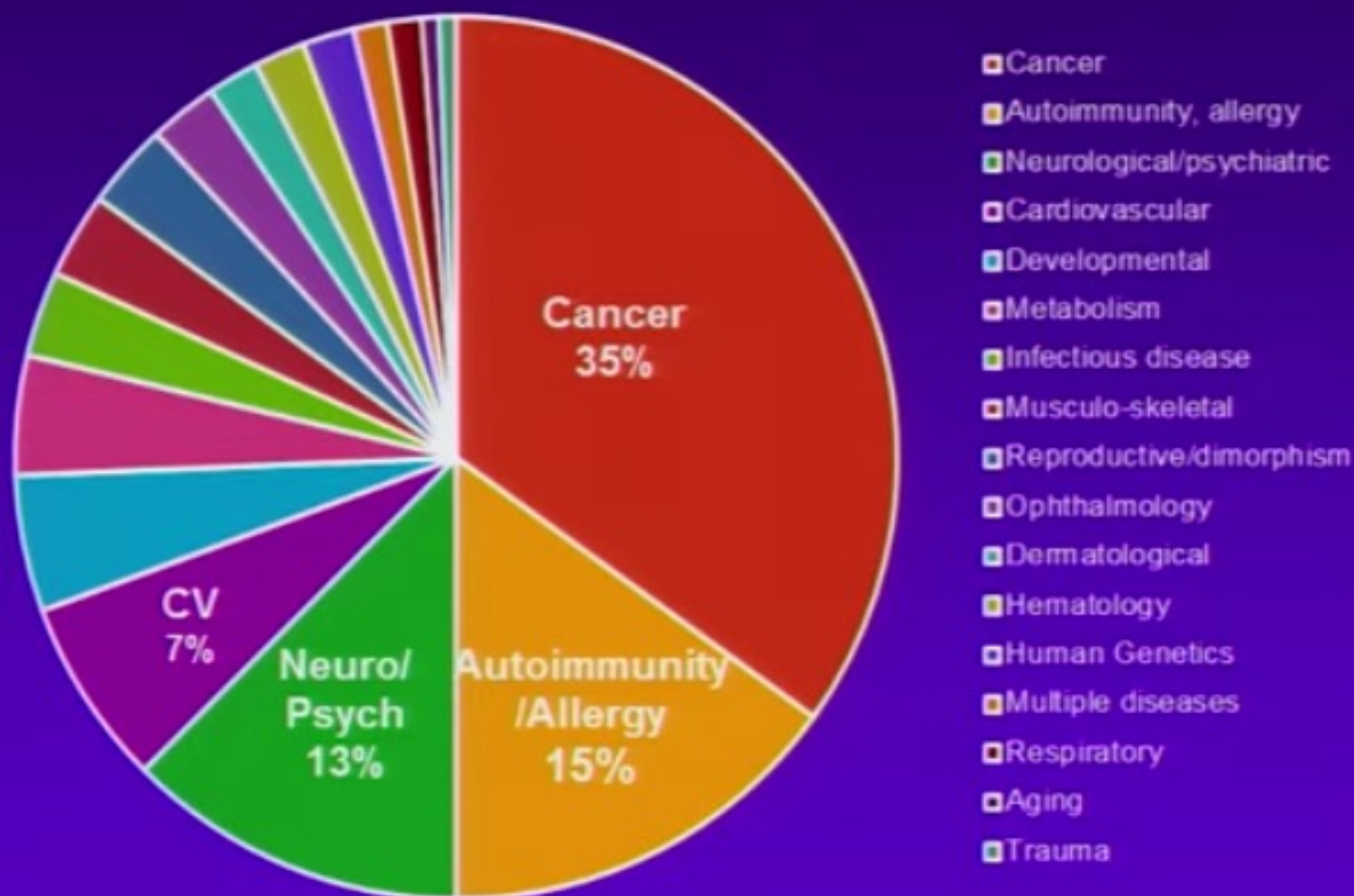


ENCODE 3 Features

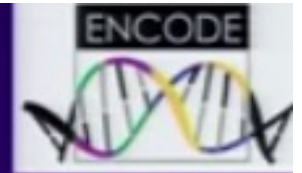
- Cloud computing
 - Data available at Amazon Web Services (AWS)
 - Uniform processing pipelines will be available at DNAnexus
 - Provides transparency about ENCODE data processing
 - Available for broader community to use on own data
- Data Interoperability
 - Coordinating ontology selection and metadata standards with related projects
- Genomics Data Sharing Policy
 - ENCODE working ahead of new GDS policy
 - Developed sample consent language for open access to genomic data
 - Currently working to obtain wide range of samples using new consent language



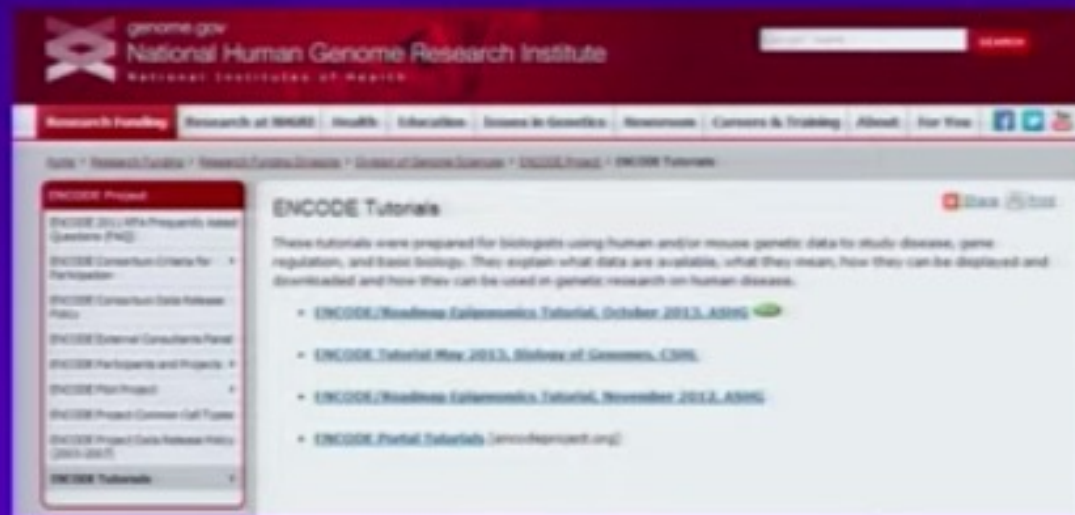
Categories of Disease-Related ENCODE Community Publications



Outreach Activities

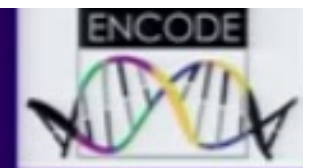


- Tutorials (<http://www.genome.gov/27553900>)



- CHARGE-ENCODE workshop
- User's meeting in 2015

Social Media



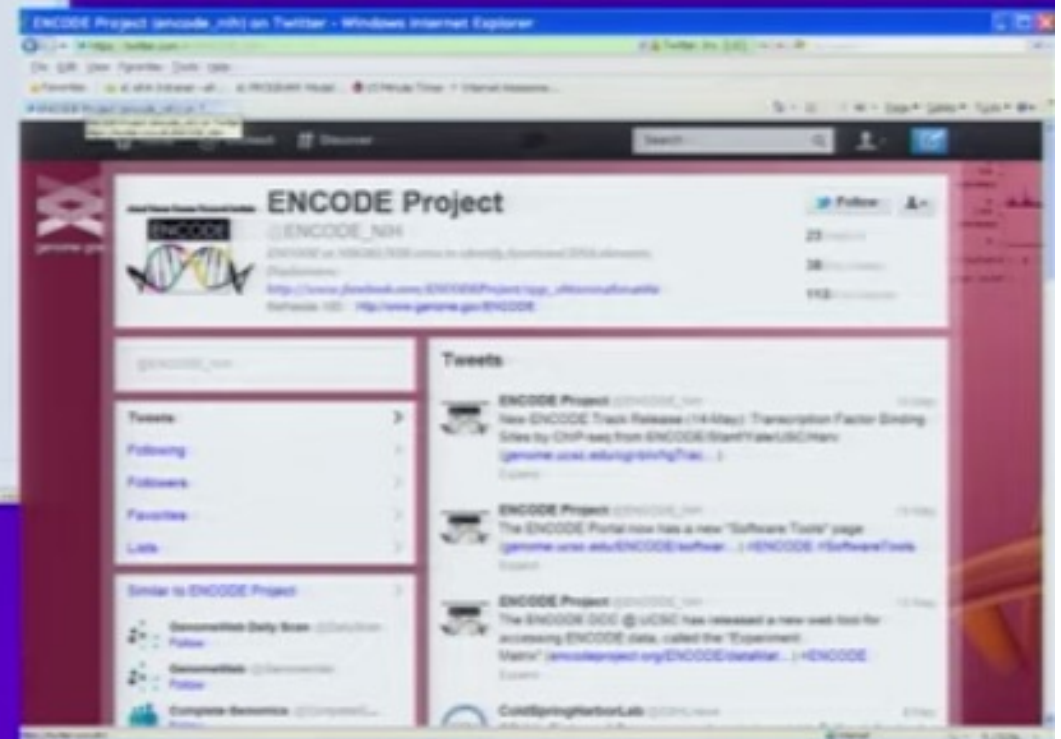
Facebook

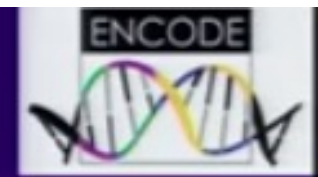
ENCODE (ENCyclopedia Of DNA Elements)



Twitter

@ENCODE_NIH





NHGRI Staff

Program Directors

Elise Feingold

Dan Gilchrist

Peter Good

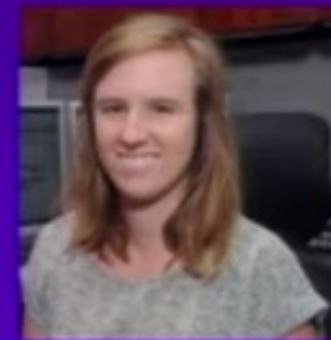
Michael Pazin



Program Analysts

Julie Coursen

Hannah Naughton



Division Director

Jeff Schloss



Phase 4

2017-2021

ENCODE 4 Participants

Mapping Center PIs

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Erez Lieberman-Aiden	Mike Snyder
Mats Ljungman	John Stamatoyannopoulos
Rick Myers Eric Mendenhall	Barbara Wold Ali Mortazavi

Computational Analysis PIs

Mike Beer	Johnathan Pritchard
Christina Leslie	Ting Wang Barak Cohen Cedric Feschotte
Alkes Price Soumya Raychaudhury	Grace Xiao

Functional Characterization PIs

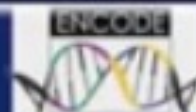
Nadav Ahituv Jay Shendure	Pardis Sabeti
Will Greenleaf Mike Bassik	Yin Shen Bing Ren
John Lis Haiyuan Yu	Tim Reddy Maria Ciofani Greg Crawford Charlie Gersbach

Data Analysis Center

Zhiping Weng	Mark Gerstein
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Data Coordination Center

Mike Cherry



ENCODE 4 Participants

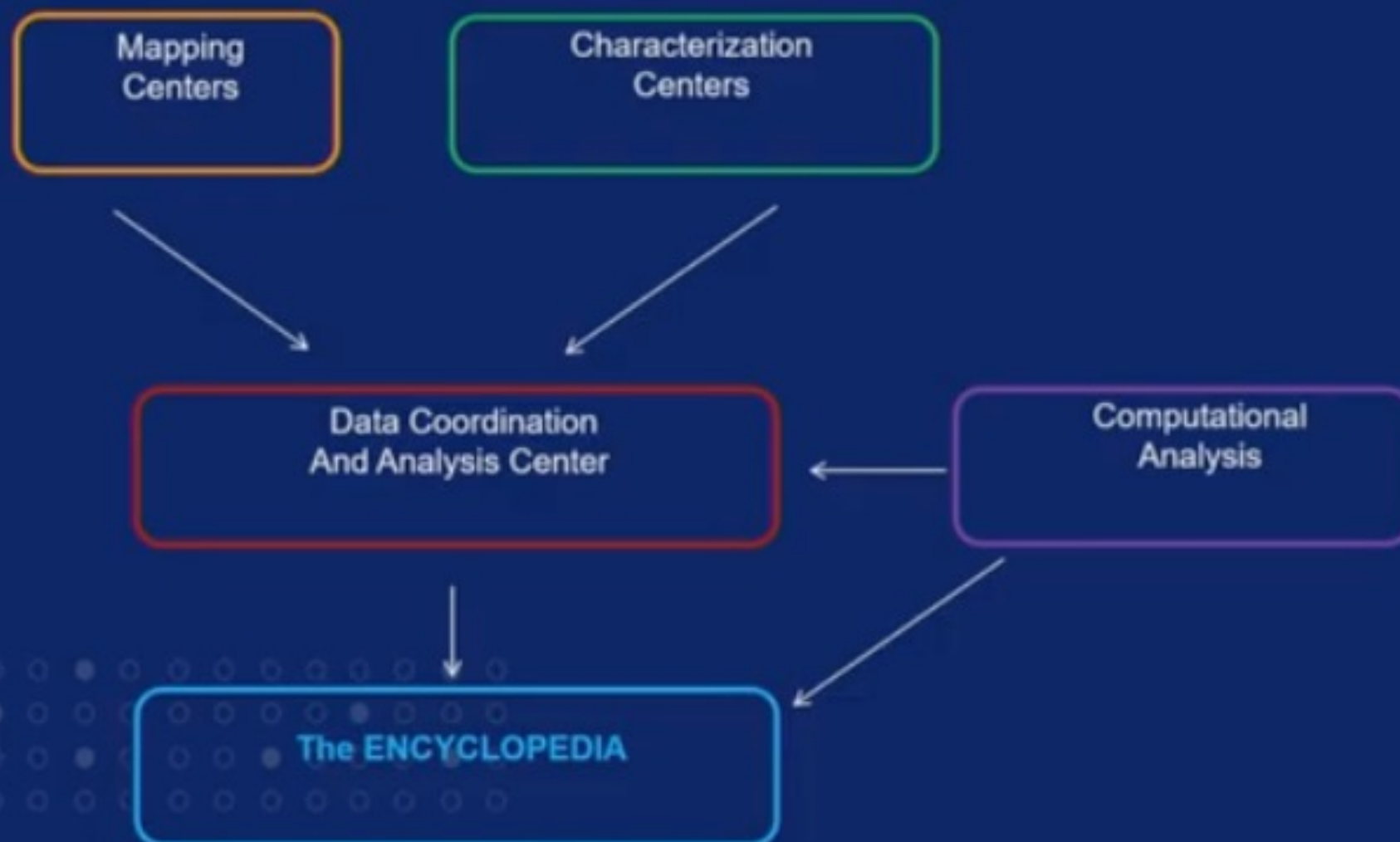
Affiliate Members

Brent Graveley
Gene Yeo

Guo-Cheng Yuan

Alex Dobin

ENCODE 4 Organization



Community Resource

- Rapid pre-publication data release
- Unrestricted access sharing of uniformly processed data, software tools and analysis pipelines
- Analysis requires development of:
 - Common data reporting formats
 - Rich, ontology-based metadata
 - Data standards
 - Uniform Processing Pipelines
 - Analytical tools
- Consortium publications
- Outreach (Online Tutorials, Users Meetings)